

Learning what we never observe

Maximum likelihood, expectation–maximization and hidden-variable models

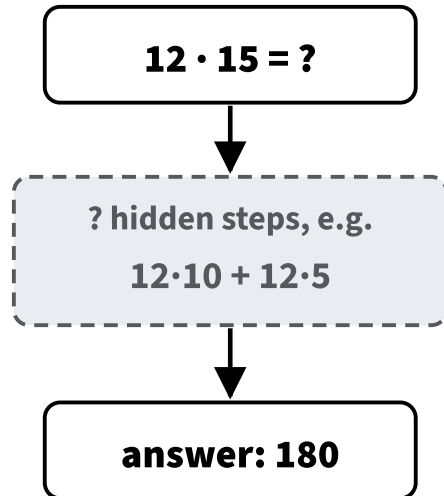
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In real data, the key variable is often never written down

 dashed box: never recorded (hidden)  solid box: in the data

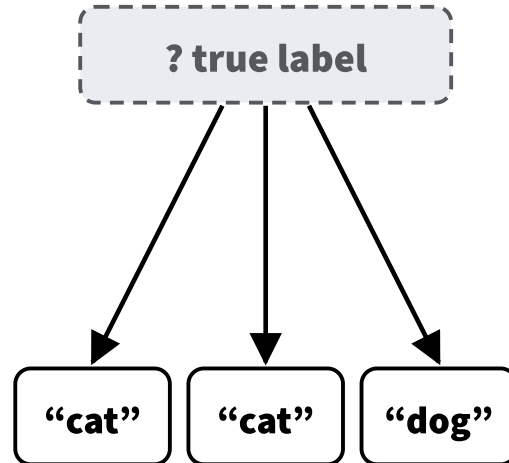
Learning to reason



data: question + answer

to learn from the data:
**good reasoning steps
from answers alone**

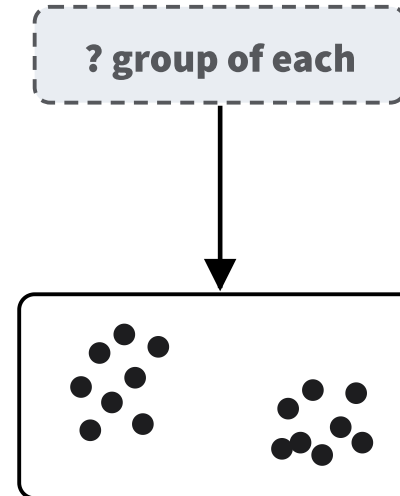
Cleaning labels



data: three annotators' labels
they disagree

to learn from the data:
**the true labels and
each annotator's skill**

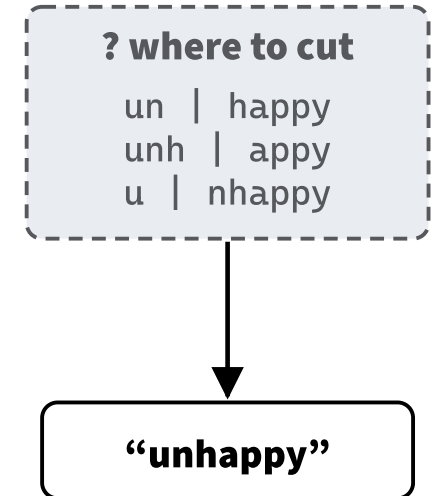
Fast vector search



data: the vectors

to learn from the data:
**the groups, so a query
checks its own group only**

Splitting words



data: the words of a text

to learn from the data:
**how often each piece
occurs, so where to cut**

All four use expectation-maximization (EM) or its hard-label form, k-means.

Two facts from earlier lectures that EM reuses

A coin shows 7 heads in 10 flips. What is our best guess of its heads chance?

1/2

7/10

3/10

Answer. 7/10: count heads, divide by flips. Today we prove this is the **maximum-likelihood** answer.



7 heads (H, blue) and 3 tails (T) in 10 flips

heads ÷ flips = 7/10

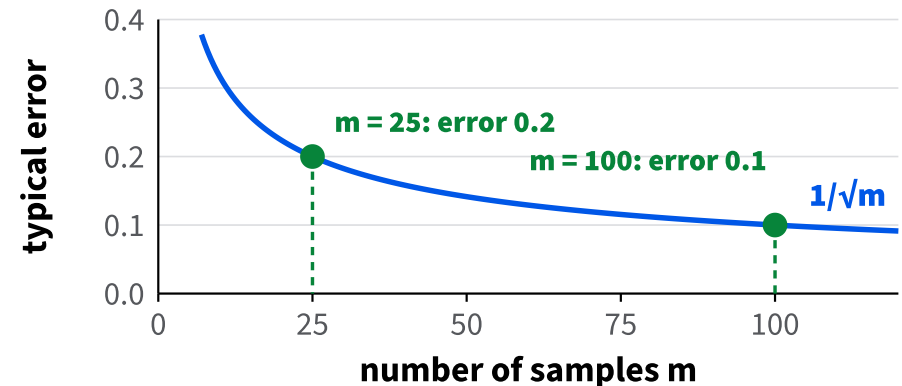
Monte Carlo (Lecture 4): with 4 times as many samples, the error shrinks by a factor of ...

4

1: no change

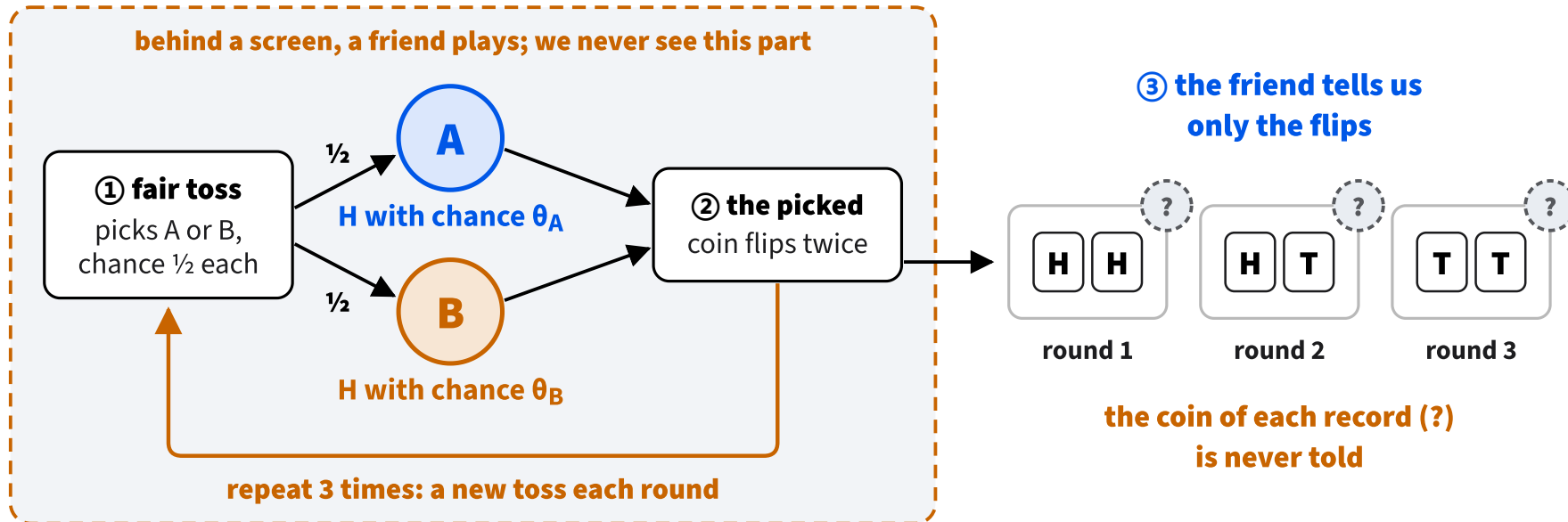
2

Answer. 2: with m samples the typical error is $\sigma/\sqrt{m}$. With spread $\sigma = 1$: $1/\sqrt{25} = 0.2$, $1/\sqrt{100} = 0.1$; so 4 times the samples halves the error, $0.2/0.1 = 2 = \sqrt{4}$.



Both return: counting in EM's maximization step, sampling for too many hidden cases.

The two-coin game: we see the flips, never the coin



We see

three records, **HH**, **HT**, **TT** (H = heads, T = tails); both flips of a record come from one coin.

Hidden

which coin made each record; a new fair toss each round, so the coins can differ.

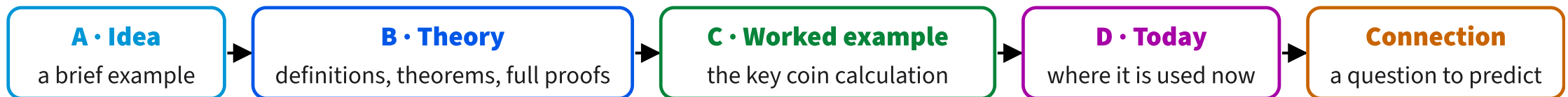
Goal

estimate the heads chances θ_A, θ_B from the three records alone.

Each section: a brief example, the theory, then the **key numbers on these three records.**

Eight questions; each answer raises the next

Section	Question on the coins	Answer
1 Counting	If we are told each coin, what are θ_A, θ_B ?	count heads, divide
2 Hidden variables	Not told: how likely is a record, and which coin made it?	a sum; Bayes' rule
3 EM	Can we still count?	yes, with soft labels: split each record by its posterior
4 Why EM increases the likelihood	Does every round improve the fit?	yes: each round raises a lower bound that touches the log-likelihood
5 Mixtures	What if coins become bell curves?	the same EM, now clustering points
6 Hidden chains	What if the coin switches between flips?	sum-product on a chain
7 Monte Carlo EM	Too many hidden cases to add up?	sample them
8 Trusting the fit	Is the answer right, and the only one?	restart from several guesses; check the data can pin down every unknown



Each section: a brief example, the whole theory, one key coin calculation, today's uses; full calculations in the Appendix.

By the end of today you can do six things

1 • Count

fit a model when nothing is hidden, and say why counting is the **maximum-likelihood** answer.

You can answer: why is 7 heads in 10 flips fitted by 7/10?

2 • Infer

compute a record's probability with a hidden variable, and the **posterior** of that variable.

You can answer: which coin probably made HH?

3 • Run EM

by hand: E-step (expectation: **soft labels**) and M-step (maximization: count and divide).

You can answer: how do soft labels become new heads chances?

4 • Prove

that no EM round lowers the log-likelihood (how well θ fits the data), with **one inequality**.

You can answer: can an EM round ever make the fit worse?

5 • Recognize

EM inside clustering for vector search, tokenizers, crowd labelling and **reasoning models**.

You can answer: where is the hidden coin in a tokenizer?

6 • Judge

a fit: a start that gets stuck, the names A and B swapped, or **more unknowns than the data can fix**.

You can answer: why may EM swap the names A and B?

Six skills, one example: three records of two coin flips.

We see x , never z , and want to learn θ_A, θ_B

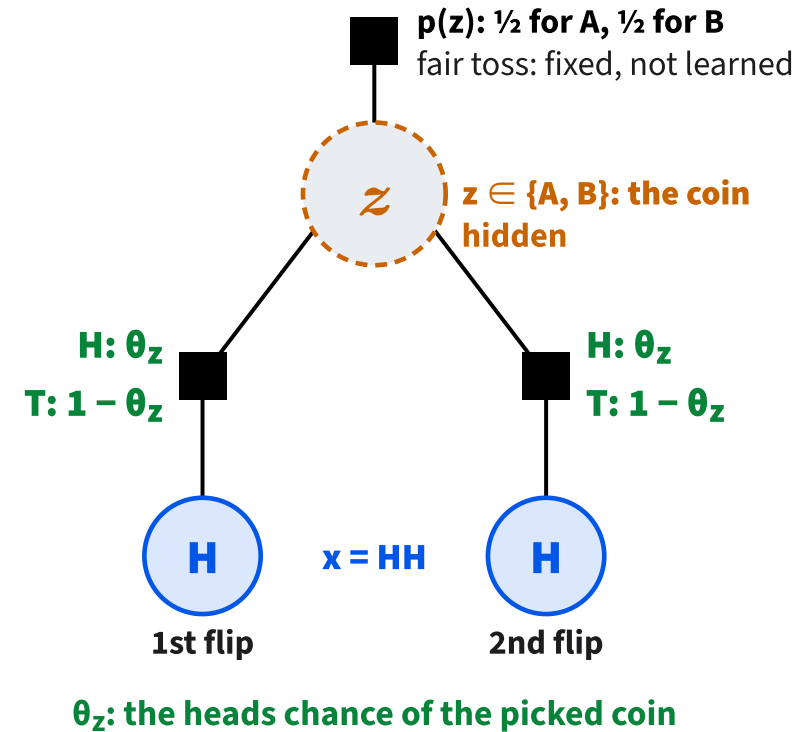
Symbol	Meaning	On the coins
x	observed record	the two flips, e.g. $x = \text{HH}$
z	hidden variable	the coin, $z \in \{A, B\}$
$\theta = (\theta_A, \theta_B)$	parameters	heads chances of A, B (unknown)
$p_\theta(x, z)$	joint probability of record and coin	$p_\theta(\text{HH}, A) = \frac{1}{2} \cdot \theta_A \cdot \theta_A$

Take a starting guess (EM will improve it later): A leans heads, $\theta_A = \frac{3}{4}$; B leans tails, $\theta_B = \frac{1}{4}$.

$$p_\theta(\text{HH}, A) = \underbrace{\frac{1}{2}}_{\text{pick A}} \cdot \underbrace{\frac{3}{4}}_{\text{1st H by A}} \cdot \underbrace{\frac{3}{4}}_{\text{2nd H by A}} = \frac{9}{32}.$$

Lecture 2's factor graph of one record

■ square: a factor (a chance) · ○ circle: a variable



The subscript θ in p_θ means "computed with these parameters"; the toss chance $\frac{1}{2}$ is fixed.

HH has chance 5/16 in all; Bayes' rule gives coin A 9/10 of it

$$p_{\theta}(x) = \sum_z p_{\theta}(x, z)$$

$$p_{\theta}(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

$$p_{\theta}(z \mid x) = \frac{p_{\theta}(x, z)}{p_{\theta}(x)}$$

$$p_{\theta}(A \mid \text{HH}) = \frac{9/32}{10/32} = \frac{9}{10}$$

Likelihood of a record: add the joint over the hidden coin.

9 HH squares of A plus 1 of B, each worth 1/32.

Posterior (Bayes' rule): which coin, given the flips.

9 of the 10 HH squares are A's.

coin A picked
chance $\frac{1}{2} \cdot \text{heads } \frac{3}{4}$

	H	H	H	T
H				
H				
H				
T				

9 squares are HH

coin B picked
chance $\frac{1}{2} \cdot \text{heads } \frac{1}{4}$

	H	T	T	T
H				
T				
T				
T				

1 square is HH

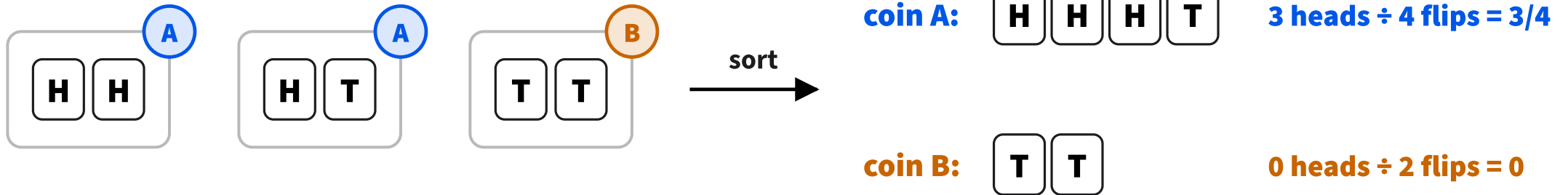
each row (1st flip) or column (2nd flip) is $\frac{1}{4}$,
so a square = $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$

Log-likelihood: each round has its own fair toss, so the records are independent given θ , and $p_{\theta}(\text{all three}) = p_{\theta}(\text{HH}) \cdot p_{\theta}(\text{HT}) \cdot p_{\theta}(\text{TT})$; the log turns the product into a sum: $\ell(\theta) = \ln p_{\theta}(\text{HH}) + \ln p_{\theta}(\text{HT}) + \ln p_{\theta}(\text{TT})$.

Maximum likelihood: pick the θ with the largest $\ell(\theta)$. First, the easy case: the coin is told.

If the friend tells us the coin, we just count

Goal: learn θ_A, θ_B , the heads chances of coins A and B, from the records HH, HT, TT. First the easy case: the friend also tells us which coin made each record.



the friend now says which coin made each record

estimate = heads ÷ flips, coin by coin

Recall the game (p. 4)

A fair toss picks coin A or B, flipped twice. Normally we see **only the flips**; for now the friend also names the coin.

Estimate

heads ÷ flips: $\theta_A = \frac{3}{4}$ and $\theta_B = \frac{0}{2} = 0$.

Two questions

Why is counting the **best** guess? Can coin B really **never** land heads?

With the coin told, learning is counting: heads ÷ flips, coin by coin.

The likelihood scores each heads chance by the flips it explains

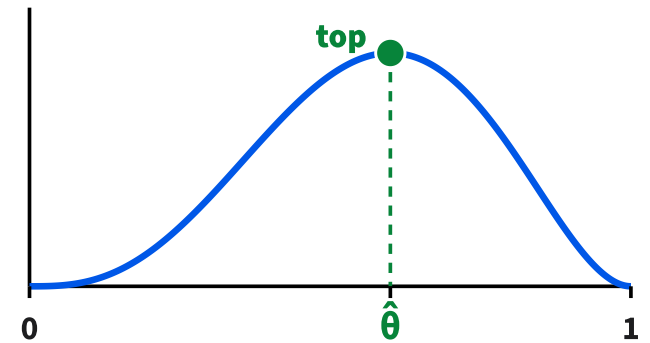
Definitions. A coin with heads chance θ shows h heads and t tails in independent flips. The **likelihood** $L(\theta)$ is the probability of these flips, read as a function of the unknown θ . The **maximum-likelihood estimate** $\hat{\theta}$ is the θ with the largest likelihood.

$$\begin{aligned}
 L(\theta) &= \underbrace{\theta \cdots \theta}_{h \text{ heads}} \cdot \underbrace{(1 - \theta) \cdots (1 - \theta)}_{t \text{ tails}} && \text{independent flips multiply} \\
 &= \theta^h \cdot (1 - \theta)^t && \text{count the factors} \\
 \ell(\theta) &= \ln L(\theta) = h \cdot \ln \theta + t \cdot \ln(1 - \theta) && \text{log of a product = sum of logs} \\
 \hat{\theta} &= \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta) && \text{ln only rises: same top}
 \end{aligned}$$

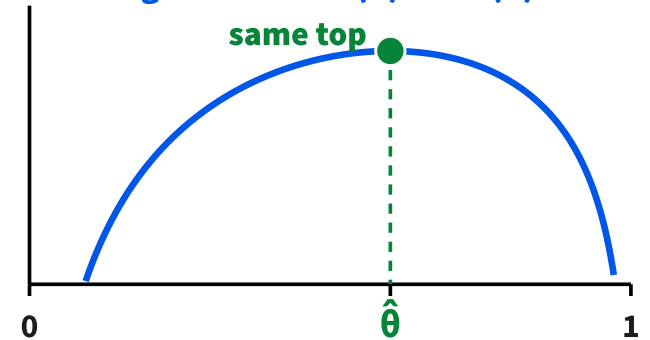
In words: the likelihood multiplies one factor per flip; its log ℓ adds them, and $\arg \max$ names the θ at the top.

Why it matters: a sum is easier to differentiate than a product, so we find $\hat{\theta}$ by setting the slope of ℓ to zero.

$$\text{likelihood } L(\theta) = \theta^h \cdot (1 - \theta)^t$$



$$\text{log-likelihood } \ell(\theta) = \ln L(\theta)$$



ln only rises: same peak $\hat{\theta}$

Maximum likelihood picks the θ that makes the flips we saw most probable.

For any coin, the counted fraction maximizes the likelihood

Theorem 1 (maximum likelihood by counting). Independent flips of a coin with heads chance θ show h heads and t tails. Then $\ell(\theta) = h \cdot \ln \theta + t \cdot \ln(1 - \theta)$ is largest at $\hat{\theta} = \frac{h}{h+t}$.

$$\ell'(\theta) = h \cdot \frac{1}{\theta} + t \cdot \frac{-1}{1-\theta}$$

$$= \frac{h}{\theta} - \frac{t}{1-\theta}$$

$$\ell'(\theta) = 0 \iff h \cdot (1-\theta) = t \cdot \theta$$

$$\iff h = (h+t) \cdot \theta$$

$$\iff \theta = \frac{h}{h+t}$$

$$\ell''(\theta) = -\frac{h}{\theta^2} - \frac{t}{(1-\theta)^2} < 0$$

$$h = 0: \ell = t \cdot \ln(1-\theta) \text{ falls}$$

$$t = 0: \ell = h \cdot \ln \theta \text{ rises}$$

$$(\ln \theta)' = 1/\theta; (\ln u)' = u'/u, \\ u = 1 - \theta, u' = -1$$

simplify

interior top ($h, t > 0$); times $\theta \cdot (1 - \theta) > 0$

expand $h - h \cdot \theta = t \cdot \theta$, add $h \cdot \theta$

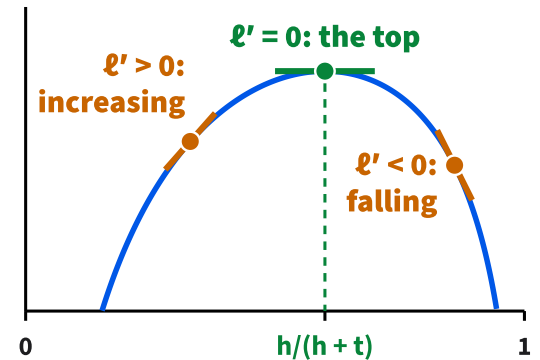
divide by $h + t$

Check: curves down: one top

Check: top at $\theta = 0 = \frac{0}{0+t}$

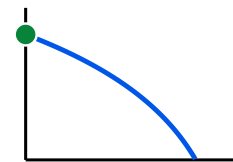
Check: top at $\theta = 1 = \frac{h}{h+0}$. ■

$$\ell(\theta) = h \cdot \ln \theta + t \cdot \ln(1 - \theta)$$

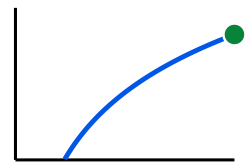


$h = 0: \ell$ falls

$t = 0: \ell$ rises



top at 0



top at 1

In words: the best heads chance is the fraction of heads seen. **Why it matters:** with each flip's coin known, learning needs no search: count, then divide.

Whatever the counts, the likelihood peaks at heads ÷ flips.

A zero count makes maximum likelihood claim certainty

↳ Theorem 1 (p. 11) with no heads ($h = 0$) in t flips gives $\hat{\theta} = \frac{0}{0+t} = 0$.

What does this fitted coin predict for a later record x that contains a head?

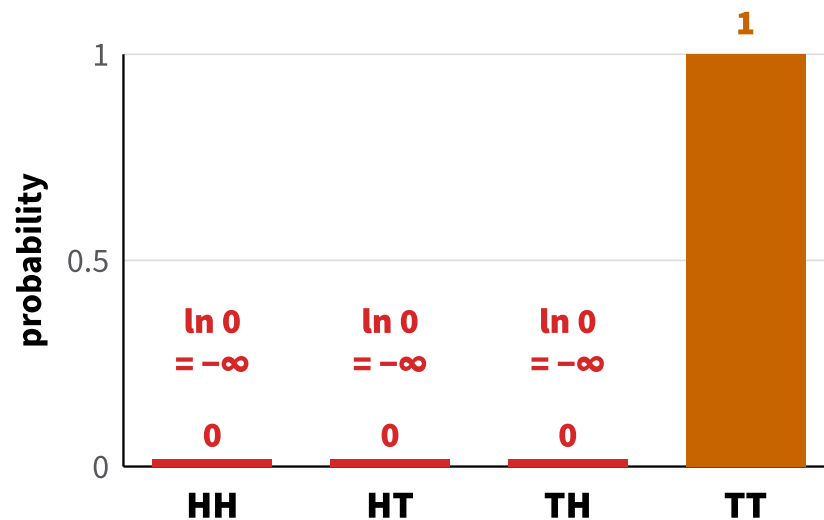
$p(H \mid \hat{\theta}) = \hat{\theta} = 0$	one head, fitted coin
$p(x \mid \hat{\theta}) = \hat{\theta} \cdot (\text{other factors}) = 0$	flips multiply
$\ln p(x \mid \hat{\theta}) = \ln 0 = -\infty$	log-likelihood of x

Weak evidence: even a fair coin shows t tails in a row with chance $(\frac{1}{2})^t$, which is not rare when t is small.

In words: "no heads seen" is read as "heads impossible".

Why it matters: one later head makes the log-likelihood $-\infty$. The fix: keep θ uncertain with a prior, and average.

fitted $\hat{\theta} = 0$: chance of the next two flips



every record with a head is called impossible;
all the probability sits on tails only

Maximum likelihood turns an unseen outcome into an impossible one.

Averaging over the posterior gives the chance of the next head

Definitions. Treat θ as unknown. The **uniform prior** makes every value in $[0, 1]$ equally likely before any flip: $p(\theta) = 1$. The **posterior** $p(\theta \mid \text{flips})$ is what we believe after the flips; the **posterior mean** is its average, $\mathbb{E}[\theta \mid \text{flips}] = \int_0^1 \theta \cdot p(\theta \mid \text{flips}) d\theta$. Below, u is an integration variable on $[0, 1]$.

$$p(\theta \mid \text{flips}) = \frac{p(\theta) \cdot L(\theta)}{\int_0^1 p(u) \cdot L(u) du}$$

Bayes' rule

$$= \frac{\theta^h \cdot (1 - \theta)^t}{\int_0^1 u^h \cdot (1 - u)^t du}$$

$$p(\theta) = 1, \\ L(\theta) = \theta^h \cdot (1 - \theta)^t$$

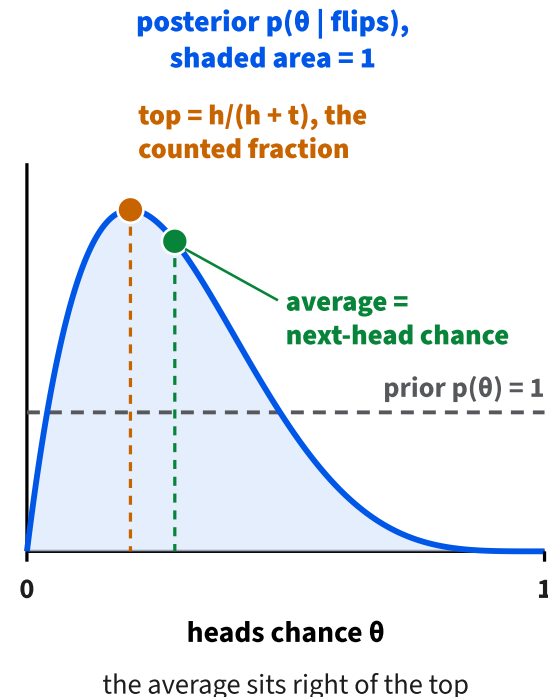
$$p(\text{next H} \mid \text{flips}) = \int_0^1 p(\text{next H} \mid \theta) \cdot p(\theta \mid \text{flips}) d\theta$$

total probability

$$= \int_0^1 \theta \cdot p(\theta \mid \text{flips}) d\theta = \mathbb{E}[\theta \mid \text{flips}] \quad p(\text{next H} \mid \theta) = \theta$$

In words: the next-head chance is the heads chance averaged over what we **still believe**.

Why it matters: every $\theta > 0$ that keeps posterior weight lifts the average above 0.



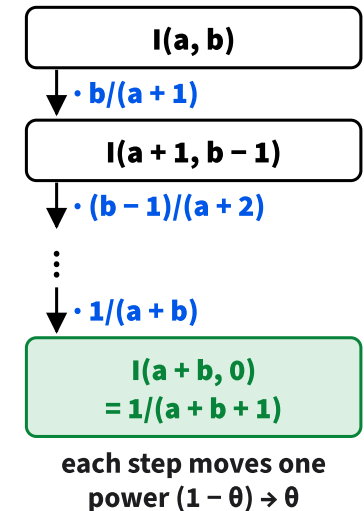
The posterior's top is the counted fraction; its average is the next-head chance.

The Beta integral turns powers of θ and $1 - \theta$ into factorials

Both integrals in the posterior mean have the form $\int_0^1 \theta^a \cdot (1 - \theta)^b d\theta$, the area under the likelihood of a heads and b tails. **Beta integral** for whole numbers $a, b \geq 0$: $I(a, b) = \int_0^1 \theta^a \cdot (1 - \theta)^b d\theta = \frac{a! b!}{(a + b + 1)!}$. If $b = 0$, $I(a, 0) = \frac{1}{a+1}$ directly; for $b \geq 1$, integrate by parts:

$$\begin{aligned}
 I(a, b) &= \left[\frac{\theta^{a+1}}{a+1} \cdot (1 - \theta)^b \right]_0^1 - \int_0^1 \frac{\theta^{a+1}}{a+1} \cdot (-b \cdot (1 - \theta)^{b-1}) d\theta \\
 &= \left[\dots \right]_0^1 + \frac{b}{a+1} \cdot I(a+1, b-1) \\
 &= \frac{b}{a+1} \cdot I(a+1, b-1) \\
 &= \frac{b}{a+1} \cdot \frac{b-1}{a+2} \cdots \frac{1}{a+b} \cdot I(a+b, 0) \\
 &= \frac{a! b!}{(a+b)!} \cdot \frac{1}{a+b+1} = \frac{a! b!}{(a+b+1)!} \blacksquare
 \end{aligned}$$

$\int u' \cdot v = [u \cdot v] - \int u \cdot v'$,
 $u' = \theta^a$, $v = (1 - \theta)^b$
 minus times minus;
 the integral is $I(a+1, b-1)$
 bracket = 0: $\theta^{a+1} = 0$ at 0,
 $(1 - \theta)^b = 0$ at 1 ($b \geq 1$)
 repeat $(a, b) \rightarrow (a+1, b-1)$,
 b times; $I(a+b, 0) = 1/(a+b+1)$
 tops: $b!$; bottoms: $(a+b)!/a!$;
 $(a+b)! (a+b+1) = (a+b+1)!$



the b factors multiply to $a! b!/(a+b)!$

In words: each step trades one factor $(1 - \theta)$ for one θ and a known fraction. **Why it matters:** both integrals in the posterior mean become factorials, which cancel.

The Beta integral turns every posterior integral into factorials.

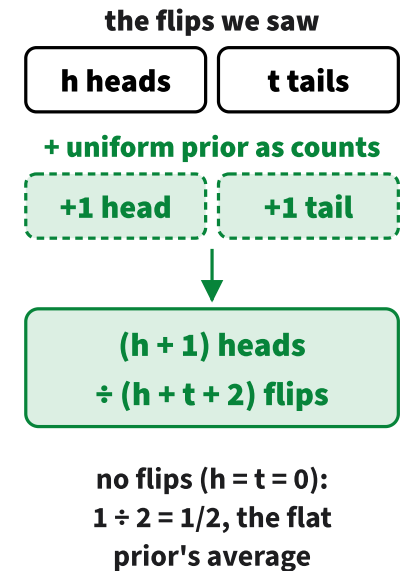
Laplace's rule: the posterior mean adds one head and one tail

Theorem 2 (Laplace's rule). Put a uniform prior on θ , then see h heads and t tails. The posterior mean is $\mathbb{E}[\theta \mid \text{flips}] = \frac{h+1}{h+t+2}$.

$$\begin{aligned}
 \mathbb{E}[\theta \mid \text{flips}] &= \int_0^1 \theta \cdot \frac{\theta^h \cdot (1-\theta)^t}{\int_0^1 u^h \cdot (1-u)^t du} d\theta && \text{posterior mean;} \\
 &= \frac{\int_0^1 \theta \cdot \theta^h \cdot (1-\theta)^t d\theta}{\int_0^1 \theta^h \cdot (1-\theta)^t d\theta} && \text{Bayes with prior 1} \\
 &= \frac{I(h+1, t)}{I(h, t)} && \text{the denominator is a constant: take it out} \\
 &= \frac{(h+1)! t! / (h+t+2)!}{h! t! / (h+t+1)!} && \theta \cdot \theta^h = \theta^{h+1}: \\
 &= \frac{(h+1)!}{h!} \cdot \frac{(h+t+1)!}{(h+t+2)!} && \text{top } (a, b) = (h+1, t) \\
 &= (h+1) \cdot \frac{1}{h+t+2} = \frac{h+1}{h+t+2} \blacksquare && \text{Beta integral, top and below}
 \end{aligned}$$

divide; $t!$ cancels

$$\frac{(h+1)!}{h!} = h+1, \quad \frac{(h+t+1)!}{(h+t+2)!} = \frac{1}{h+t+2}$$



In words: add one imagined head and one imagined tail, then count. **Why +1 and +1?** With no flips ($h = t = 0$): $\frac{0+1}{0+2} = \frac{1}{2}$, the flat prior's average, so **the imagined head and tail are the uniform prior written as counts.**

Laplace's rule is counting with one imagined head and one imagined tail.

Laplace's rule pulls the count toward 1/2 by a weight ε

Corollary of Theorem 2. Over $n = h + t \geq 1$ flips, Laplace's value is a mix of the **counted fraction** $\frac{h}{n}$ and the even guess $\frac{1}{2}$:

$$\frac{h+1}{n+2} = (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2}, \quad \varepsilon = \frac{2}{n+2}.$$

$$\begin{aligned} (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2} &= \frac{n}{n+2} \cdot \frac{h}{n} + \frac{2}{n+2} \cdot \frac{1}{2} & 1 - \varepsilon &= \frac{n}{n+2} \\ &= \frac{h}{n+2} + \frac{1}{n+2} & (1 - \varepsilon) \cdot \frac{h}{n} &= \frac{h}{n+2}, \\ &= \frac{h+1}{n+2} & \varepsilon \cdot \frac{1}{2} &= \frac{1}{n+2} \end{aligned}$$

same denominator

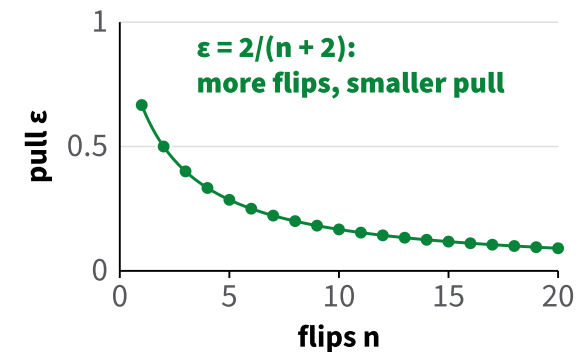
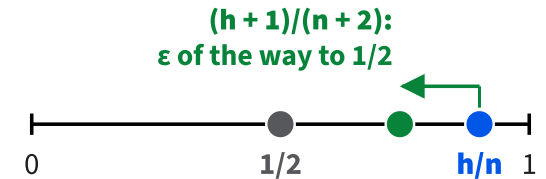
$$\begin{aligned} \frac{h+1}{n+2} - \frac{h}{n} &= (1 - \varepsilon) \cdot \frac{h}{n} + \varepsilon \cdot \frac{1}{2} - \frac{h}{n} & \text{the mix equals } (h+1)/(n+2); \\ & & \text{subtract } h/n \\ &= \varepsilon \cdot \left(\frac{1}{2} - \frac{h}{n} \right) & (1 - \varepsilon) \cdot h/n - h/n &= -\varepsilon \cdot h/n; \\ & & \text{factor out } \varepsilon \end{aligned}$$

In words: Laplace's value $\frac{h+1}{n+2}$ sits ε of the way from the count to $\frac{1}{2}$.

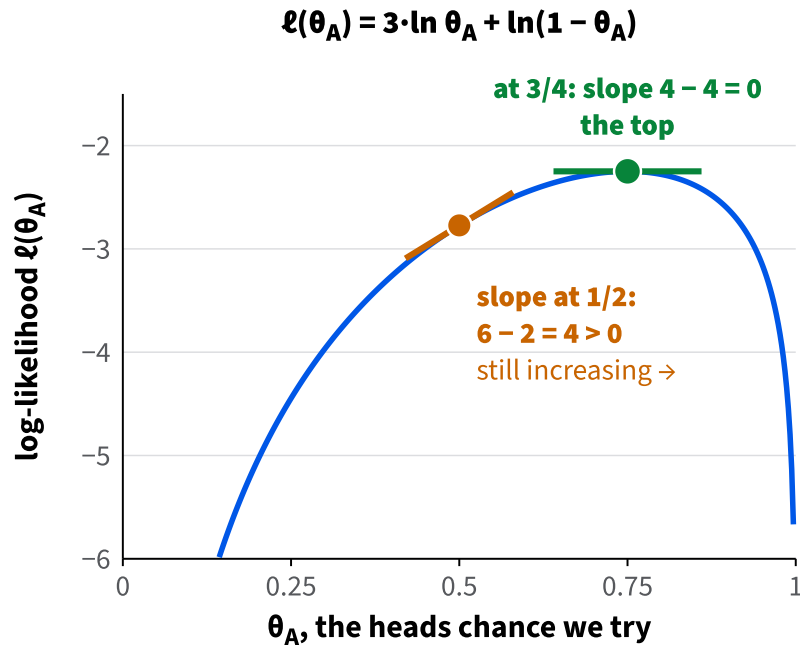
Why it matters: more flips shrink ε , so the data soon **outweigh** the two imagined flips.

Laplace moves the count $\varepsilon = \frac{2}{n+2}$ of the way to $\frac{1}{2}$; more flips, smaller pull.

one coin: n flips, h heads



Coin A's log-likelihood has slope zero at $\theta_A = 3/4$



↳ Theorem 1 (p. 11) applied to coin A, which gave H, H, H, T ($h = 3, t = 1$); independent flips multiply, so $L(\theta_A) = \theta_A^3 \cdot (1 - \theta_A)$. Theorem 1's slope: $\ell'(\theta) = \frac{h}{\theta} - \frac{t}{1-\theta}$.

$$\hat{\theta}_A = \frac{h}{h+t} = \frac{3}{3+1} = \frac{3}{4} \quad \text{Theorem 1, } h = 3, t = 1$$

$$\ell'\left(\frac{3}{4}\right) = \frac{3}{3/4} - \frac{1}{1/4} = 4 - 4 = 0 \quad \text{check: slope zero at the top}$$

Left of the top the log-likelihood still increases:

$$\ell'\left(\frac{1}{2}\right) = \frac{3}{1/2} - \frac{1}{1/2} = 6 - 2 = 4 > 0.$$

All steps: Appendix, Section 1 (p. 138).

Maximum likelihood gives $\theta_A = \frac{3}{4}$, the counted fraction 3 of 4.

Counting says coin B can never land heads

counting only

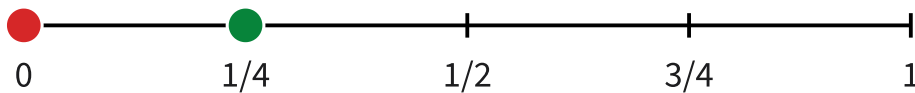
$$\boxed{T} \boxed{T} \longrightarrow 0 \text{ heads} \div 2 \text{ flips} = 0$$

with one imagined head and one imagined tail

$$\boxed{T} \boxed{T} + \boxed{H} \boxed{T} \longrightarrow 1 \text{ head} \div 4 \text{ flips} = 1/4$$

0: heads impossible

1/4: heads rare, still possible



estimate of θ_B , coin B's heads chance

↳ Theorem 1's zero-count case (p. 12) on coin B, which showed T, T: $\theta_B = \frac{0}{0+2} = 0$.

Too sure. Suppose B is flipped again and shows HT:

$$p(\text{HT} \mid B) = \theta_B \cdot (1 - \theta_B) = 0 \cdot 1 = 0.$$

Its log-likelihood is $\ln 0 = -\infty$: the model calls a record we could easily see **impossible**.

Weak evidence: a fair coin shows TT with chance $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Fix (Theorem 2, Laplace's rule) (p. 15): add one imagined head and one imagined tail (dashed):

$$\theta_B = \frac{0 + 1}{2 + 2} = \frac{1}{4}, \quad p(\text{HT} \mid B) = \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}.$$

Two tails cannot prove heads impossible; imagined flips keep every outcome possible.

Laplace's rule on the coins: B rises from 0 to $1/4$, A drops to $2/3$

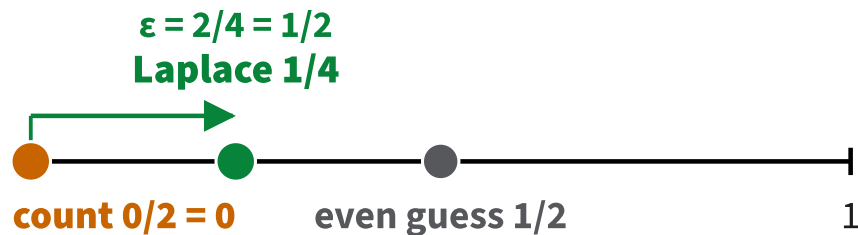
↳ Theorem 2 (p. 15) and its corollary (p. 16) applied to both coins:
Laplace's value $\frac{h+1}{n+2}$ is the count $\frac{h}{n}$ moved the weight $\varepsilon = \frac{2}{n+2}$ of the way to $\frac{1}{2}$.

	coin B: T, T	coin A: H, H, H, T
count $\frac{h}{n}$	$\frac{0}{2} = 0$	$\frac{3}{4}$
weight $\varepsilon = \frac{2}{n+2}$	$\frac{2}{2+2} = \frac{1}{2}$	$\frac{2}{4+2} = \frac{1}{3}$
Laplace $\frac{h+1}{n+2}$	$\frac{0+1}{2+2} = \frac{1}{4}$	$\frac{3+1}{4+2} = \frac{2}{3}$

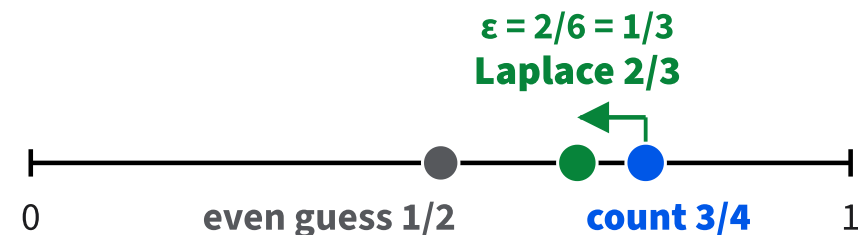
Coin A has more flips (4, not 2), so its pull is smaller: $\frac{1}{3} < \frac{1}{2}$.

All steps: Appendix, Section 1 (p. 139).

coin B: T, T (n = 2 flips)



coin A: H, H, H, T (n = 4 flips)



Laplace moves the count ε of the way
toward the even guess $1/2$

Two flips: Laplace moves coin B halfway to $\frac{1}{2}$. Four flips: coin A only a third.

Label smoothing gives every class an imagined count

An image classifier learns from photos labelled cat, dog or fox. **Label smoothing** gives every class a small imagined share of each label, as Laplace's rule gave heads and tails imagined flips.

COIN GAME

counted fraction of one flip (H:

$$\frac{1}{1} = 1)$$

$\varepsilon \cdot \frac{1}{2}$ added to heads and to tails (
 $K = 2$ outcomes)

$\varepsilon = \frac{2}{n+2}$, shrinks with more flips

even guess $\frac{1}{2}$

IMAGE CLASSIFIER

hard target (1, 0, 0) of one cat
 photo

$\Leftrightarrow \varepsilon/K$ added to every class

$\Leftrightarrow \varepsilon$ fixed by hand (0.1 in practice)

$\Leftrightarrow$ even guess $\frac{1}{K}$

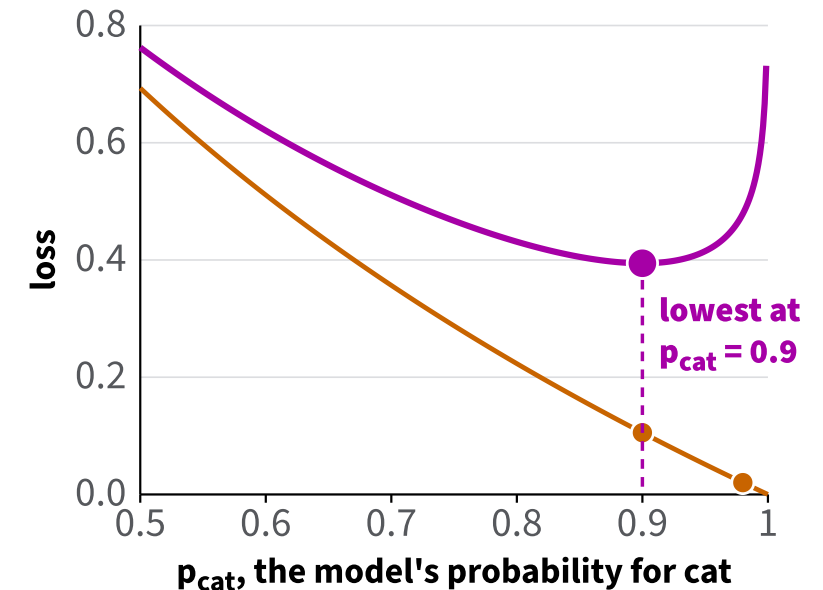
Hard target: training pushes p_{cat} toward 1. Smoothed target: the loss is **lowest at $p_{\text{cat}} = 0.9$** , not at 1 (figure).

Szegedy et al., "Rethinking the Inception Architecture for Computer Vision," CVPR = Conference on Computer Vision and Pattern Recognition, 2016: $\varepsilon = 0.1$ with $K = 1000$ classes.

Worked example: Appendix, Section 1 (p. 140).

smoothed target (0.9, 0.05, 0.05)

hard target (1, 0, 0)



Label smoothing is Laplace's rule for classes: no class gets target 0, so no forced certainty.

Next: the friend stops telling us the coin

we are here

Counting

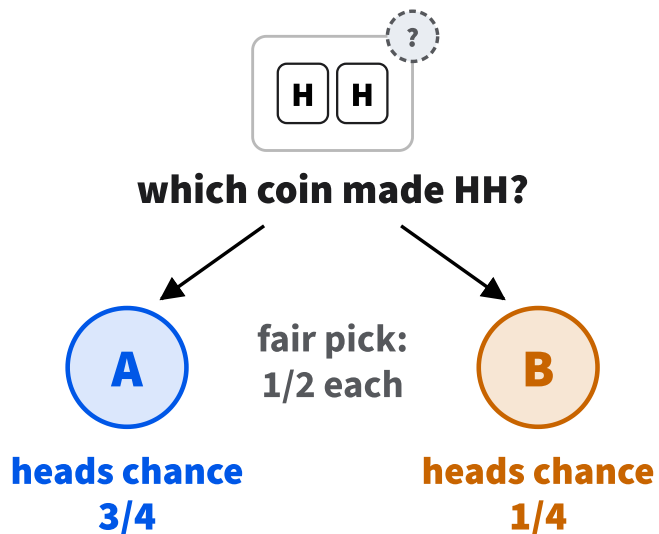
Hidden
variables

next

EM

Why EM
increases ℓ

Mixtures

Hidden
chainsMonte Carlo
EMTrusting the
fit

Counting needs each flip's coin. Now the friend reports only "HH".

Working guess from now on: $\theta_A = \frac{3}{4}$ (A's counted fraction (p. 17), kept because 4 flips make the pull small), $\theta_B = \frac{1}{4}$ (B's Laplace value (p. 19): heads stays possible), fair pick. How likely is HH?

9/16

5/16

1/16

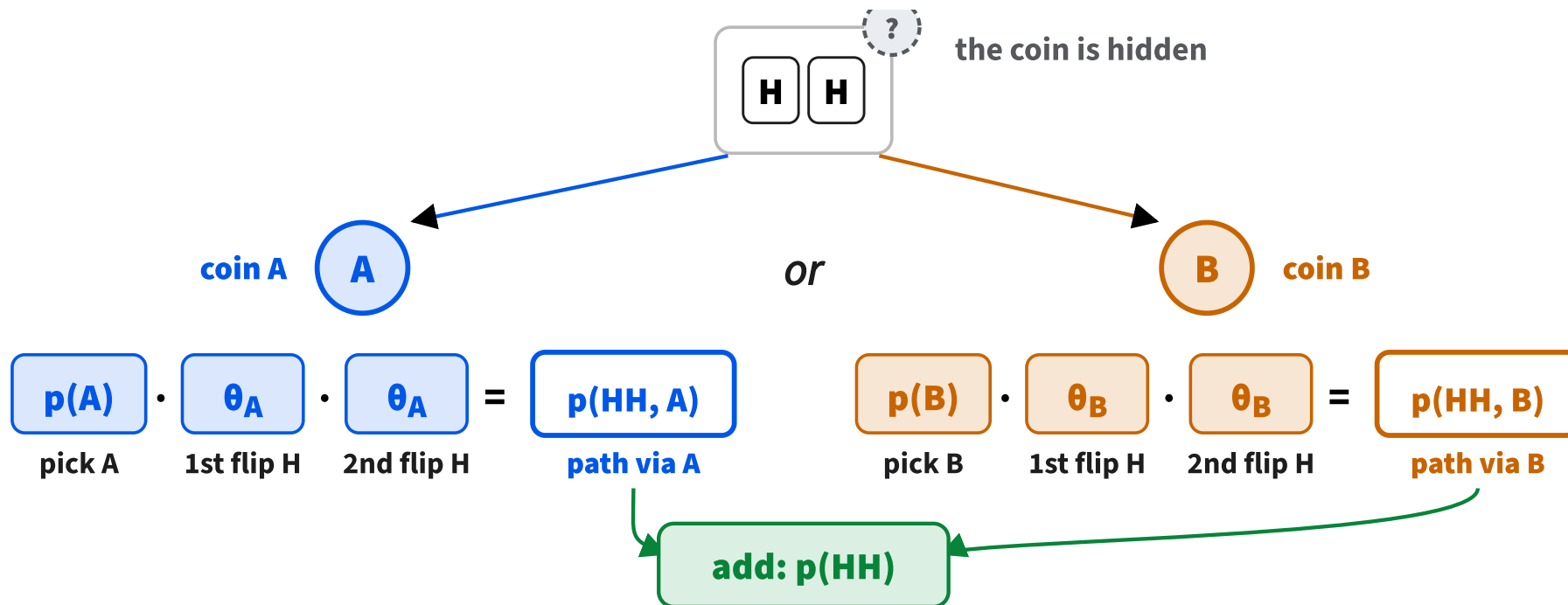
Answer. $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$: a sum over the two coins, each weighted by its pick chance $\frac{1}{2}$.

A hidden coin turns one probability into a sum over the coins.

With the coin hidden, the chance of HH adds two paths

Goal: learn θ_A, θ_B from HH, HT, TT, now that the friend no longer says which coin was used.

Recall the game (p. 4): a fair toss picks coin A or coin B; θ_A, θ_B are their heads chances. Working guess: $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$.



Why add? HH came from A **or** from B, never both, so the chances of the two paths add.

$p(HH)$ = path via A + path via B: a sum over the hidden coin.

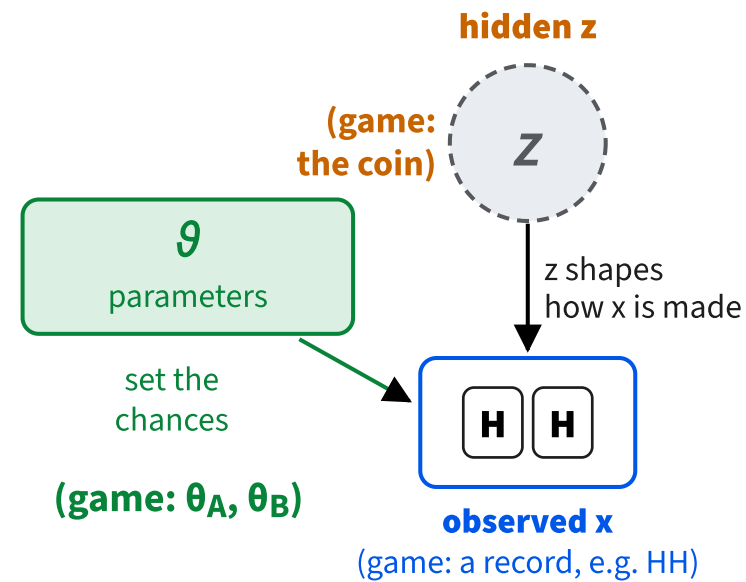
In general: an observed x , a hidden z , parameters θ

Definition (hidden-variable model). A model $p_\theta(x, z)$ gives the chance of an observed record x **together with** a hidden variable z ; θ are its parameters. Each value of z is one **path** to x . The data are N independent records $x^{(1)}, \dots, x^{(N)}$.

name	formula	what it does
marginal likelihood	$p_\theta(x) = \sum_z p_\theta(x, z)$	adds the paths
log-likelihood	$\ell(\theta) = \sum_{n=1}^N \ln p_\theta(x^{(n)})$	adds the records' logs
posterior	$p_\theta(z x) = \frac{p_\theta(x, z)}{p_\theta(x)}$	path ÷ total

Why "marginal"? Adding over z is called marginalizing z out: what is left is the chance of x alone.

Goal: the θ with the largest $\ell(\theta)$, the **maximum-likelihood estimate**.



$$p_\theta(x, z) = p(z) \cdot p_\theta(x | z)$$

$p(z)$: chance of the hidden value

$p_\theta(x | z)$: chance of x given z

one value of z = one path to x

Three formulas: add the paths, take logs, divide for the posterior.

With z told, the log splits into one term per parameter

↳ Recall Theorem 1 (counting) (p. 11): a coin with h heads and t tails has best heads chance $\frac{h}{h+t}$. It needs each coin's flips **separately**. In the game, $z \in \{A, B\}$ is the coin and $\theta = (\theta_A, \theta_B)$.

z told: the log of the complete data

$$\sum_n \ln p_\theta(x^{(n)}, z^{(n)}) = \sum_n [\ln p(z^{(n)}) + \ln p_\theta(x^{(n)} | z^{(n)})]$$

log of the product $p(z) \cdot p_\theta(x | z)$

$$= \sum_n \ln p(z^{(n)}) + \sum_{z^{(n)}=A} \ln p_\theta(x^{(n)} | A) + \sum_{z^{(n)}=B} \ln p_\theta(x^{(n)} | B)$$

group the records by coin

$$= \underbrace{\sum_n \ln p(z^{(n)})}_{\text{no } \theta} + \underbrace{\sum_{z^{(n)}=A} \ln p_{\theta_A}(x^{(n)} | A)}_{\text{only } \theta_A} + \underbrace{\sum_{z^{(n)}=B} \ln p_{\theta_B}(x^{(n)} | B)}_{\text{only } \theta_B}$$

each coin uses only its own chance:

$$p_\theta(x | A) = p_{\theta_A}(x | A), \\ p_\theta(x | B) = p_{\theta_B}(x | B)$$



In words: with z told, each parameter has its own sum of logs, so each is maximized alone.

Why it matters: this is why Theorem 1 works coin by coin: θ_A is fitted to A's flips only, θ_B to B's flips only.

z told $\Rightarrow$ separate terms $\Rightarrow$ count each coin's flips (Theorem 1).

A log of a sum does not split, so counting no longer works

↳ Now z is hidden: record n enters through $p_\theta(x) = \sum_z p_\theta(x, z)$, with paths $\underbrace{p(A) \cdot p_{\theta_A}(x^{(n)} | A)}_{a_n: \text{ only } \theta_A}$ via A and $\underbrace{p(B) \cdot p_{\theta_B}(x^{(n)} | B)}_{b_n: \text{ only } \theta_B}$ via B.

z hidden: one log holds every parameter

$$\begin{aligned} \ell(\theta) &= \sum_n \ln p_\theta(x^{(n)}) && \text{definition of } \ell \\ &= \sum_n \ln(a_n + b_n) && \text{marginal likelihood: add the paths} \\ &\neq \sum_n \ln(a_n \cdot b_n) && \text{a sum is not a product} \\ &= \sum_n [\ln a_n + \ln b_n] && \text{a log splits products: } \ln a + \ln b = \ln(a \cdot b) \end{aligned}$$

z told: one path per record, a product

$$\ln(a \cdot b) = \ln a + \ln b$$

splits: one term per parameter

vs

z hidden: two paths per record, a sum

$$\ln(a + b) \neq \ln a + \ln b$$

one log holds θ_A and θ_B

In words: with z hidden, all parameters share one log of a sum, and it does not split.

Why it matters: ℓ has no term with θ_A alone, so Theorem 1's per-coin counting (p. 11) has nothing to act on.

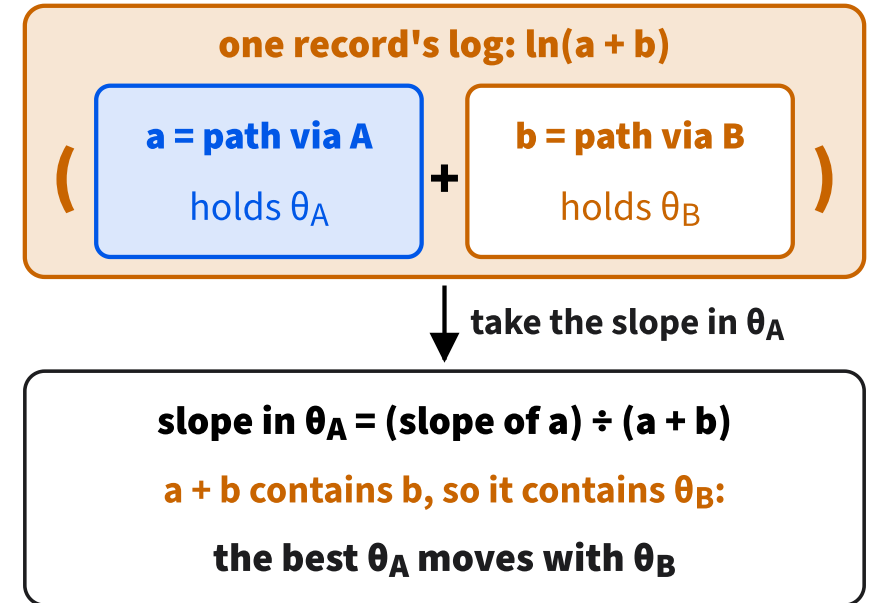
A log of a sum is not a sum of logs: ℓ has no separate term per parameter.

The slope of ℓ in θ_A still holds θ_B , so the parameters are tied

↳ Recall $\ell(\theta) = \sum_n \ln(a_n + b_n)$ (p. 25): path a_n holds only θ_A , path b_n only θ_B . To increase ℓ , take its slope in θ_A .

z hidden: the slope of a log of a sum

$$\begin{aligned} \frac{\partial \ell}{\partial \theta_A} &= \sum_n \frac{1}{a_n + b_n} \cdot \frac{\partial(a_n + b_n)}{\partial \theta_A} && \text{chain rule } (\ln f)' = \frac{f'}{f} \\ &= \sum_n \frac{\partial a_n / \partial \theta_A + \partial b_n / \partial \theta_A}{a_n + b_n} && \text{slope of a sum = sum of slopes} \\ &= \sum_n \frac{\partial a_n / \partial \theta_A}{a_n + b_n} && b_n \text{ holds only } \theta_B, \text{ so } \partial b_n / \partial \theta_A = 0; \\ &&& \text{the denominator still holds } \theta_B \\ &= \sum_n \underbrace{\frac{a_n}{a_n + b_n}}_{p_\theta(A|x^{(n)})} \cdot \frac{\partial \ln a_n}{\partial \theta_A} && a' = a \cdot (\ln a)'; \\ &&& \text{path } \div \text{ total} = \text{posterior} \end{aligned}$$



In words: the slope keeps the whole sum $a_n + b_n$ in its denominator.

Why it matters: the slope in θ_A contains θ_B , so the best θ_A depends on θ_B and Theorem 1 cannot be used coin by coin.

Hidden $z \Rightarrow$ a log of a sum $\Rightarrow$ the parameters are tied together: no counting formula.

The slope of a log of a sum is a posterior-weighted slope

↳ Recall (p. 26): $\frac{\partial \ell}{\partial \theta_A} = \sum_n p_\theta(A | x^{(n)}) \cdot \frac{\partial \ln a_n}{\partial \theta_A}$, the posterior of A · the slope of $\ln a_n$. Theorem 3 says this holds for every model.

Theorem 3 (Fisher's identity). Let θ_j be any one entry of θ (in the game, θ_A or θ_B). Then

$$\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z).$$

In words: the slope of the log of a sum is each path's slope, weighted by its posterior.

Why a slope? To increase ℓ toward its maximum, we need its slope in each parameter.

Why it matters: one path's log splits, $\ln p_\theta(x, z) = \ln p(z) + \ln p_\theta(x | z)$, so its slope is as easy as counting; the weights are the posteriors, which we can compute.

slope of $\ln p_\theta(x)$ in one parameter
a log of a sum: hard to handle directly

=

$p_\theta(z_1 | x)$
· slope of
 $\ln p_\theta(x, z_1)$
one path, no sum
weight · easy slope

+

$p_\theta(z_2 | x)$
· slope of
 $\ln p_\theta(x, z_2)$
one path, no sum
weight · easy slope

+ ... (one term per value of z)

one path splits into separate logs:

$$\ln p_\theta(x, z) = \ln p(z) + \ln p_\theta(x | z)$$

its slope is as easy as counting

game: $z_1 = \text{coin A}$, $z_2 = \text{coin B}$

Slope of a log of a sum = posterior-weighted single-path slopes; those split like counting.

Proof of Theorem 3: chain rule, then Bayes' rule

Proof. One record x . Recall the marginal likelihood (p. 23) $p_\theta(x) = \sum_z p_\theta(x, z)$, posterior $p_\theta(z | x) = p_\theta(x, z) / p_\theta(x)$.

$$\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \frac{1}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} p_\theta(x)$$

$$\text{chain rule } (\ln f)' = \frac{f'}{f}$$

$$= \frac{1}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} \sum_z p_\theta(x, z)$$

$$\text{marginal likelihood } f = p_\theta(x) = \sum_z p_\theta(x, z)$$

$$= \frac{1}{p_\theta(x)} \cdot \sum_z \frac{\partial}{\partial \theta_j} p_\theta(x, z)$$

the slope of a sum is the sum of the slopes

$$= \frac{1}{p_\theta(x)} \cdot \sum_z p_\theta(x, z) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

$$\text{for each path } g = p_\theta(x, z): (\ln g)' = \frac{g'}{g}, \text{ so } g' = g \cdot (\ln g)'$$

$$= \sum_z \frac{p_\theta(x, z)}{p_\theta(x)} \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

move $\frac{1}{p_\theta(x)}$ inside the sum

$$= \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$$

Bayes: path $\div$ total = posterior. ■

In words: differentiate the log, write each path's slope via its own log; path $\div$ total is the posterior.

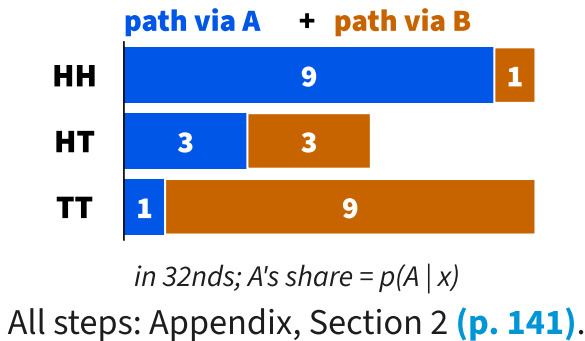
Why it matters: the right side differentiates only **one-path logs**, never a log of a sum; expectation–maximization (EM) builds on it.

Six steps, each a rule from calculus or probability: the identity holds for every model $p_\theta(x, z)$.

Two paths give $p(\text{HH}) = 5/16$; Bayes' rule gives coin A 9/10

↳ Definitions (p. 23) and Theorem 3 (p. 27) applied to the coins, at the working guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$: a path is $\frac{1}{2}$ (pick the coin) · the chance of each flip; $p(x)$ adds the two paths; the posterior is path ÷ total; ℓ adds the logs.

record	path via A	path via B	$p(x)$	$p(A x)$	$\ln p(x)$
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{10}{32}$	$\frac{9}{10}$	-1.163
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{6}{32}$	$\frac{3}{6} = \frac{1}{2}$	-1.674
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{10}{32}$	$\frac{1}{10}$	-1.163
total	add the logs: $\ell = -4.000$				



$$p(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32} = \frac{5}{16}, \quad p(A | \text{HH}) = \frac{9}{32} \div \frac{10}{32} = \frac{9}{10}$$
$$\frac{\partial}{\partial \theta_A} \ln p(\text{HH}) = \frac{9}{10} \cdot \underbrace{\frac{2}{\theta_A}}_{\substack{\text{slope of } \ln \frac{1}{2} + 2 \cdot \ln \theta_A \\ = 8/3}} + \frac{1}{10} \cdot \underbrace{0}_{\text{no } \theta_A} = 2.4$$
$$= \frac{\theta_A}{p(\text{HH})} = \frac{3/4}{10/32} = 2.4 \checkmark$$

add the paths; path ÷ total

Theorem 3

directly, chain rule

Two paths: 5/16; Bayes: 0.9 A / 0.1 B; these weights turn the path slopes into 2.4 (Theorem 3).

Our guess scores $\ell = -4.000$, just below a top of the map

↳ Definition (p. 23) applied to the coins, with $N = 3$ records:

$$\ell(\theta) = \sum_{n=1}^N \ln p_{\theta}(x^{(n)}).$$

Every pair (θ_A, θ_B) gets its own **log-likelihood** ℓ ; the map colours them all. A higher ℓ means the data are more likely.

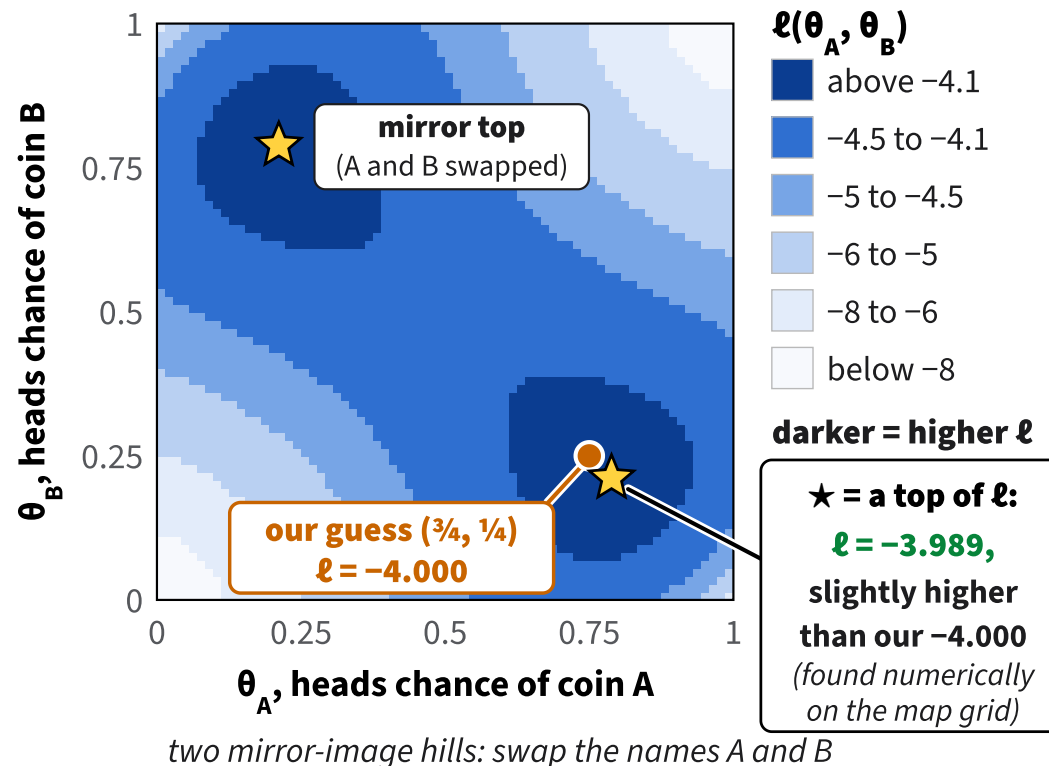
our guess $(\frac{3}{4}, \frac{1}{4})$

$\ell = -4.000$

a top ★ $(0.789, 0.211)$

$\ell = -3.989$

Learning = finding the (θ_A, θ_B) at a top of the map (★), the **maximum-likelihood estimate**. Swapping the names A and B gives the mirror top.



Ours scores $\ell = -4.000$, close to a top but not on it. How do we increase ℓ ?

Told, counting fits θ_A alone; hidden, the best θ_A moves with θ_B

↳ The theory's two chains applied to the coins. Recall Theorem 1 (counting) (p. 11): best heads chance $\frac{h}{h+t}$ from h heads and t tails of one coin.

Coin told: the log splits

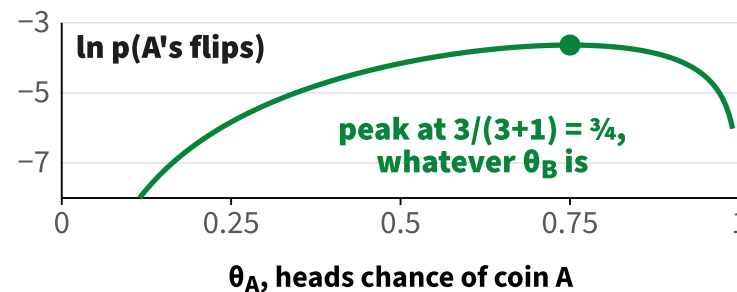
Say HH and HT came from A, so A's flips are H, H, H, T. Their log is $\ln p(\text{HH}, A) + \ln p(\text{HT}, A)$
 $= 2 \cdot \ln \frac{1}{2} + 3 \cdot \ln \theta_A + \ln(1 - \theta_A)$: no θ_B , so **counting** gives $\theta_A = \frac{3}{3+1} = \frac{3}{4}$, whatever θ_B is.

Coin hidden: one log of a sum

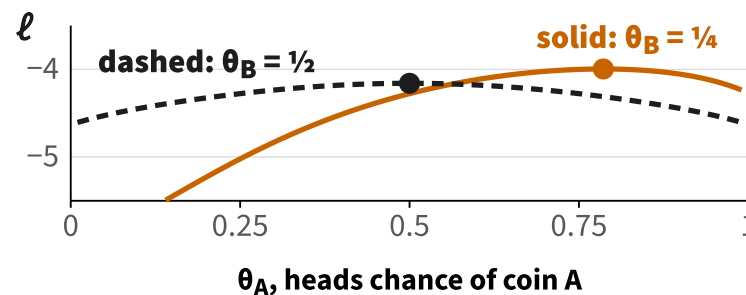
ℓ contains $\ln p(\text{HH}) = \ln \left(\frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot \theta_B^2 \right)$, one log holding both parameters, so ℓ does not split. In the figure (ℓ against θ_A), changing θ_B from $\frac{1}{4}$ to $\frac{1}{2}$ moves the best θ_A (dot) from about 0.79 to 0.5.

All steps: Appendix, Section 2 (p. 142).

Coin told (say HH and HT were from A):
A's flips are H, H, H, T

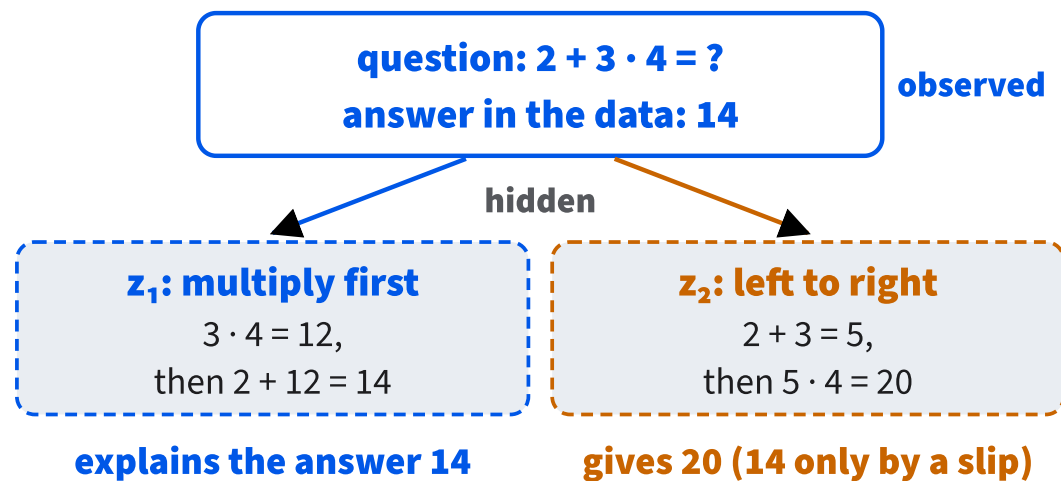


Coin hidden: the best θ_A (dot) moves with θ_B



Hidden coin $\Rightarrow$ a log of a sum $\Rightarrow$ the best θ_A depends on θ_B : no counting formula.

A reasoning trace is a hidden coin



Bayes' rule, given the answer 14:

posterior of z₁ (multiply first) = 0.81

the trace that explains the answer gets most weight

A language model is trained on **question** → **answer** pairs. The reasoning written in between, the **trace** z , is never in the data.

COIN GAME

record HH (seen)

coin z (hidden)

chance of picking coin A

$p(\text{HH})$: sum over coins

posterior $p(A | \text{HH})$

REASONING MODEL

question + answer 14 (seen)

trace z (hidden)

chance of writing trace z_1

$p(14)$: sum over traces

posterior $p(z_1 | 14)$

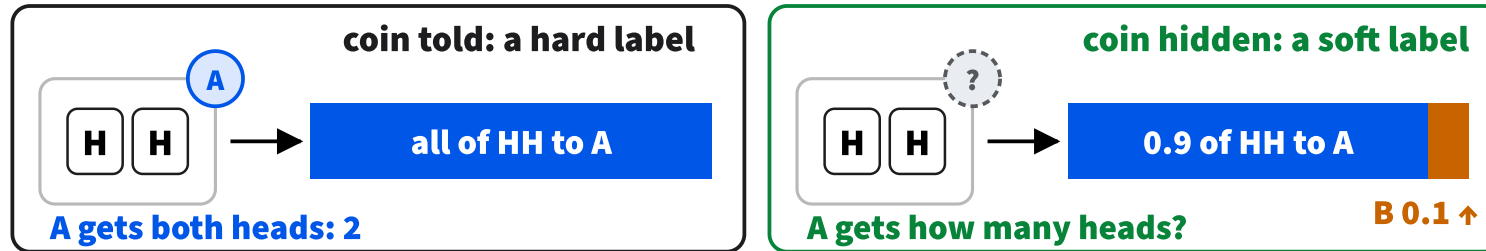
Given the answer 14, the trace "multiply first" has **posterior 0.81**, so training on the answer mostly strengthens that trace (Theorem 3) (p. 27).

Zelikman et al., "STaR: Bootstrapping Reasoning With Reasoning," Neural Information Processing Systems (NeurIPS) 2022. Phan et al., "Training Chain-of-Thought via Latent-Variable Inference," NeurIPS 2023.

Worked example: Appendix, Section 2 (p. 143).

Same sum over paths, same Bayes' rule; only the hidden variable is a text.

Next: count with soft labels



The posterior says HH is 0.9 coin A (p. 29). How many heads should HH give to coin A?

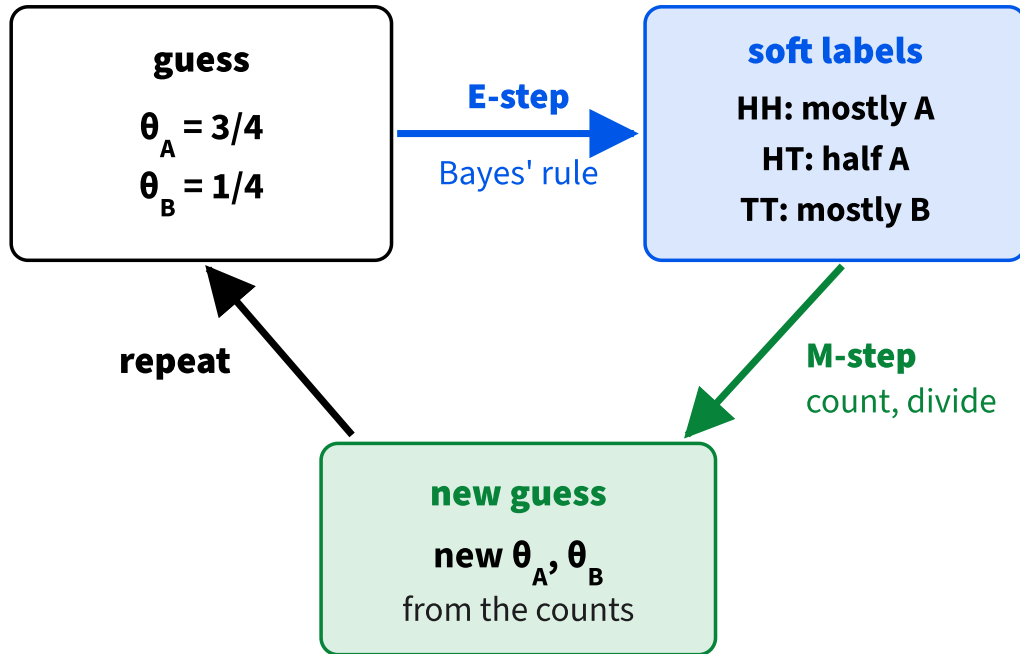
1.8 2 0.9

Answer. $1.8 = 0.9 \cdot 2$ heads. Computing the 0.9 is the E-step (expectation); counting fractional heads and dividing is the M-step (maximization) of expectation–maximization (EM).

Soft-label every record, count fractions, divide: expectation–maximization.

EM: guess the coins, soft-label the records, count, repeat

one EM round on the records HH, HT, TT



the flips never change; only the guess does

Goal: the θ_A, θ_B with the largest **log-likelihood** $\ell = \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT})$, where $p(\text{HH})$ is the model's chance of record HH.

A circle: with θ we could label the records; with labels we could count θ . **Break the circle with a guess.**

E-step (expectation): Bayes' rule turns the guess into **soft labels**, each record's chance of coin A.

M-step (maximization): **count heads ÷ flips** for each coin, a record counting only by its share.

Why a guess is fine: each round corrects the guess using only the flips; whether a round can ever make the fit worse is the question this section ends on.

Each round needs only the flips and the last guess, never the true coin.

EM for any hidden-variable model: posterior, then weighted fit

Definition (expectation-maximization; Dempster, Laird & Rubin, 1977). Records $x^{(1)}, \dots, x^{(N)}$, hidden z , model $p_\theta(x, z)$. Start from a guess θ_0 ; for $t = 0, 1, 2, \dots$:

$$q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)}) = \frac{p_{\theta_t}(x^{(n)}, z)}{\sum_{z'} p_{\theta_t}(x^{(n)}, z')}$$

$$\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z)$$

E-step: posterior soft labels, record n 's chance of each value of z . **Bayes' rule:** one path ÷ all paths of the record.

M-step: maximize each path's log, weighted by its soft label.

SYMBOL		IN THE COIN GAME
$x^{(n)}$	↔	record n : HH, HT, TT
z	↔	the coin, A or B
θ_t	↔	guess after t rounds; θ_0 the start
$q_t^{(n)}(z)$	↔	soft label of record n
$p_{\theta}(x, z)$	↔	one path: pick coin z , then its flips

Why "expectation": the M-step maximizes $\sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_{\theta}(x^{(n)}, z)]$, the **expected complete-data log-likelihood** (record and coin together). It is not ℓ itself but an easy stand-in: every term is the log of one path, with no log of a sum, so it splits into counts. Whether raising it always raises ℓ is Section 4's question.

E: infer the hidden part. M: fit as if it were seen, with weights.

The M-step objective collects expected heads and tails

↳ Soft labels $q^{(n)}(z)$ held fixed (round index dropped). Record $x^{(n)}$ has $h^{(n)}$ heads, $2 - h^{(n)}$ tails; θ_z means θ_A if $z = A$, θ_B if $z = B$. A path is $p_\theta(x^{(n)}, z) = \frac{1}{2} \cdot \theta_z^{h^{(n)}} \cdot (1 - \theta_z)^{2-h^{(n)}}$: pick z , then its flips (pick chance held at $\frac{1}{2}$ for now).

$$\begin{aligned}
 \sum_n \sum_z q^{(n)}(z) \cdot \ln p_\theta(x^{(n)}, z) &= \sum_n \sum_z q^{(n)}(z) \cdot \ln \left(\frac{1}{2} \cdot \theta_z^{h^{(n)}} \cdot (1 - \theta_z)^{2-h^{(n)}} \right) && \text{(insert the path)} \\
 &= \sum_n \sum_z q^{(n)}(z) \cdot \left[\ln \frac{1}{2} + h^{(n)} \cdot \ln \theta_z + (2 - h^{(n)}) \cdot \ln(1 - \theta_z) \right] && \text{(product} \rightarrow \text{sum of logs)} \\
 &= \sum_n \sum_z q^{(n)}(z) \cdot \ln \frac{1}{2} + \sum_n \sum_z q^{(n)}(z) \cdot h^{(n)} \cdot \ln \theta_z && \text{(multiply out the bracket)} \\
 &\quad + \sum_n \sum_z q^{(n)}(z) \cdot (2 - h^{(n)}) \cdot \ln(1 - \theta_z) \\
 &= \ln \frac{1}{2} \cdot \sum_n \sum_z q^{(n)}(z) + \sum_z \ln \theta_z \cdot \sum_n q^{(n)}(z) \cdot h^{(n)} && \text{(swap the order of the sums;} \\
 &\quad + \sum_z \ln(1 - \theta_z) \cdot \sum_n q^{(n)}(z) \cdot (2 - h^{(n)}) && \text{ln } \theta_z \text{ does not depend on } n) \\
 &= N \cdot \ln \frac{1}{2} + \sum_z [H_z \cdot \ln \theta_z + T_z \cdot \ln(1 - \theta_z)] && (\sum_z q^{(n)}(z) = 1; \text{ name sums)} \\
 &= \text{const} + \underbrace{H_A \cdot \ln \theta_A + T_A \cdot \ln(1 - \theta_A)}_{\text{coin A's part: only } \theta_A} + \underbrace{H_B \cdot \ln \theta_B + T_B \cdot \ln(1 - \theta_B)}_{\text{coin B's part: only } \theta_B} && \text{(write out } z = A, B)
 \end{aligned}$$

$H_A = \sum_n q^{(n)}(A) \cdot h^{(n)}$, $T_A = \sum_n q^{(n)}(A) \cdot (2 - h^{(n)})$
 expected heads and tails **credited to A**; H_B, T_B the same
 with $q^{(n)}(B)$.

In words: Theorem 1's form (p. 11), **heads**·ln θ + **tails**·ln(1 - θ), once per coin, with fractional counts.

Why it matters: θ_A, θ_B share no term: each coin is fitted **on its own**.

No log of a sum remains: the M-step objective is expected counts times logs.

Slope zero: each coin's part peaks at expected heads ÷ flips

↳ Coin A's part: $f(\theta_A) = H_A \cdot \ln \theta_A + T_A \cdot \ln(1 - \theta_A)$; H_A, T_A = A's expected heads, tails.

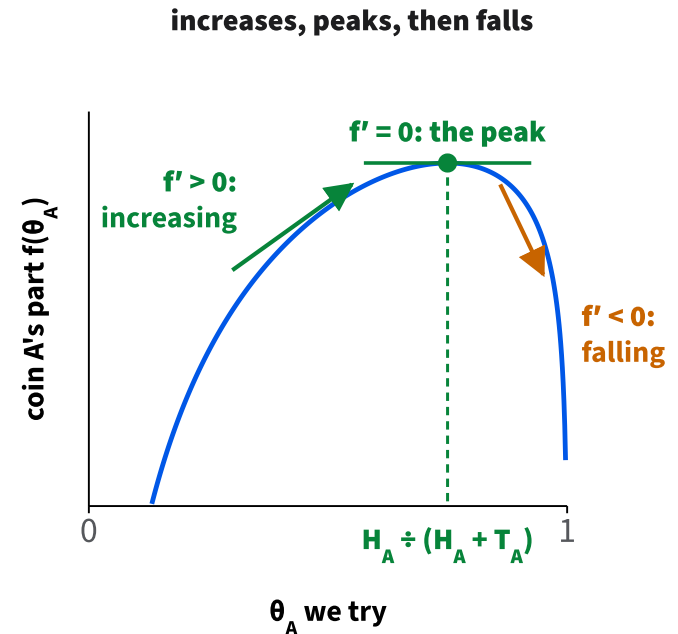
$$\begin{aligned}
 f'(\theta_A) &= \frac{H_A}{\theta_A} - \frac{T_A}{1 - \theta_A} && ((\ln \theta)' = \frac{1}{\theta}, (\ln(1 - \theta))' = \frac{-1}{1 - \theta}) \\
 &= \frac{H_A \cdot (1 - \theta_A) - T_A \cdot \theta_A}{\theta_A \cdot (1 - \theta_A)} && (\text{common denominator}) \\
 &= \frac{H_A - (H_A + T_A) \cdot \theta_A}{\theta_A \cdot (1 - \theta_A)} && (\text{multiply out the top}) \\
 &= \frac{H_A + T_A}{\theta_A \cdot (1 - \theta_A)} \cdot \left(\frac{H_A}{H_A + T_A} - \theta_A \right) && (\text{factor out } H_A + T_A)
 \end{aligned}$$

The first factor is positive, so $f' > 0$ left of $\frac{H_A}{H_A + T_A}$ and $f' < 0$ right of it: f **increases, then falls**.

$$\theta_A^{\text{new}} = \frac{H_A}{H_A + T_A} = \frac{\text{expected heads by A}}{\text{expected flips by A}}, \quad \theta_B^{\text{new}} = \frac{H_B}{H_B + T_B}$$

In words: Theorem 1 (p. 11) with $h = H_A, t = T_A$: **no search needed**.

The M-step is count-and-divide, with soft labels as fractional counts.



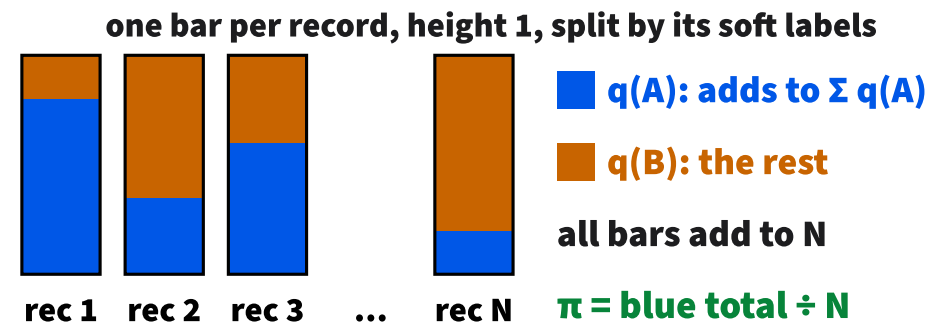
The pick chance π is counted too: records by A ÷ records

↳ So far the pick chance was held at $\frac{1}{2}$, which gave the constant $N \cdot \ln \frac{1}{2}$; learning $\pi = p(A)$ replaces that constant. A path's log splits, $\ln p_\theta(x^{(n)}, z) = \ln p(z) + \ln p_\theta(x^{(n)} | z)$, and only the pick factor $p(z)$, π if $z = A$ and $1 - \pi$ if $z = B$, holds π . Soft labels $q^{(n)}(z)$ held fixed.

$$\begin{aligned}
 \sum_n \sum_z q^{(n)}(z) \cdot \ln p(z) &= \sum_n [q^{(n)}(A) \cdot \ln \pi + q^{(n)}(B) \cdot \ln(1 - \pi)] && \text{(write out } z = A, B: p(A) = \pi, p(B) = 1 - \pi) \\
 &= \left(\sum_n q^{(n)}(A) \right) \cdot \ln \pi + \left(\sum_n q^{(n)}(B) \right) \cdot \ln(1 - \pi) && \text{(swap the sums)} \\
 \Rightarrow \pi &= \frac{\sum_n q^{(n)}(A)}{\sum_n q^{(n)}(A) + \sum_n q^{(n)}(B)} && \text{(slope zero, as for } \theta_A \text{ in Theorem 1's form,} \\
 &= \frac{\sum_n q^{(n)}(A)}{N} && \text{with } h = \sum_n q^{(n)}(A), t = \sum_n q^{(n)}(B)) \\
 &&& (\sum_z q^{(n)}(z) = 1 \text{ for each of the } N \text{ records})
 \end{aligned}$$

In words: π = **expected records by A ÷ records**, the same count-and-divide as θ_A .

Why it matters: the θ terms do not hold π , so θ_A, θ_B keep their formulas; **every table**, root included, is count-and-divide.



Learning the pick chance adds one more count: soft labels summed, divided by N .

On a tree model, the M-step is count-and-divide with fractions

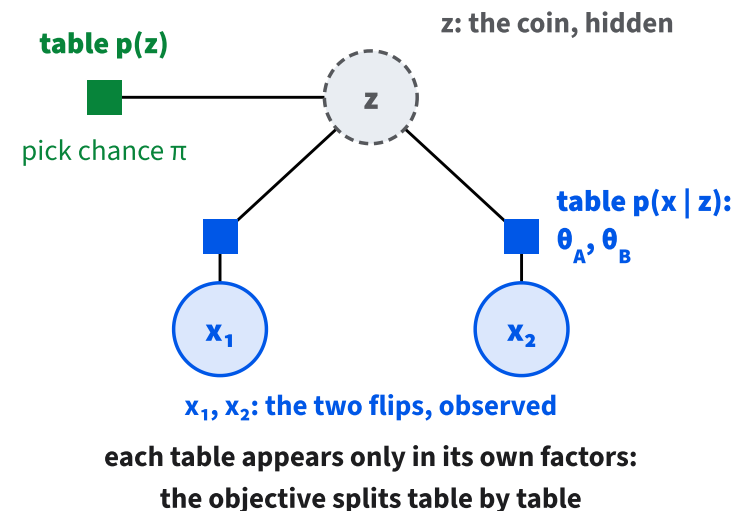
↳ Every table we have met, θ_A , θ_B and π , is count-and-divide. In general:

Corollary (of Theorem 1) (p. 11). If $p_\theta(x, z)$ is a product of tables on a tree factor graph, the M-step sets each table entry to

$$\theta_{\text{entry}} = \frac{\text{expected count of its case}}{\text{expected count of its condition}}.$$

In words: each record adds soft label • (times the case occurs in it).

entry of the coin model	case ÷ condition	formula
$\theta_A = p(H \mid A)$	flips by A showing H ÷ flips by A	$\frac{H_A}{H_A + T_A}$
$\theta_B = p(H \mid B)$	flips by B showing H ÷ flips by B	$\frac{H_B}{H_B + T_B}$
$\pi = p(A)$, pick chance	records by A ÷ records	$\frac{\sum_n q^{(n)}(A)}{N}$



Why: each table, a **conditional distribution** $p(\text{child} \mid \text{parent})$, appears only in its own factors and sums to 1 on its own, so the objective splits table by table.

Why it matters: a chain is a tree, so the hidden chain of Section 6 is refilled the same way.

Every table of a tree model is refilled by counting, with soft labels as fractions.

Where EM stops: a fixed point, $\theta^* = M(\theta^*)$

The round map. One EM round turns a guess θ' into

$$M(\theta') = \arg \max_{\theta} \sum_n \sum_z p_{\theta'}(z \mid x^{(n)}) \cdot \ln p_{\theta}(x^{(n)}, z),$$

so $\theta_{t+1} = M(\theta_t)$. A **fixed point** is a θ^* with $M(\theta^*) = \theta^*$.

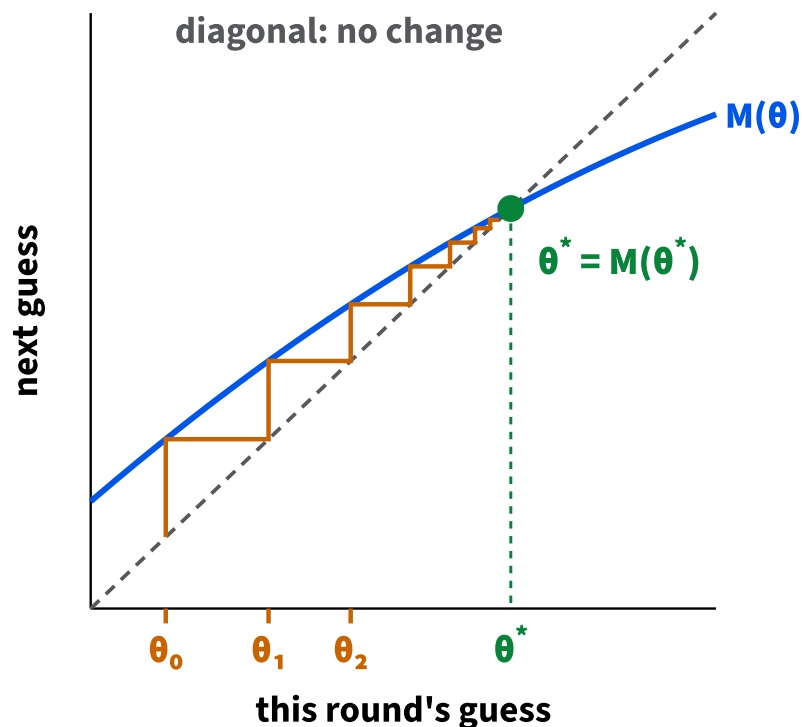
Claim. If the rounds settle, $\theta_t \rightarrow \theta^*$, and M is continuous, then θ^* is a fixed point:

$$\begin{aligned} \theta^* &= \lim_{t \rightarrow \infty} \theta_{t+1} && \text{(the rounds settle at } \theta^*) \\ &= \lim_{t \rightarrow \infty} M(\theta_t) && \text{(one round: } \theta_{t+1} = M(\theta_t)) \\ &= M\left(\lim_{t \rightarrow \infty} \theta_t\right) = M(\theta^*) && (M \text{ continuous}) \end{aligned}$$

In words: EM stops where one more round gives back the same guess.

Why it matters: to find the end, solve the equation $\theta = M(\theta)$ instead of running rounds.

round map M: the staircase stops at the diagonal



EM's end point solves $\theta = M(\theta)$: one more round changes nothing.

At a fixed point inside, every slope of ℓ is zero

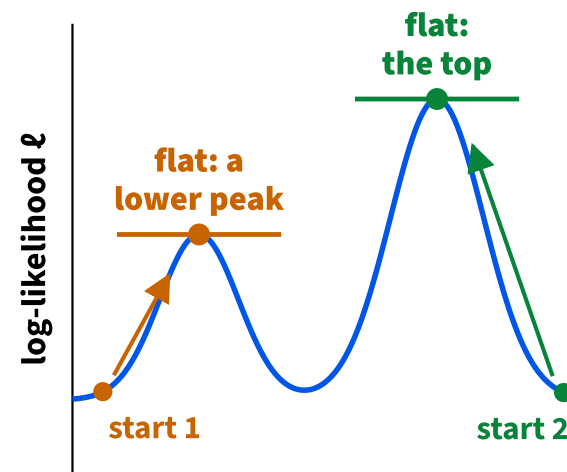
↳ $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$ is the log-likelihood; θ_j is any entry of θ . Let $\theta^* = M(\theta^*)$ lie inside the range of θ , with soft labels $q_*^{(n)}(z) = p_{\theta^*}(z | x^{(n)})$. Recall Theorem 3 (p. 27) (Fisher's identity): $\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$.

$$\begin{aligned}
 \frac{\partial \ell}{\partial \theta_j}(\theta^*) &= \sum_n \frac{\partial}{\partial \theta_j} \ln p_\theta(x^{(n)}) \Big|_{\theta^*} && \text{(slope of a sum)} \\
 &= \sum_n \sum_z p_{\theta^*}(z | x^{(n)}) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x^{(n)}, z) \Big|_{\theta^*} && \text{(Fisher's identity, each record)} \\
 &= \frac{\partial}{\partial \theta_j} \left[\sum_n \sum_z q_*^{(n)}(z) \cdot \ln p_\theta(x^{(n)}, z) \right]_{\theta^*} && \text{(labels } q_*^{(n)} \text{ held fixed)} \\
 &= 0 && \text{(its peak is } M(\theta^*) = \theta^*, \text{ inside)}
 \end{aligned}$$

In words: at EM's end, ℓ is **flat in every direction**.

Why it matters: flat is **not always the top**: a lower peak is flat too, so the start guess can decide where EM ends.

EM increases ℓ up to the flat point near its start



parameter θ
zero slope: a peak,
not always the highest

EM stops at a point where the log-likelihood has zero slope.

When is a flat point the top? Gibbs' inequality can certify it

↳ A flat point need not be the top. When ℓ is an **average of logs of shares**, one inequality settles it.

Gibbs' inequality. For distributions p and s on the same values k (every $p_k > 0$),

$$\sum_k p_k \cdot \ln s_k \leq \sum_k p_k \cdot \ln p_k, \quad \text{with equality only when } s = p.$$

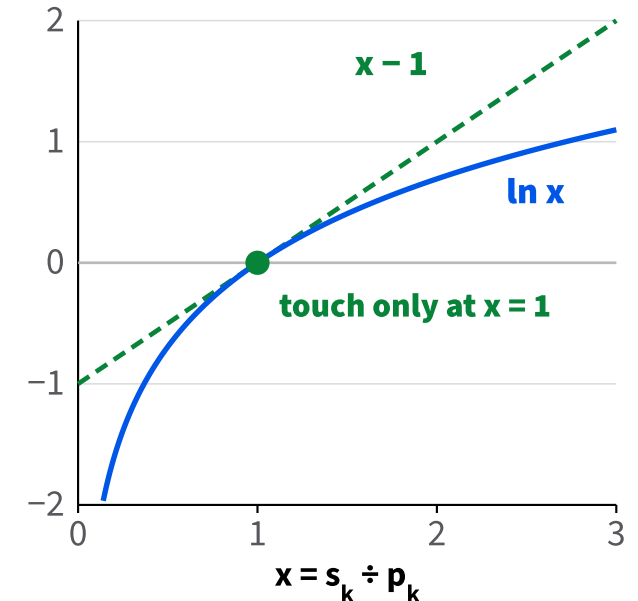
$$\begin{aligned} \sum_k p_k \cdot \ln s_k - \sum_k p_k \cdot \ln p_k &= \sum_k p_k \cdot \ln \frac{s_k}{p_k} && \text{(log of a quotient)} \\ &\leq \sum_k p_k \cdot \left(\frac{s_k}{p_k} - 1 \right) && (\ln x \leq x - 1; \text{ equal only at } x = 1) \\ &= \sum_k s_k - \sum_k p_k && \text{(multiply out)} \\ &= 1 - 1 = 0 && \text{(both sum to 1)} \end{aligned}$$

In words: the average log of a model share is **largest when the model's shares are the data's shares**.

Why it matters: it can certify that a flat point is the **global top**, not a lower peak.

Weighted by the data's shares, an average log share peaks only at those shares.

$\ln x$ never rises above $x - 1$



so equality needs $s_k = p_k$ for every k

E-step by hand: Bayes' rule splits each record between the coins

↳ **Definition (E-step) (p. 35)** on the coins. Start guess $\theta_0 = (\frac{3}{4}, \frac{1}{4})$: $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$; the friend picks each coin with chance $\frac{1}{2}$.

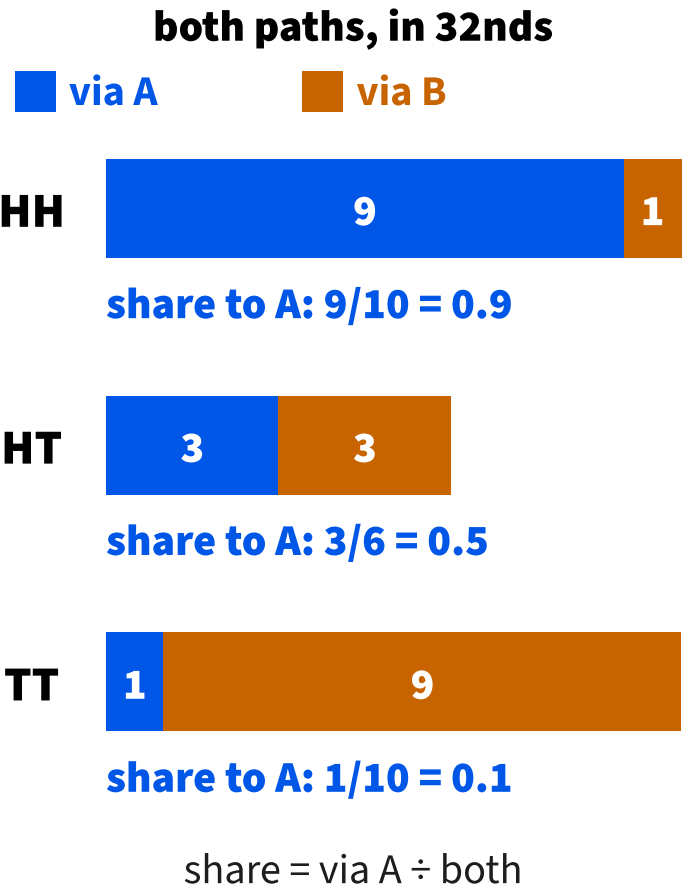
Bayes' rule, from the definition:

$$\underbrace{p(A \mid \text{record})}_{\text{share to A, the posterior}} = \frac{\text{path via A}}{\text{path via A} + \text{path via B}}$$

A path multiplies its steps, e.g. HH via A:

$$\underbrace{\frac{1}{2}}_{\text{pick A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} \cdot \underbrace{\frac{3}{4}}_{\text{H by A}} = \frac{9}{32}$$

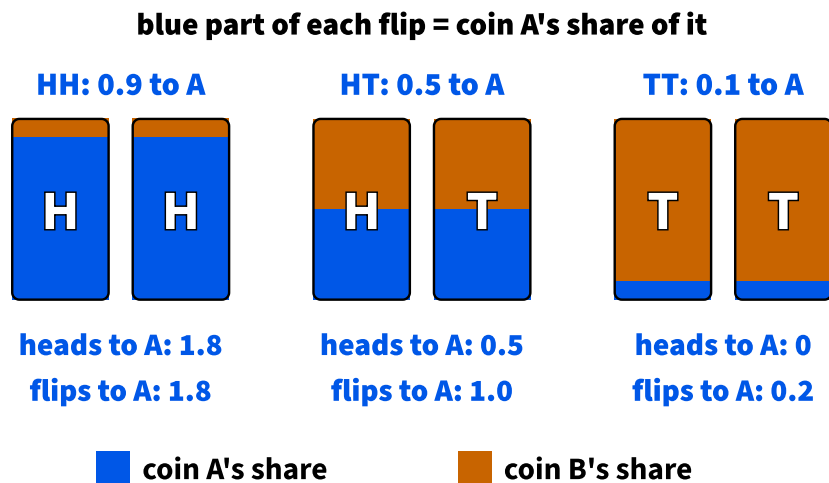
record	path via A	path via B	share to A
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{9}{9+1} = 0.9$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{3}{3+3} = 0.5$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{1+9} = 0.1$



E-step output: soft labels 0.9, 0.5, 0.1 for coin A; coin B gets the rest.

M-step by hand: the guess moves to (0.767, 0.233)

↳ Count-and-divide (p. 37), $\theta_A = \frac{H_A}{H_A + T_A}$, on the coins: A is **credited** its share (0.9, 0.5, 0.1) (p. 43) of each record's heads and flips.



blue totals: heads 2.3, flips 3

record	heads to A share · heads	flips to A share · 2
HH	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$0.5 \cdot 1 = 0.5$	$0.5 \cdot 2 = 1.0$
TT	$0.1 \cdot 0 = 0$	$0.1 \cdot 2 = 0.2$
total	2.3 = H_A	3.0 = $H_A + T_A$

Coin B is credited the other shares, $1 - 0.9$, $1 - 0.5$, $1 - 0.1 = 0.1$, 0.5 , 0.9 .

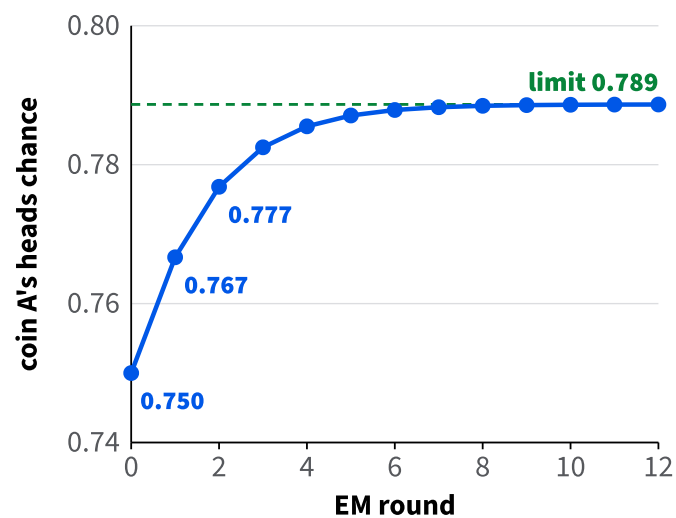
$$\theta_A = \frac{\text{heads credited to A}}{\text{flips credited to A}} = \frac{2.3}{3} = 0.767$$

$$\theta_B = \frac{0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0}{2 \cdot (0.1 + 0.5 + 0.9)} = \frac{0.7}{3} = 0.233$$

All steps: Appendix, Section 3 (p. 144).

One round moved the guess from $(\frac{3}{4}, \frac{1}{4})$ to (0.767, 0.233).

Each round, θ_A increases: 0.750, 0.767, 0.777, ..., 0.789



↳ The round map (p. 40) $\theta_{t+1} = M(\theta_t)$ on the coins, again and again.

By symmetry $\theta_B = 1 - \theta_A$, HT's share stays 0.5 and A is credited 3 flips every round. One round:

$$\theta_A^{\text{new}} = \frac{\overbrace{2 \cdot q_{\text{HH}}}^{\text{HH's heads}} + \overbrace{1 \cdot 0.5}^{\text{HT's}}}{3}, \quad q_{\text{HH}} = \frac{\theta_A^2}{\theta_A^2 + \theta_B^2}$$

round	θ_A	q_{HH} , share to A	new θ_A
1	$\frac{3}{4}$	$\frac{9}{9+1} = 0.900$	$\frac{2 \cdot (0.900) + 0.5}{3} = 0.767$
limit	$\frac{3+\sqrt{3}}{6} = 0.789$	$\frac{0.789^2}{0.789^2 + 0.211^2} = 0.933$	$\frac{2 \cdot (0.933) + 0.5}{3} = 0.789$

At the limit a round returns the same θ_A : EM has reached a **fixed point**. Is it the best possible fit, or only a flat point?

All steps: Appendix, Section 3 (p. 145).

round 12 · $\theta_A = 0.789$ · $\theta_B = 0.211$

At 0.789 a round gives back 0.789: EM has reached a fixed point.

Why 0.789 is the global top: no pair of coins scores above -3.989

↳ **Gibbs' inequality (p. 42) applied to the coins.** Write $s_2 = p(\text{HH})$, $s_1 = p(\text{HT}) + p(\text{TH}) = 2 \cdot p(\text{HT})$, $s_0 = p(\text{TT})$: the model's chances of 2, 1, 0 heads, which add to 1. The data have one record of each, so the data's shares are $p_k = \frac{1}{3}$.

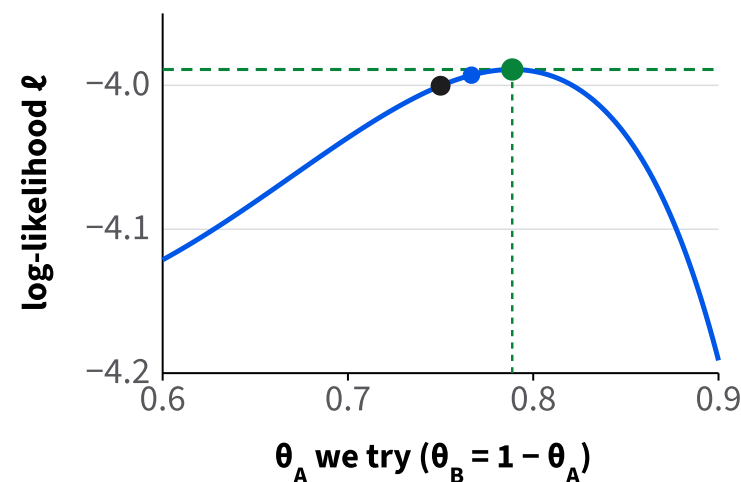
$$\begin{aligned}
 \ell(\theta) &= \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT}) && \text{(the three records)} \\
 &= \ln s_2 + \ln \frac{s_1}{2} + \ln s_0 && (p(\text{HT}) = \frac{s_1}{2}) \\
 &= 3 \cdot \left(\frac{1}{3} \cdot \ln s_2 + \frac{1}{3} \cdot \ln s_1 + \frac{1}{3} \cdot \ln s_0 \right) - \ln 2 && \text{(pull out 3 and } \ln 2) \\
 &\leq 3 \cdot \left(\frac{1}{3} \cdot \ln \frac{1}{3} + \frac{1}{3} \cdot \ln \frac{1}{3} + \frac{1}{3} \cdot \ln \frac{1}{3} \right) - \ln 2 && \text{(Gibbs, } p_k = \frac{1}{3}) \\
 &= 3 \cdot \ln \frac{1}{3} - \ln 2 = -3.989 && \text{(for every } \theta)
 \end{aligned}$$

At the limit $\theta_A = \frac{3+\sqrt{3}}{6} = 0.789$, $\theta_B = \frac{3-\sqrt{3}}{6} = 0.211$: $\theta_A \cdot \theta_B = \frac{9-3}{36} = \frac{1}{6}$, so

$$\begin{aligned}
 s_1 &= 2 \cdot \theta_A \cdot \theta_B = \frac{1}{3}, \\
 s_2 &= s_0 = \frac{1}{2} \cdot (\theta_A^2 + \theta_B^2) = \frac{1}{2} \cdot ((\theta_A + \theta_B)^2 - 2 \cdot \theta_A \cdot \theta_B) = \frac{1}{2} \cdot \left(1 - \frac{1}{3}\right) = \frac{1}{3}.
 \end{aligned}$$

Every share equals the data's share, so the " $\leq$ " is an equality: $\ell = -3.989$ **reaches the ceiling** that no pair of coins can pass.

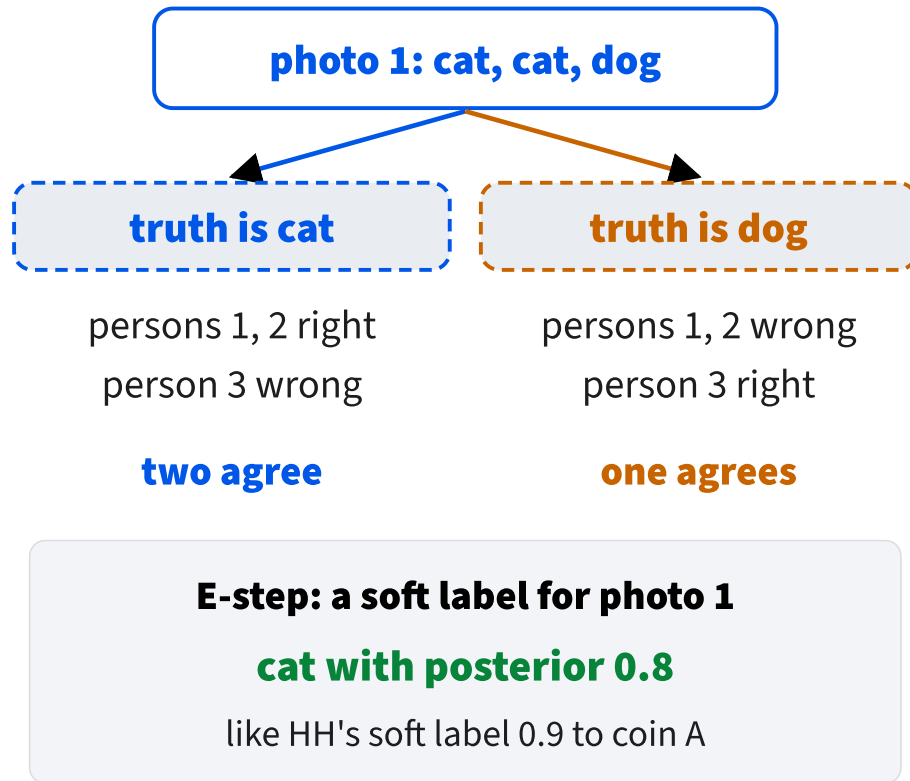
ℓ increases to the bound and touches it



- start 0.750: $\ell = -4.000$
- round 1, 0.767: $\ell = -3.993$
- fixed point 0.789: $\ell = -3.989$
- Gibbs bound $3 \cdot \ln(1/3) - \ln 2$

Gibbs' inequality gives a ceiling for every θ ; at 0.789 EM touches it, so the fixed point is the global top.

Crowd labelling: a photo's true label is the hidden coin



Three people label photo 1: **cat, cat, dog**. The truth is hidden. Person k says the truth with an unknown chance a_k , their **accuracy**. EM learns the truths and the accuracies together.

COIN GAME

coin A or B

two flips

θ_A : chance a flip shows H,
given the coin

soft label 0.9

CROWD LABELLING

↔ **truth cat or dog**

↔ **three people's labels**

↔ **a_k : chance person k is right,
given the truth**

↔ **posterior of cat 0.8**

M-step: each accuracy a_k becomes person k 's **expected right answers ÷ photos labelled**, counted like $\theta_A = \text{heads} \div \text{flips}$. Person 3, outvoted, is trusted less.

Worked example: Appendix, Section 3 ([p. 146](#)).

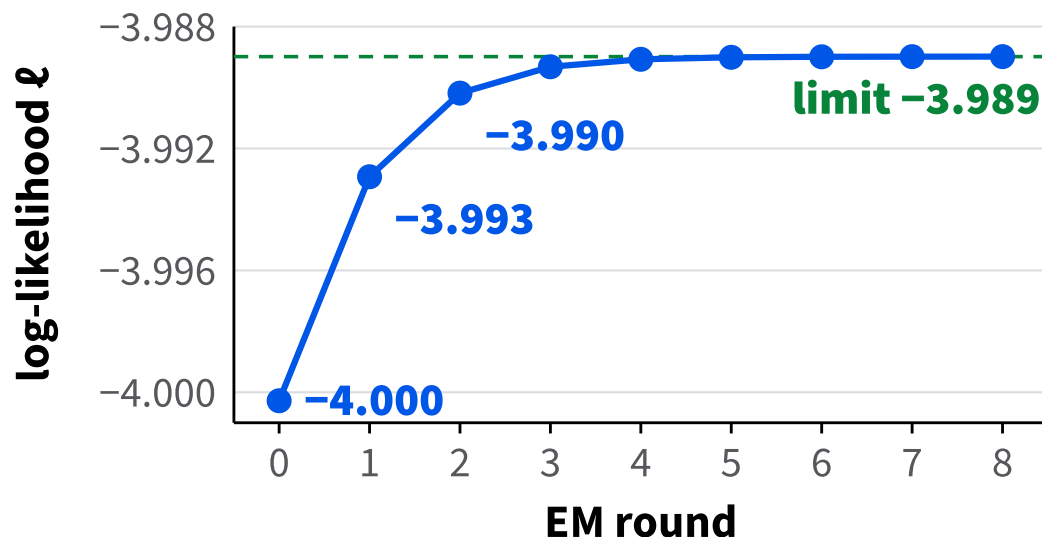
E-step: the votes give soft labels (cat 0.8), not hard majority votes; EM learns whom to trust.

Dawid & Skene, Maximum likelihood estimation of observer error-rates using the EM algorithm, *J. Royal Statistical Society C* 28(1):20–28, 1979; used today to merge annotators' noisy labels.

Next: does every EM round improve the fit?



ℓ after each round on HH, HT, TT



On the coins, ℓ rose in every round (figure), from -4.000 to -3.989 where $\theta_A = 0.789$ (p. 46).

All steps: Appendix, Section 3 (p. 145).

For any model and any start, can an EM round lower ℓ ?

yes, sometimes

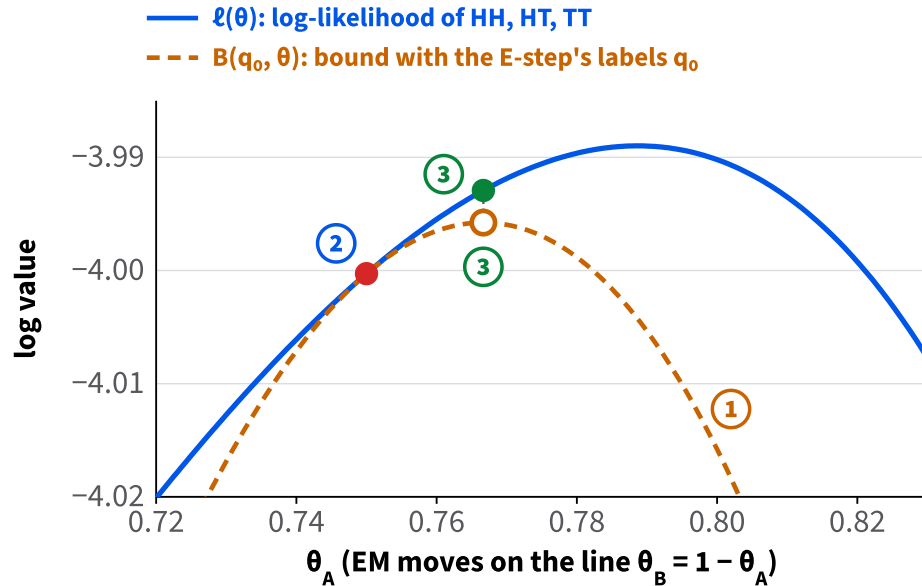
never

only at the start

Answer. Never. Section 4 proves it with one picture and one inequality.

A guarantee, not luck: EM increases the likelihood.

EM builds a floor under ℓ , then jumps to the floor's top



round 1: θ_A moves from the red dot to the circle ($\theta_A = 0.767$)

Goal: show that EM's log-likelihood ℓ , the sum of $\ln p(\text{record})$, never goes down.

1 · The bound B (dashed curve)

With the E-step's labels $q_0(z)$ (chance of coin A: HH 0.9, HT 0.5, TT 0.1) (p. 43), define $B(q_0, \theta) = \sum_{\text{records } x} \sum_{z=A,B} q_0(z) \cdot \ln \frac{p_\theta(x, z)}{q_0(z)}$, with $p_\theta(x, z)$ = chance of picking coin z and seeing x . It is a **floor**: $B \leq \ell$ (Lemma 1).

2 · Touch (red dot)

At the current guess θ_0 the labels are the posteriors, and the bound **touches** ℓ .

3 · Below, then top

Elsewhere B lies **below** ℓ . The M-step jumps to B 's top (circle); ℓ is **higher still** (green dot).

Touch at the current estimate, below everywhere else, jump to the top: ℓ can only go up.

Lemma 1: for every guess, $\ln p(x) = \text{bound} + \text{gap}$

↳ Symbols: record x ; hidden z (the coin); guess $q(z)$, any chances over z adding to 1; posterior $p_\theta(z | x) = p_\theta(x, z)/p_\theta(x)$; log-likelihood $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$; bound $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$.

$$\ln p_\theta(x) = \ln p_\theta(x, z) - \ln p_\theta(z | x)$$

$$= \sum_z q(z) \cdot [\ln p_\theta(x, z) - \ln p_\theta(z | x)]$$

$$= \sum_z q(z) \cdot \left[\ln \frac{p_\theta(x, z)}{q(z)} + \ln \frac{q(z)}{p_\theta(z | x)} \right]$$

$$= \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)} + \sum_z q(z) \cdot \ln \frac{q(z)}{p_\theta(z | x)}$$

$$= \underbrace{B(q, \theta)}_{\text{bound}} + \underbrace{D_{\text{KL}}(q \| p_\theta(\cdot | x))}_{\text{gap}}$$

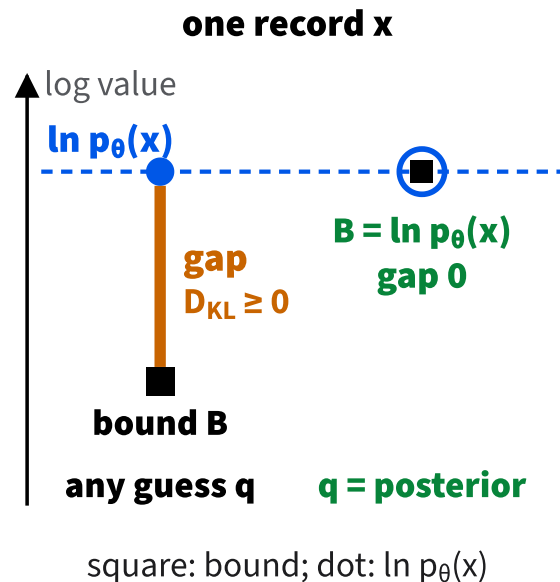
posterior's definition, in logs;
true for every z

no z on the left: its q -average
(weights adding to 1) is itself

subtract and add $\ln q(z)$

split the sum in two

name the two sums



The second sum is the **Kullback–Leibler divergence** of the guess from the posterior. **In words:** for **any** guess, the log-likelihood splits exactly into the bound and a gap.

Why it matters: the bound uses only $p_\theta(x, z)$, one path at a time, so it is easy to maximize; the gap is what we give up by using it.

For any guess, $\ln p(x)$ splits exactly into bound + gap.

The gap is never negative because $\ln y \leq y - 1$

↳ To show: $D_{\text{KL}} \geq 0$. Write $y = p_\theta(z | x) / q(z)$, posterior ÷ guess.

$$\begin{aligned}
 -D_{\text{KL}} &= -\sum_z q(z) \cdot \ln \frac{q(z)}{p_\theta(z | x)} && \text{the gap's definition} \\
 &= \sum_z q(z) \cdot \ln \frac{p_\theta(z | x)}{q(z)} && \text{flip fraction and sign} \\
 &\leq \sum_z q(z) \cdot \left(\frac{p_\theta(z | x)}{q(z)} - 1 \right) && \ln y \leq y - 1; q(z) \geq 0 \\
 &= \sum_z (p_\theta(z | x) - q(z)) && q(z) \text{ cancels} \\
 &= \sum_z p_\theta(z | x) - \sum_z q(z) \leq 1 - 1 = 0 && \text{posterior adds to } \leq 1, \text{ guess to } 1
 \end{aligned}$$

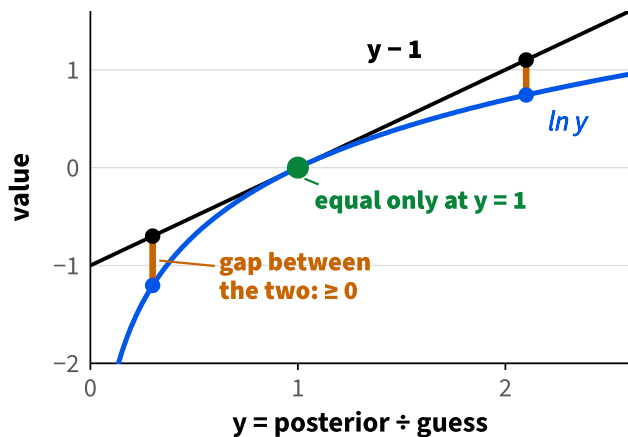
Equality needs every $y = 1$, that is $q = \text{posterior}$. ■ (Sums skip z with $q(z) = 0$, which contribute 0; over the rest the posterior adds to at most 1.)

Same fact, picture form (Jensen's inequality: $\ln$ bends down, so mean of logs $\leq$ log of mean): $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)} \leq \ln \sum_z q(z) \cdot \frac{p_\theta(x, z)}{q(z)} = \ln p_\theta(x)$.

In words: gap ≥ 0 , 0 only at the posterior. **Why it matters:** for every guess, the bound is a **floor** under $\ln p_\theta(x)$.

Lemma 1 (bound and gap). For every guess q : (i) $\ln p_\theta(x) = B(q, \theta) + D_{\text{KL}}(q \| p_\theta(\cdot | x))$; (ii) $D_{\text{KL}} \geq 0$, and $D_{\text{KL}} = 0$ exactly when $q = p_\theta(z | x)$. Hence $B(q, \theta) \leq \ln p_\theta(x)$ for every guess; with one guess $q^{(n)}$ per record, $\sum_n B(q^{(n)}, \theta) \leq \ell(\theta)$.

$\ln y$ stays below its tangent line $y - 1$



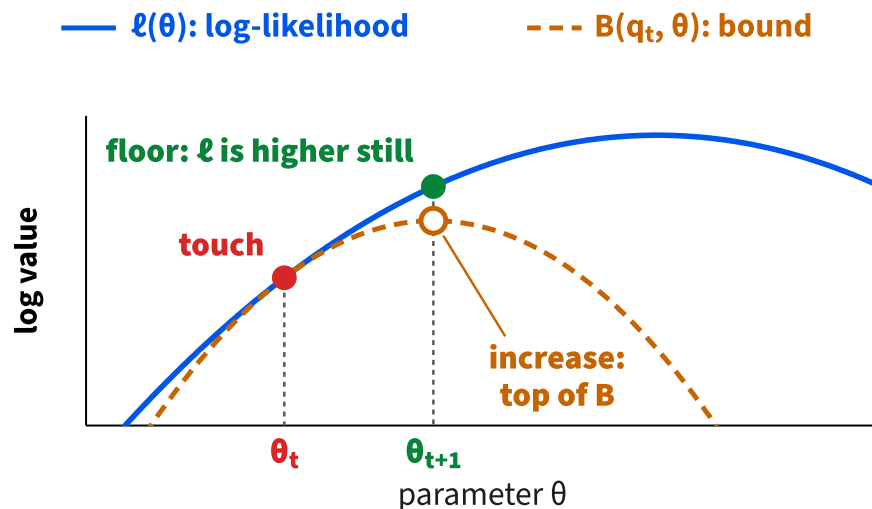
$\ln$ lies below its tangent, so the gap is ≥ 0 and vanishes only at the posterior: Lemma 1 holds.

Theorem 4: no EM round lowers the log-likelihood

Recall EM (p. 35), round t , records $x^{(n)}$:

E-step $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$, the posterior.

M-step $\theta_{t+1} = \arg \max_{\theta} \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z)$.



Why it matters: EM needs no step size and never overshoots; we can stop when ℓ stops rising.

Theorem 4 (EM increases the likelihood; Dempster, Laird & Rubin, 1977). With Lemma 1's bound (p. 50) at the E-step's q_t ,

$$B(q_t, \theta) = \sum_n \sum_z q_t^{(n)}(z) \cdot \ln \frac{p_{\theta}(x^{(n)}, z)}{q_t^{(n)}(z)},$$

every round t satisfies

$$\ell(\theta_{t+1}) \underset{\text{floor}}{\geq} B(q_t, \theta_{t+1}) \underset{\text{increase}}{\geq} B(q_t, \theta_t) \underset{\text{touch}}{=} \ell(\theta_t).$$

In words: Read right to left: equal at θ_t , the M-step raises B , ℓ stays above B .

Predict: if the M-step only raises its objective, without maximizing it, can ℓ fall?

yes no

Answer. No: the increase link only needs B to rise. This variant is **generalized EM**.

Touch, increase, floor: every EM round keeps or raises ℓ .

Proof of Theorem 4: touch, increase, floor

↳ Touch and floor use Lemma 1 (p. 50); increase uses the M-step's definition. Write $c_t = \sum_n \sum_z q_t^{(n)}(z) \cdot \ln q_t^{(n)}(z)$: it does not depend on θ .

$$\begin{aligned}
 \ell(\theta_{t+1}) &= B(q_t, \theta_{t+1}) + \sum_n D_{\text{KL}}(q_t^{(n)} \parallel p_{\theta_{t+1}}(\cdot \mid x^{(n)})) \\
 &\geq B(q_t, \theta_{t+1}) \\
 &= \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_{t+1}}(x^{(n)}, z) - c_t \\
 &\geq \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_t}(x^{(n)}, z) - c_t \\
 &= B(q_t, \theta_t) \\
 &= \ell(\theta_t) - \sum_n D_{\text{KL}}(q_t^{(n)} \parallel p_{\theta_t}(\cdot \mid x^{(n)})) \\
 &= \ell(\theta_t) \quad \blacksquare
 \end{aligned}$$

Lemma 1 at θ_{t+1} , guess q_t ,
summed over n

floor: each $D_{\text{KL}} \geq 0$

$$\ln \frac{p}{q} = \ln p - \ln q$$

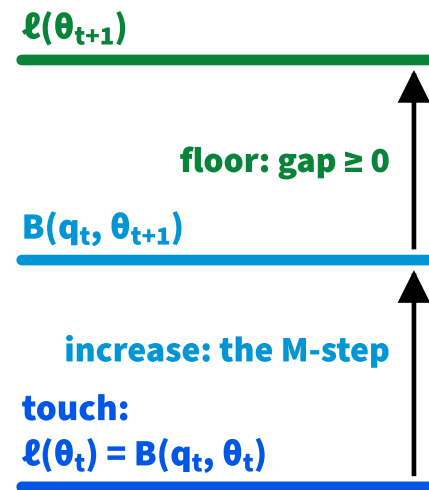
increase: the sum's max, at θ_{t+1} ,
 $\geq$ its value at θ_t

the same split, backwards

Lemma 1 at θ_t

touch: $q_t = \text{posterior}$, so $D_{\text{KL}} = 0$

read bottom to top



each link is $\geq$ or $=$

In words: start equal at θ_t ; the M-step raises the bound; ℓ stays above the bound. **Why it matters:** Lemma 1 holds for **every** guess, also the old q_t at the new θ_{t+1} .

Chain the three lines: $\ell(\theta_{t+1}) \geq B(q_t, \theta_{t+1}) \geq B(q_t, \theta_t) = \ell(\theta_t)$.

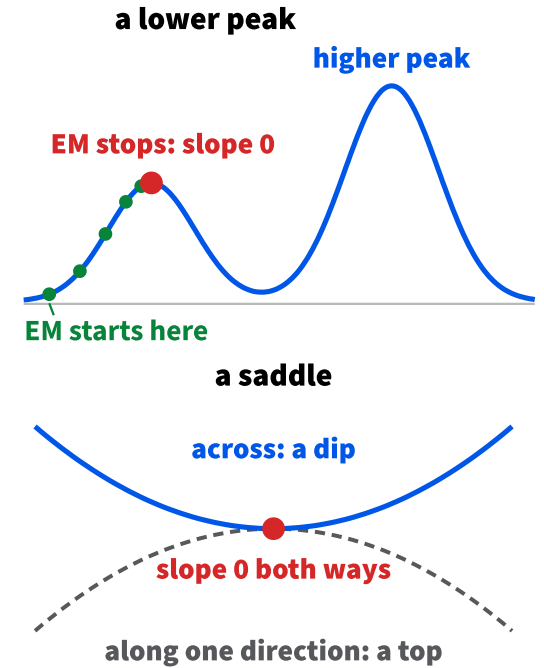
Where EM stops, ℓ is flat, but not always at a peak

↳ EM **stops** at θ_t when the M-step returns it: $\theta_{t+1} = \theta_t$, inside the allowed range. Take any one entry θ_j of θ . Theorem 3 (Fisher's identity) (p. 27): $\frac{\partial}{\partial \theta_j} \ln p_\theta(x) = \sum_z p_\theta(z | x) \cdot \frac{\partial}{\partial \theta_j} \ln p_\theta(x, z)$.

$$\begin{aligned} \left. \frac{\partial \ell}{\partial \theta_j} \right|_{\theta_t} &= \sum_n \sum_z p_{\theta_t}(z | x^{(n)}) \cdot \left. \frac{\partial}{\partial \theta_j} \ln p_{\theta_t}(x^{(n)}, z) \right|_{\theta_t} && \text{Theorem 3, each record} \\ &= \sum_n \sum_z q_t^{(n)}(z) \cdot \left. \frac{\partial}{\partial \theta_j} \ln p_{\theta_t}(x^{(n)}, z) \right|_{\theta_t} && \text{E-step: } q_t^{(n)} = \text{posterior at } \theta_t \\ &= \frac{\partial}{\partial \theta_j} \left[\sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta_t}(x^{(n)}, z) \right]_{\theta_t} && q_t \text{ is fixed numbers (no } \theta), \\ & && \text{so } \partial \text{ moves outside the sums} \\ &= 0 && \theta_t \text{ maximizes the M-step sum} \\ & && \text{inside the allowed range: slope 0} \end{aligned}$$

Where EM stops. A point that EM returns to itself is a **stationary point** of ℓ : every slope is 0. It may be the highest peak, a lower peak, or a **saddle**: a top in one direction, a dip in another.

In words: EM stops where ℓ is flat. **Why it matters:** Theorem 4 (p. 52) forbids going down, not stopping early; start EM from several different guesses.



Increasing ℓ is guaranteed; reaching the highest peak is not.

On HH every guess is a floor; only the posterior touches $\ln p(\text{HH})$

↳ Lemma 1 (p. 50) applied to HH at $\theta = (\frac{3}{4}, \frac{1}{4})$. Recall its paths (p. 29) $\frac{9}{32}$ via A and $\frac{1}{32}$ via B, so $\ln p(\text{HH}) = \ln \frac{5}{16} = -1.163$.

$$B(q) = q(A) \cdot \ln \frac{9/32}{q(A)} + q(B) \cdot \ln \frac{1/32}{q(B)}$$

Predict: which guess $q(A)$ gives HH the highest floor?

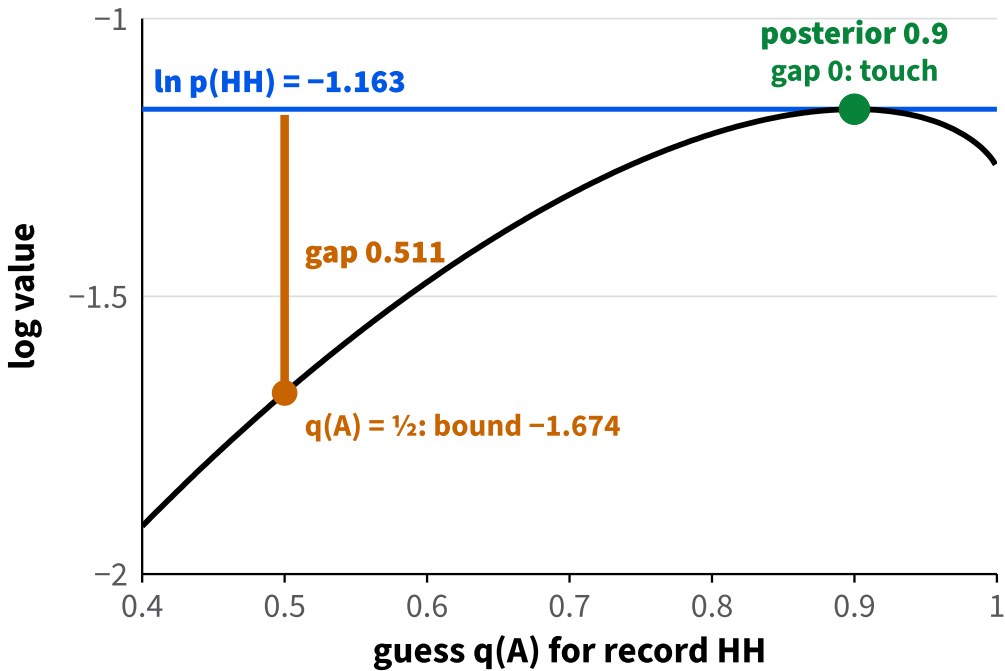
Answer. **0.9, HH's posterior**: its floor reaches $\ln p(\text{HH})$ itself.

guess $q(A)$	bound B	gap = $-1.163 - B$
$\frac{1}{2}$	-1.674	0.511
0.9 (the posterior)	-1.163	0

Why **0.9 touches**: path ÷ posterior = $p(\text{HH})$ for **each** coin, so $B = \ln p(\text{HH})$.

All steps: Appendix, Section 4 (p. 147).

black: bound $B(q)$ for every guess $q = (q(A), 1 - q(A))$



guess $q(A) = 0.90$: bound -1.163 + gap 0.000 = -1.163

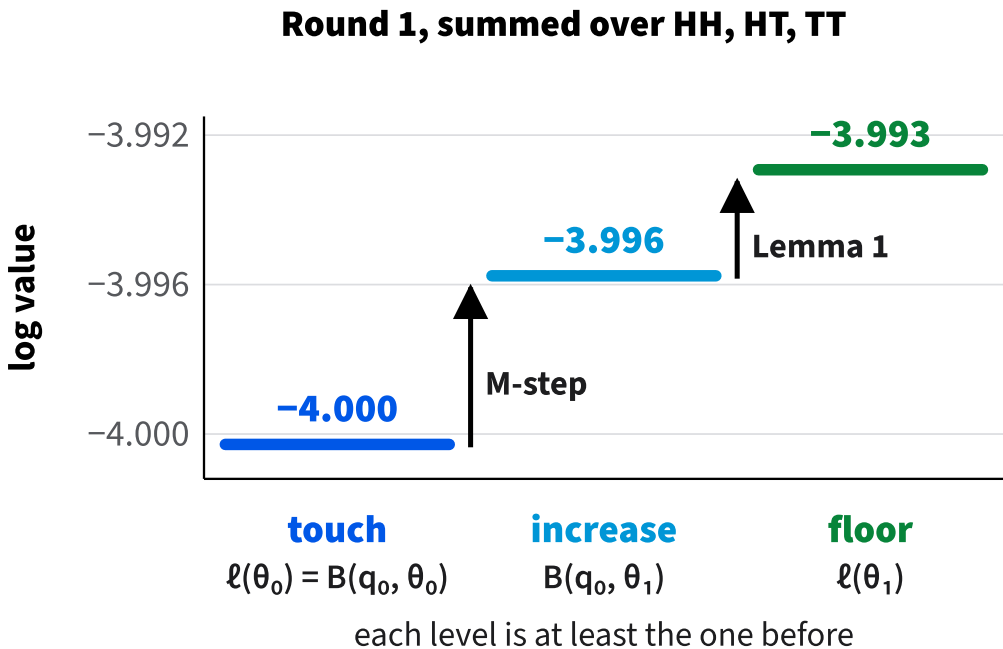
Every guess is a floor; only the posterior touches, and that is the E-step's guess.

Round 1 on the coins: ℓ goes up from -4.000 to -3.993

↳ Theorem 4 (p. 52) on one EM round from $\theta_0 = (\frac{3}{4}, \frac{1}{4})$ to $\theta_1 = (0.767, 0.233)$ (both from Section 3) (p. 44).

step	value
start: $\ell(\theta_0)$; the floor touches here	-4.000
M-step: the top of the floor	-3.996
new $\ell(\theta_1)$, above the floor	-3.993

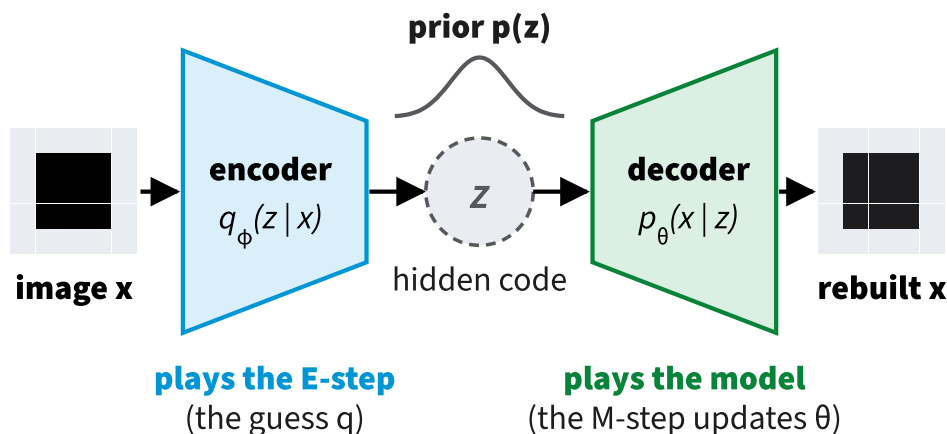
All steps: Appendix, Section 4 (p. 148).



Each step is at least the one before, so ℓ went up.

A variational autoencoder learns the E-step's guess

one network pass gives the guess for any image



training raises the ELBO in both θ and ϕ

Lemma 1 for one image: $\ln p_{\theta}(x) = \text{ELBO} + \text{gap}$

ELBO

gap

gap = $\text{KL}(\text{guess} \parallel \text{posterior}) \geq 0$

For an image x , the hidden code z is many real numbers, so the posterior cannot be computed exactly. A **variational autoencoder (VAE)** (Kingma & Welling, International Conference on Learning Representations (ICLR) 2014) trains an **encoder** with weights ϕ to output the guess $q_{\phi}(z | x)$.

TWO-COIN GAME

record HH

coin A or B

fair pick, chance $\frac{1}{2}$

E-step: exact posterior

M-step: count

bound $B(q, \theta)$

VARIATIONAL AUTOENCODER

an image x

code z , many numbers

prior $p(z)$, a standard bell curve

encoder's guess q_{ϕ}

gradient step on θ

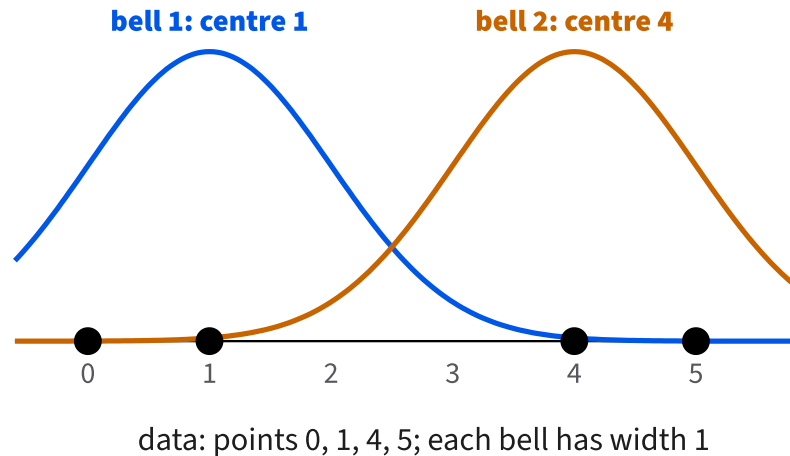
evidence lower bound (ELBO)

If the encoder guesses 0.8 for HH instead of the exact posterior 0.9, the bound falls **0.0444 short** of $\ln p(\text{HH})$: Lemma 1's gap (p. 50), which training shrinks.

Worked example: Appendix, Section 4 (p. 149).

EM uses the exact posterior; a VAE learns a guess that scales to images.

Next: replace the coins by bell curves



Nothing in the proofs of Lemma 1 (p. 51) and Theorem 4 (p. 53) used coins: only an observed x , a hidden z , and a model $p_{\theta}(x, z)$.

The points 0, 1, 4, 5 come from two bell curves. In EM, what plays the hidden coin?

each point

each point's bell

the bell's width

Answer. The bell: hidden, picked first, then the point is drawn from it. This is a Gaussian mixture.

Same E-step, same M-step, same increase in ℓ : now EM clusters points.

Same game, new records: numbers on a line instead of coin flips

Goal: find hidden groups in data with the same EM.

A friend picks **group 1 or group 2** by a fair toss, then draws a number from that group's **bell**: a normal (Gaussian) distribution with standard deviation (width)

1. We see only the numbers **0, 1, 4, 5**.

COIN GAME

coin A or B, chance $\frac{1}{2}$

record HH

θ_A , guess $\frac{3}{4}$

hidden: which coin

↔

↔

↔

↔

MIXTURE

group 1 or 2, chance $\frac{1}{2}$

one number, e.g. 1

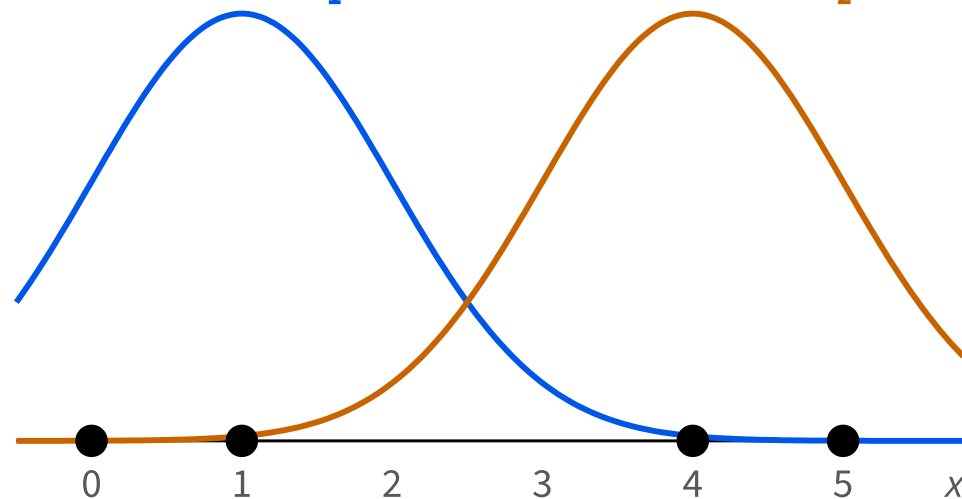
centre μ_1 , guess 1

hidden: which group

Unknowns: the centres μ_1, μ_2 . Working guess $\mu_1 = 1, \mu_2 = 4$, as $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ for the coins.

group 1: centre $\mu_1 = 1$

group 2: centre $\mu_2 = 4$



data: points 0, 1, 4, 5

which group made each point is hidden
like the coins: pick a group ($\frac{1}{2}, \frac{1}{2}$),
then draw a point from its bell

Coin ↔ group, flips ↔ a number: each EM step carries over one by one.

EM for mixtures: Bayes' rule, then weighted averages

Gaussian mixture. Pick group k with chance π_k ; draw x from the bell $\mathcal{N}(x \mid \mu_k, \sigma^2) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-(x-\mu_k)^2/2\sigma^2}$ (centre μ_k , standard deviation, or width, σ), for $k = 1, \dots, K$. Data $x_1, \dots, x_N$; the hidden group of point n is z_n , as the hidden coin was.

$$p(x) = \sum_k p(z = k) \cdot p(x \mid z = k) \quad \text{sum over the hidden group}$$

$$= \sum_k \pi_k \cdot \mathcal{N}(x \mid \mu_k, \sigma^2) \quad \text{pick chance} \cdot \text{bell}$$

$$r_{nk} = p(z_n = k \mid x_n) = \frac{\pi_k \cdot \mathcal{N}(x_n \mid \mu_k, \sigma^2)}{\sum_j \pi_j \cdot \mathcal{N}(x_n \mid \mu_j, \sigma^2)}$$

E-step: Bayes' rule

$$= \frac{e^{-(x_n - \mu_k)^2/2\sigma^2}}{\sum_j e^{-(x_n - \mu_j)^2/2\sigma^2}}$$

if all π_j are equal: π_j , $1/(\sqrt{2\pi} \cdot \sigma)$ cancel

$$\mu_k = \frac{\sum_n r_{nk} \cdot x_n}{\sum_n r_{nk}}, \quad \pi_k = \frac{1}{N} \cdot \sum_n r_{nk}$$

M-step: weighted averages

In words: r_{nk} = posterior chance that group k made point n ; new centre = share-weighted average.

Why it matters: both steps are closed-form, one division per point and one average per group, so a round is cheap.

The coins' two steps, with bell heights in place of $\theta^h \cdot (1 - \theta)^t$ (h heads, t tails).

The weighted average maximizes the weighted log-likelihood

↳ Group 1's weighted log-likelihood $\ell_1(\mu) = \sum_n r_n \cdot \ln \mathcal{N}(x_n | \mu, 1)$: point n counts with its share $r_n = r_{n1}$ (group 1's share of point n). **EM's M-step maximizes this.** For simplicity every bell has width $\sigma = 1$, so $\ln \mathcal{N}(x | \mu, 1) = -\frac{1}{2} \cdot \ln 2\pi - \frac{1}{2} \cdot (x - \mu)^2$.

$$\ell_1(\mu) = \sum_n r_n \cdot \left(-\frac{1}{2} \cdot \ln 2\pi - \frac{1}{2} \cdot (x_n - \mu)^2 \right)$$

log of each bell

$$= c - \frac{1}{2} \cdot \sum_n r_n \cdot (x_n - \mu)^2$$

$$c = -\frac{1}{2} \cdot \ln 2\pi \cdot \sum_n r_n: \text{no } \mu$$

$$\frac{d\ell_1}{d\mu} = \sum_n r_n \cdot (x_n - \mu)$$

$$\frac{d}{d\mu} \left[-\frac{1}{2} \cdot (x_n - \mu)^2 \right] = x_n - \mu$$

$$= \sum_n r_n \cdot x_n - \mu \cdot \sum_n r_n$$

split the sum

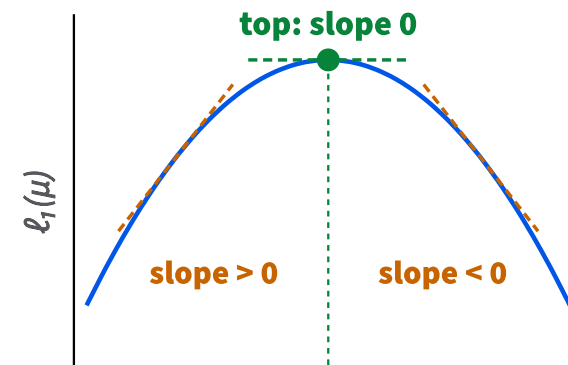
$$0 = \sum_n r_n \cdot x_n - \mu \cdot \sum_n r_n$$

at the top the slope is 0

$$\mu = \frac{\sum_n r_n \cdot x_n}{\sum_n r_n}$$

divide by $\sum_n r_n > 0$;

$$\frac{d^2\ell_1}{d\mu^2} = -\sum_n r_n < 0: \text{a top}$$



$$\mu = \sum r_n x_n \div \sum r_n$$

a downward parabola in μ :
its one top is where the slope is 0

In words: the best centre is the average of the points, each point counted r_n times.

Why it matters: the same algebra on $h \cdot \ln \theta + t \cdot \ln(1 - \theta)$ gives the coins' $\theta = h/(h + t)$ (p. 11): weighted heads $\div$ weighted flips.

The M-step is maximum likelihood with fractional counts, for coins and bells alike.

k-means: join the nearest centre, then move to the average

k-means (Lloyd, 1982), the most widely used clustering method, repeats two steps until nothing changes. Its score is $J = \sum_n (x_n - \mu_{k(n)})^2$, the sum of (point – its centre)², where $k(n)$ is the label of point n ; smaller is tighter.

Assign

each point joins the **nearest** centre,
 $k(n) = \arg \min_k |x_n - \mu_k|$: a hard label, weight 1 for that centre and 0 for the others, never a fraction.

Assign: hard labels



dashed: each point joins its nearest centre (triangle)

Move

each centre goes to the **plain average** of the points that joined it: $\mu_k = \frac{1}{|G_k|} \cdot \sum_{n \in G_k} x_n$, with G_k the set of points labelled k .

Move: plain averages



each centre (triangle) moves to the average of its points (diamond)

Two hard steps: a nearest-centre test for each point, then a plain average for each group.

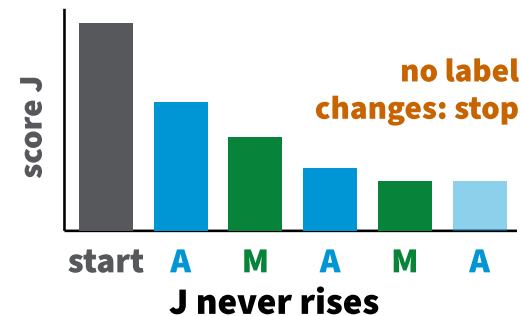
Neither k-means step raises J, so k-means always stops

↳ **Descent result.** $J(\text{labels}, \text{centres}) = \sum_n (x_n - \mu_{k(n)})^2$. One round: assign (new labels), then move (new centres).

$$\begin{aligned} J(\text{old labels}, \text{old centres}) &\geq J(\text{new labels}, \text{old centres}) && \text{assign: each point's term made smallest} \\ &\geq J(\text{new labels}, \text{new centres}) && \text{move: each group's sum smallest at its average} \end{aligned}$$

Why the average: for a fixed group G_k ,

$$\begin{aligned} \frac{d}{d\mu} \sum_{n \in G_k} (x_n - \mu)^2 &= -2 \cdot \sum_{n \in G_k} (x_n - \mu) && \text{derivative of each square} \\ 0 &= \sum_{n \in G_k} x_n - |G_k| \mu && \text{slope 0; split the sum, divide by } -2 \\ \mu &= \frac{1}{|G_k|} \cdot \sum_{n \in G_k} x_n && \begin{aligned} &\text{divide by } |G_k| > 0; \\ &\text{2nd derivative } 2|G_k| > 0: \text{ a minimum} \end{aligned} \end{aligned}$$



A = assign · M = move

Why it stops: after each move, J depends only on the labelling and strictly falls at every label change (ties keep the old label), so **no labelling repeats**; only K^N labellings exist.

In words: each step tightens the groups or leaves them alone, and k-means stops after finitely many rounds.

Why it matters: like EM's log-likelihood (Theorem 4) (p. 52), J only improves; but "stops" is not "best": a poor start can stop at a poor answer. In practice, use several starts or k-means++ (Arthur & Vassilvitskii, 2007), then run EM from k-means' centres.

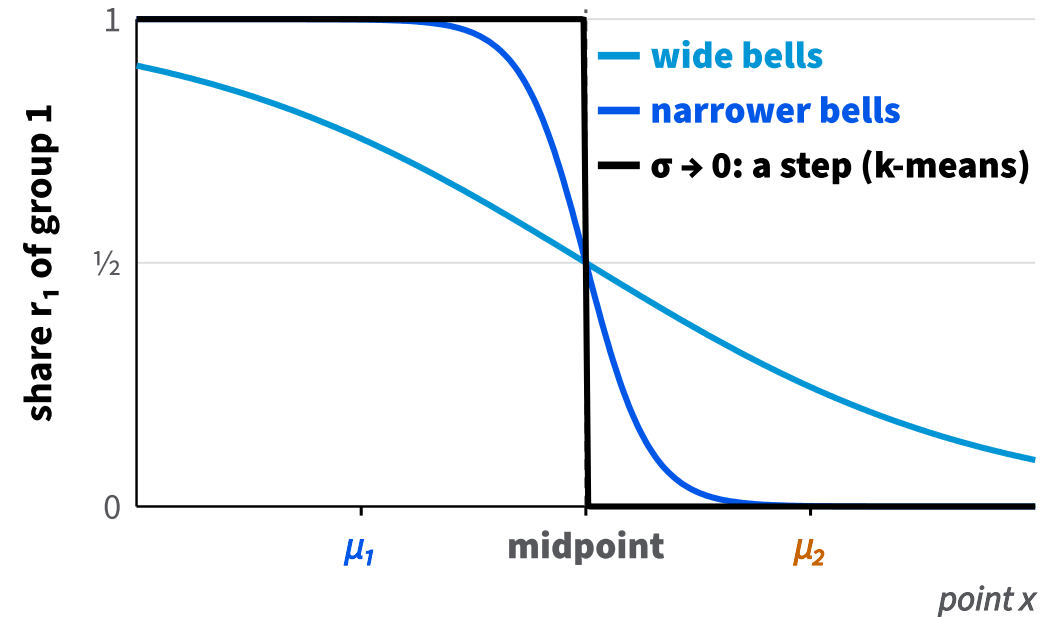
Each step lowers J or leaves it; only finitely many labellings exist, so k-means stops.

Theorem 5: k-means is EM with bells of width zero

Theorem 5 (k-means as a limit). Take a Gaussian mixture with equal weights π_k and one shared width σ , **held fixed** (not re-estimated), and let $\sigma \rightarrow 0$. Then EM's E-step becomes "assign each point to its nearest centre" and its M-step becomes "move each centre to the plain average of its points": EM becomes k-means.

In words: a share is a soft vote; shrinking every bell turns it into a hard vote for the nearest centre.

Why it matters: the most used clustering method is a special case of EM, so EM's two steps explain k-means, and EM is k-means that keeps its doubt.



nearer μ_1 : share $\rightarrow 1$ · nearer μ_2 : share $\rightarrow 0$
midpoint: $\frac{1}{2}$

Soft labels harden into hard ones; weighted averages become plain averages.

Proof: write the share as $1/(1 + e^{-u})$, then let σ shrink

↳ **Step 1: two groups.** Equal weights π_k and the factor $1/(\sqrt{2\pi} \cdot \sigma)$ cancel in the share. Distances $d_k = |x - \mu_k|$; $\Delta = d_2^2 - d_1^2$.

$$\begin{aligned} r_1 &= \frac{e^{-d_1^2/2\sigma^2}}{e^{-d_1^2/2\sigma^2} + e^{-d_2^2/2\sigma^2}} \\ &= \frac{1}{1 + e^{-d_2^2/2\sigma^2} / e^{-d_1^2/2\sigma^2}} \\ &= \frac{1}{1 + e^{-(d_2^2 - d_1^2)/2\sigma^2}} \\ &= \frac{1}{1 + e^{-u}}, \quad u = \frac{\Delta}{2 \cdot \sigma^2} \end{aligned}$$

$$\xrightarrow{\sigma \rightarrow 0} \begin{cases} 1/(1+0) = 1, & \Delta > 0 \\ 1/(1+\infty) = 0, & \Delta < 0 \\ \frac{1}{2} \text{ for every } \sigma, & \Delta = 0 \end{cases}$$

E-step share

divide by the first height

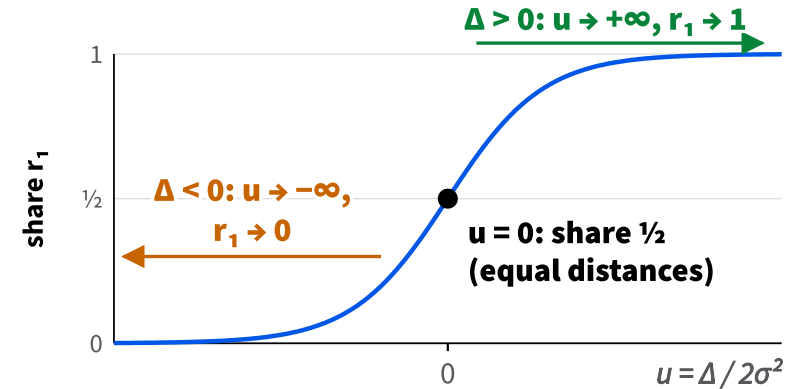
$$e^a / e^b = e^{a-b}$$

$$\Delta = d_2^2 - d_1^2$$

$$e^{-u} \rightarrow 0 \text{ if } u \rightarrow +\infty;$$

$$e^{-u} \rightarrow \infty \text{ if } u \rightarrow -\infty;$$

$$u = 0 \text{ if } \Delta = 0$$



$\sigma \rightarrow 0$: $|u| = |\Delta| / 2\sigma^2$ grows without bound,
so the share is pushed to a hard 1 or 0

$\Delta > 0$ means $d_1 < d_2$: **centre 1 is nearer**, so **E-step** → **nearest centre** (and $r_1 \rightarrow 0$ when $\Delta < 0$).

In words: narrow bells make the distance gap Δ count $1/2\sigma^2$ times over, so the nearer centre takes the whole point.

Why it matters: only the distance gap Δ decides the share, so **the nearer centre wins outright** once the bells are narrow; the same division works for any number of groups K .

E-step half: the soft share hardens into a hard vote for the nearer centre.

Proof, step 2: the E-step share becomes the nearest centre

$$\begin{aligned}
 r_1 &= \frac{\pi_1 \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-d_1^2/2\sigma^2}}{\sum_{j=1}^K \pi_j \cdot \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot e^{-d_j^2/2\sigma^2}} \\
 &= \frac{e^{-d_1^2/2\sigma^2}}{\sum_{j=1}^K e^{-d_j^2/2\sigma^2}} \\
 &= \frac{e^{-d_1^2/2\sigma^2}}{e^{-d_1^2/2\sigma^2} + \sum_{j=2}^K e^{-d_j^2/2\sigma^2}} \\
 &= \frac{1}{1 + \sum_{j=2}^K e^{-d_j^2/2\sigma^2} / e^{-d_1^2/2\sigma^2}} \\
 &= \frac{1}{1 + \sum_{j=2}^K e^{-(d_j^2 - d_1^2)/2\sigma^2}} \\
 &\xrightarrow{\sigma \rightarrow 0} \frac{1}{1 + 0} = 1
 \end{aligned}$$

E-step: group 1's height
 $\div$ all heights (Bayes' rule)

equal weights $\pi_j = 1/K$ and
 the common $1/(\sqrt{2\pi} \cdot \sigma)$ cancel

take the $j = 1$ term
 out of the sum

divide top and bottom
 by the first height

$$e^a / e^b = e^{a-b}$$

each $d_j^2 - d_1^2 > 0$: each
 exponent $\rightarrow -\infty$, each term $\rightarrow 0$

↳ **Step 2: E-step $\rightarrow$ nearest centre, any K .** Fix $\sigma > 0$ first, then let $\sigma \rightarrow 0$. Fix one point $x = x_n$; $d_j = |x - \mu_j|$ is its distance to centre j . $r_1 = r_{n1}$ is the share of group 1 for this one point (not a point, and not the centre with the smallest μ). Number the centres so that centre 1 is the one nearest to x (any centre can be called 1); then $d_1 < d_j$ for every $j \neq 1$. The sum runs over the K centres, not over points; G_k (a set of points) appears only in step 3.

Divide by the nearest centre's height: every other term is a ratio that vanishes as $\sigma \rightarrow 0$.

Proof, step 2 (continued): every share becomes 1 or 0

↳ **Step 2 continued. Step 3: M-step with the 1-or-0 weights.** Assume no point is equidistant from two centres, and every G_k is non-empty ($|G_k| > 0$), where G_k is the set of points nearest centre k and $|G_k|$ = how many. Recall (p. 66) $r_1 = \frac{1}{1 + \sum_{j=2}^K e^{-(d_j^2 - d_1^2)/2\sigma^2}}$ for one point $x = x_n$.

Why for every j : centre 1 was chosen as the centre nearest to x , so $d_1 < d_j$ for each $j \geq 2$ (no ties assumed). Distances are ≥ 0 , so squaring keeps the order: $d_1^2 < d_j^2$, i.e. $d_j^2 - d_1^2 > 0$. **Example:** $x = 1$, centres at 1 and 4: $d_1 = 0$, $d_2 = 3$, $d_2^2 - d_1^2 = 9 > 0$, so $e^{-9/2\sigma^2} \rightarrow 0$.

So each exponent $-(d_j^2 - d_1^2)/2\sigma^2 \rightarrow -\infty$ and each of the $K - 1$ terms $\rightarrow 0$; a finite sum of terms that $\rightarrow 0$ also $\rightarrow 0$, so $r_1 \rightarrow 1$: the **Assign** step.

Consequence: $r_{n1} \rightarrow 1$ for $n \in G_1$; for a point nearer another centre, the same argument with that centre in place of 1 gives $r_{n1} \rightarrow 0$.

In words: narrow bells: the far bell's height falls much faster than the near one's, so their ratio $e^{-(d_j^2 - d_1^2)/2\sigma^2} \rightarrow 0$ and the nearest centre takes the whole point.

Every other centre is farther, so every other term vanishes: the share becomes 1 or 0.

Proof, step 3: weighted averages become plain averages

↳ Recall step 2 (p. 67): as $\sigma \rightarrow 0$, $r_{n1} \rightarrow 1$ for $n \in G_1$ and $r_{n1} \rightarrow 0$ otherwise.

M-step → plain average:

$$\begin{aligned} \mu_1 &= \frac{\sum_n r_{n1} \cdot x_n}{\sum_n r_{n1}} && \text{M-step} \\ &\xrightarrow{\sigma \rightarrow 0} \frac{\sum_{n \in G_1} 1 \cdot x_n + \sum_{n \notin G_1} 0 \cdot x_n}{\sum_{n \in G_1} 1 + \sum_{n \notin G_1} 0} && \text{weights 1 or 0} \\ &= \frac{\sum_{n \in G_1} x_n}{|G_1|} && \text{drop the zeros} \end{aligned}$$

The weights 1 or 0 come from the E-step; $|G_1| > 0$, so the division is safe. The same holds for every centre k : μ_k becomes the plain average of its set G_k , the **Move** step of k-means.

Both EM steps become the two k-means steps, for every K . ■

In words: with hard weights, a weighted average is the plain average of the points that carry weight 1.

Why it matters: k-means needs no new theory: it is EM's two steps in the limit of very narrow bells.

Both EM steps turn into the two k-means steps.

One EM round: share 0.989, then centres 1, 4 → 0.52, 4.48

↳ **Definition (p. 60)** applied to points 0, 1, 4, 5: centres 1 and 4, width $\sigma = 1$, weights $\pi_k = \frac{1}{2}$ (cancel); bell height $e^{-d^2/2}$, d = distance to the centre.

point x	bell 1 $e^{-(x-1)^2/2}$	bell 2 $e^{-(x-4)^2/2}$	share r of group 1	$r \cdot x$
0	$e^{-1/2}=0.607$	$e^{-8}=0.000335$	0.9994	0
1	$e^0=1$	$e^{-9/2}=0.0111$	0.9890	0.989
4	$e^{-9/2}=0.0111$	$e^0=1$	0.0110	0.044
5	$e^{-8}=0.000335$	$e^{-1/2}=0.607$	0.0006	0.003
total			2.000	1.036

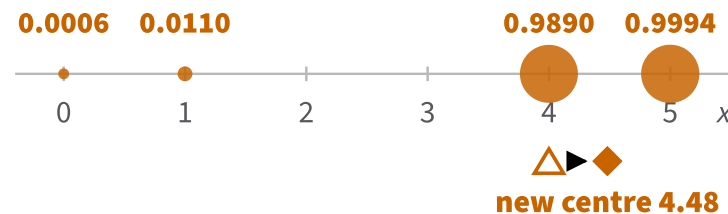
E-step, share = bell 1 ÷ (bell 1 + bell 2). Point 1: $r = \frac{1}{1 + 0.0111} = 0.989$, as HH gave coin A $\frac{9}{9+1} = 0.9$ (p. 29).

M-step, weighted average: $\mu_1 = \frac{1.036}{2.000} = 0.518 \approx 0.52$; the mirror $x \mapsto 5 - x$ swaps the groups, so $\mu_2 = 5 - 0.518 \approx 4.48$.

group 1: dot size = share r (number above)



group 2: dot size = share r (number above)



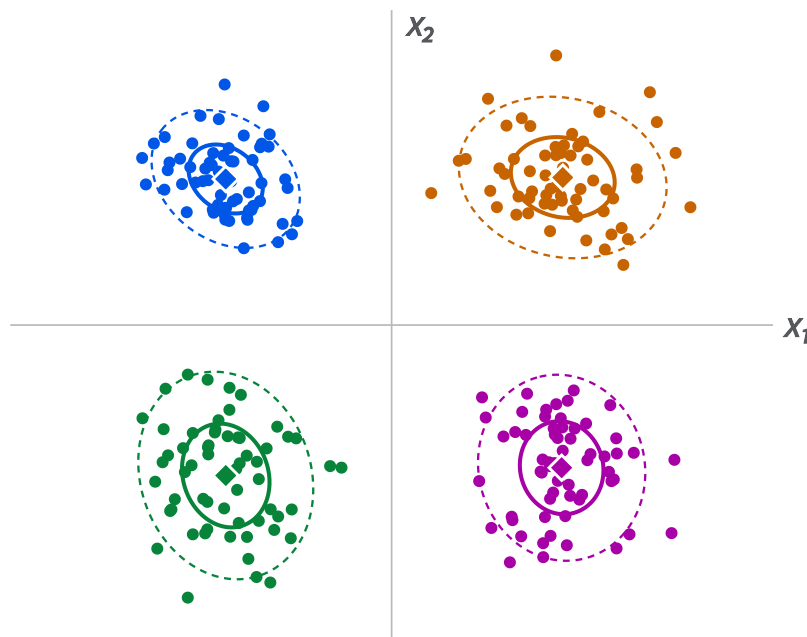
hollow: old centre

diamond: weighted average = new centre

Bayes' rule gives soft labels; weighted averages pull the centres to 0.52 and 4.48.

The same EM round finds four groups in 240 points

↳ Definition (p. 60) applied in the plane: $K = 4$ groups; each group's width (oval) is re-estimated too.



ovals: 1 and 2 widths from each hill's centre
diamonds: centres

Points (x_1, x_2) in the plane, four groups. Each bell is now a hill whose contours are ovals.

Step 1: each hill's height at each point.

Step 2 (E): colour each point by its shares; pale = unsure.

Step 3 (M): move each centre to its share-weighted average; refit each oval from the share-weighted spread of its points.

Average log-likelihood per point **ris**es every **round**, from -4.309 to -2.901 at round 40 (Theorem 4 (p. 52): an EM round never lowers the likelihood).

round **40** • average log-likelihood per point **-2.901** • colour: the group with the largest share, paler when unsure

Recolour, move, repeat: clustering is EM.

k-means stops at J = 1; EM with narrow bells ends there too

↳ **Descent result (p. 63) applied:** k-means on 0, 1, 4, 5 from a **poor guess**, centres 0 and 1; J = sum of (point – its centre)². **Theorem 5 (p. 64) applied:** EM from the working guess, centres 1 and 4, with the bells narrowed.

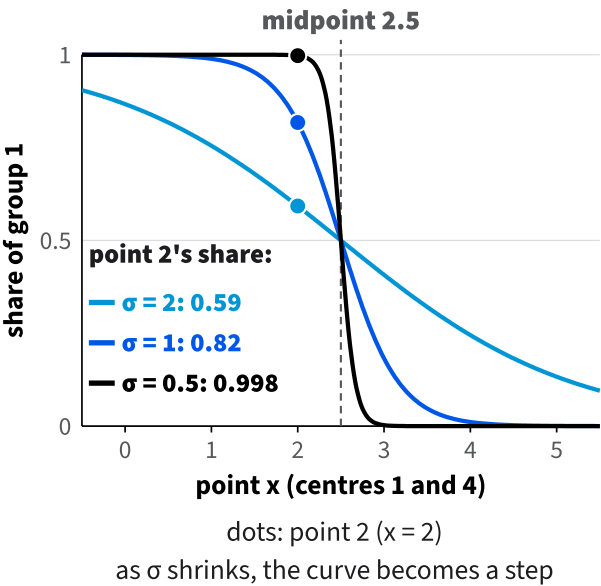
step	centres	groups	J
assign	0, 1	{0} · {1, 4, 5}	$0 + 0 + 9 + 16 = 25$
move	$0, \frac{1+4+5}{3} = \frac{10}{3}$	{0} · {1, 4, 5}	$(\frac{7}{3})^2 + (\frac{2}{3})^2 + (\frac{5}{3})^2 = \frac{49 + 4 + 25}{9} = 8.67$
assign	$0, \frac{10}{3}$	{0, 1} · {4, 5}	$1^2 + (\frac{2}{3})^2 + (\frac{5}{3})^2 = 4.22$
move	$\frac{0+1}{2}, \frac{4+5}{2} = 0.5, 4.5$	{0, 1} · {4, 5}	$4 \cdot 0.5^2 = 1$

Distances of 1, 4, 5 to the centre $\frac{10}{3}$: $\frac{7}{3}, \frac{2}{3}, \frac{5}{3}$.

A new point at $x = 2$ (distances 1 and 2 to centres 1 and 4) gets the share $\frac{e^{-1/2\sigma^2}}{e^{-1/2\sigma^2} + e^{-2/\sigma^2}}$, which **tends to 1** as $\sigma \rightarrow 0$ (the figure's dots: 0.59, 0.82, 0.998 at $\sigma = 2, 1, 0.5$).

With every share 1 or 0, EM's M-step gives $\frac{0+1}{2} = 0.5$ and $\frac{4+5}{2} = 4.5$ (groups {0, 1} and {4, 5}): **k-means' centres.**

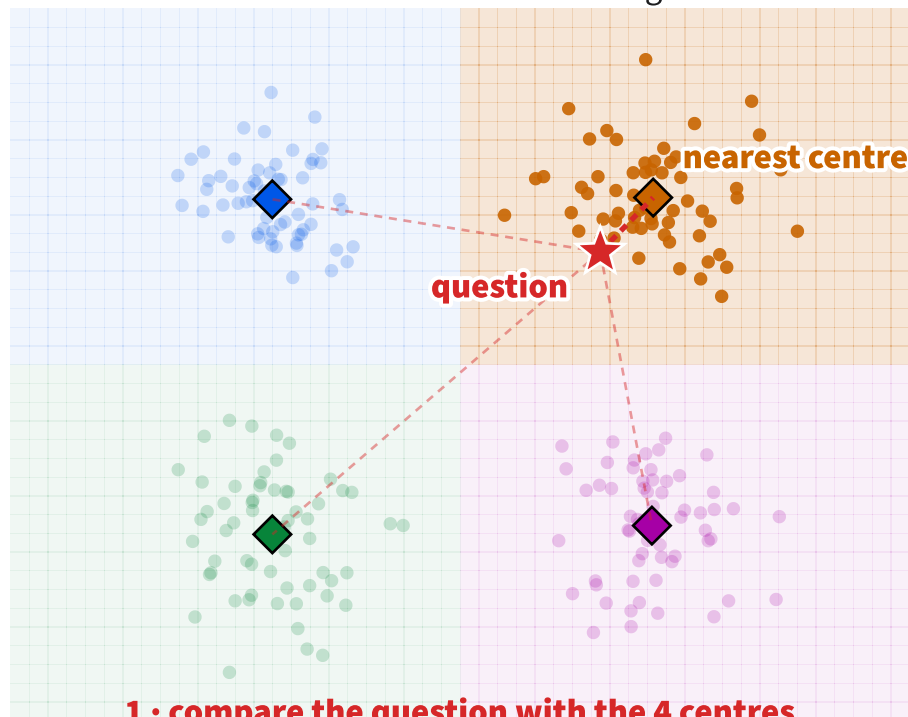
All steps: Appendix, Section 5 (p. 150).



One family: EM is k-means that admits doubt; k-means is EM that refuses it.

Vector search files texts in k-means cells and searches one cell

diamonds: 4 k-means centres • shading: their cells



1 • compare the question with the 4 centres

2 • search only the nearest centre's cell

A chatbot answers from a library of texts, each stored as a **vector** (similar meanings lie close). For a question it must find the **nearest** stored vectors, then answer with those texts (**retrieval-augmented generation**).

COIN GAME		MIXTURE		VECTOR SEARCH
record HH	↔	point x	↔	one text's vector
coin A or B	↔	group	↔	cell of similar texts
θ_A	↔	centre μ_k	↔	cell centre
which coin?	↔	nearest centre	↔	file a text in its cell

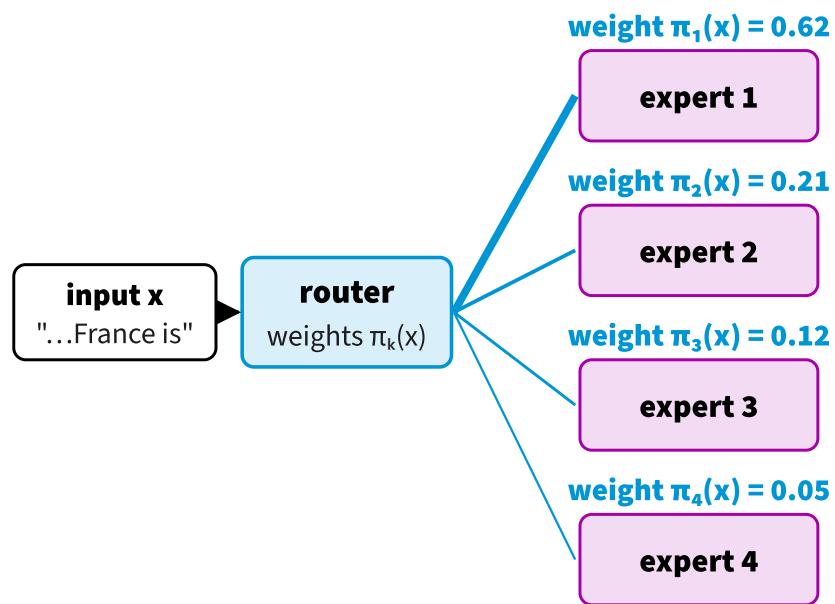
For a million texts this needs about **500 times fewer** distance computations than comparing with every vector. **Price:** a neighbour just across a cell border can be missed.

Jégou, Douze & Schmid, "Product Quantization for Nearest Neighbor Search," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2011.

Worked example: Appendix, Section 5 (p. 151).

k-means' nearest-centre step, reused as an index: search a cell, not the library.

A mixture of experts: mixture weights that depend on the input



arrow width = router weight for this token

A language model reads text one **token** (a word or a piece of one) at a time. A layer holds several smaller networks, the **experts**. A small **router** gives each expert a weight that depends on the text read so far, and only the top few experts run.

COIN GAME		MIXTURE		MIXTURE OF EXPERTS
coin A or B	↔	group k	↔	expert k
chance $\frac{1}{2}$	↔	weight π_k	↔	router weight $\pi_k(x)$
$p(\text{HH} \mid A)$	↔	bell $\mathcal{N}(x \mid \mu_k, \sigma^2)$	↔	prediction $p_k(y \mid x)$
record HH	↔	point x	↔	next token y

Why bother: the model holds many experts' parameters, yet each token pays only for the few it runs. Mixtral runs **2 of its 8 experts** per token (Jiang et al., "Mixtral of Experts," 2024).

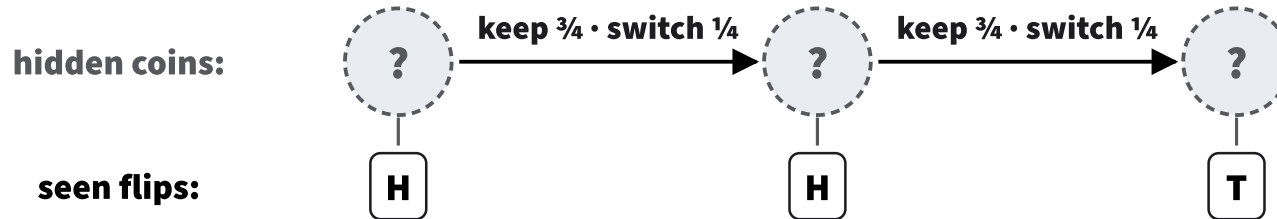
Worked example: Appendix, Section 5 (p. 152).

Same sum as a mixture (Jacobs et al., 1991); only the weights now depend on the token.

Next: the coin may switch between flips



So far one hidden group per record. Now the friend may switch coins between flips: **one hidden coin per flip**, kept with chance $\frac{3}{4}$, switched with $\frac{1}{4}$.



Three flips, each with its own hidden coin. How many coin sequences are there?

2 6 8

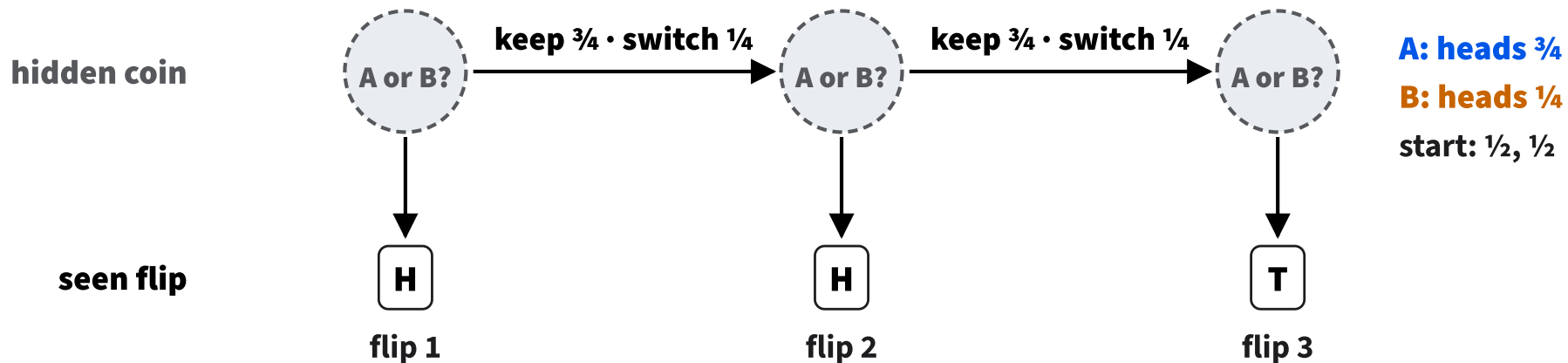
Answer. $8 = 2 \cdot 2 \cdot 2$. For 100 flips, 2^{100} : too many to list.

Hidden chains: the sum over sequences, done step by step.

If the coin can switch between flips, the coins form a hidden chain

Goal: the posterior of each flip's coin, and $p(\text{flips})$, without listing every sequence.

↳ In the two-coin game one hidden coin made both flips of a record. Now the friend may switch coins between flips (keep chance $\kappa = \frac{3}{4}$: keep $\frac{3}{4}$, switch $\frac{1}{4}$), so **every flip has its own hidden coin**.



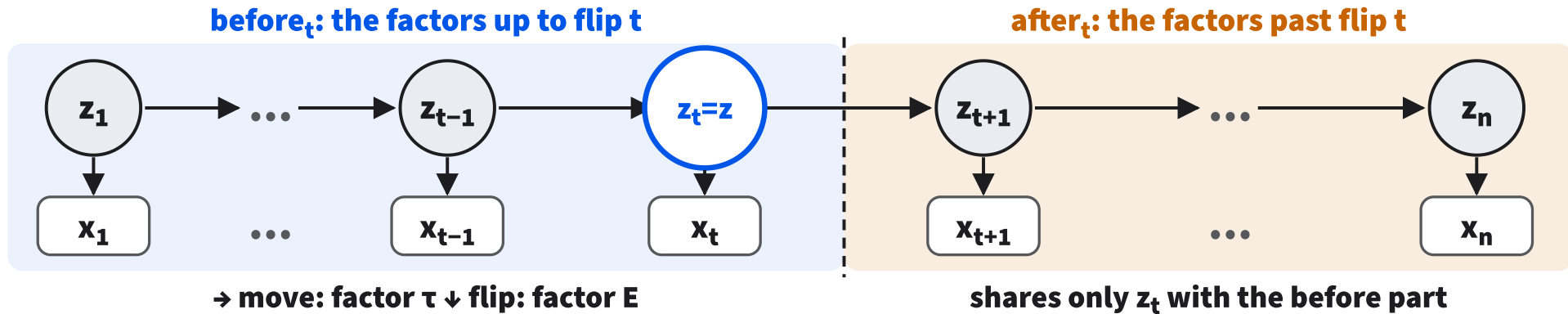
A **hidden Markov model**: each coin depends only on the coin before (Lecture 2's chain), and each flip only on its own coin. Early speech recognizers were built on it.

A hidden chain links the coins flip to flip; on H, H, T we will find flip 2's coin and $p(\text{H}, \text{H}, \text{T})$.

A hidden Markov model multiplies one factor per arrow

↳ Symbols: $z = (z_1, \dots, z_n)$ = the hidden coins, $x = (x_1, \dots, x_n)$ = the flips; $\pi(z)$ = chance the first coin is z ; $\tau(z' \rightarrow z)$ = chance the coin moves from z' to z : κ if $z = z'$ (keep), $1 - \kappa$ otherwise (switch), with keep chance $\kappa = \frac{3}{4}$ here; $E(z, x)$ = chance coin z shows x .

$$p(x, z) = \underbrace{\pi(z_1) \cdot E(z_1, x_1)}_{\text{start, flip 1}} \cdot \prod_{s=2}^n \underbrace{\tau(z_{s-1} \rightarrow z_s)}_{\text{move}} \cdot \underbrace{E(z_s, x_s)}_{\text{flip } s}$$



Cut at flip t : $p(x, z) = \text{before}_t \cdot \text{after}_t$, where

$$\text{before}_t = \pi(z_1) \cdot E(z_1, x_1) \cdot \prod_{s=2}^t \tau(z_{s-1} \rightarrow z_s) \cdot E(z_s, x_s), \quad \text{after}_t = \prod_{s=t+1}^n \tau(z_{s-1} \rightarrow z_s) \cdot E(z_s, x_s)$$

One factor per arrow; the part past flip t touches the earlier coins only through z_t .

Forward-backward works on any chain of n flips

↳ Recall (p. 76): $\pi(z)$ first coin, $\tau(z' \rightarrow z)$ move, $E(z, x)$ flip chance (one factor per arrow); flips $x = (x_1, \dots, x_n)$; z_t = hidden coin at flip t .

$\alpha_t(z)$: **forward**

chance of flips 1 to t **and** coin z at flip t

$$\alpha_1(z) = \pi(z) \cdot E(z, x_1), \quad \alpha_t(z) = \left(\sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \right) \cdot E(z, x_t)$$

$$\beta_n(z) = 1, \quad \beta_t(z) = \sum_{z'} \tau(z \rightarrow z') \cdot E(z', x_{t+1}) \cdot \beta_{t+1}(z')$$

$$p(x) = \sum_z \alpha_t(z) \cdot \beta_t(z), \quad p(z_t = z \mid x) = \frac{\alpha_t(z) \cdot \beta_t(z)}{p(x)}$$

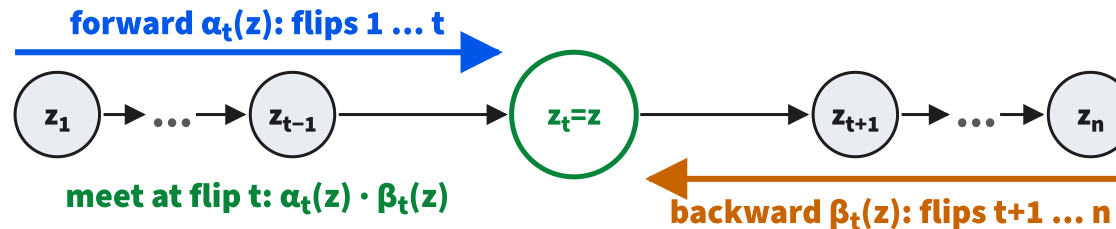
$\beta_t(z)$: **backward**

chance of flips $t+1$ to n , **given** coin z at flip t

Forward, left to right: add the ways in, then the flip.

Backward, right to left: after flip n nothing remains.

Combine at any flip t : add for the total, share for the posterior.



Each flip, each of K coins adds K ways in: about $n \cdot K^2$ steps, not K^n .

Proof, part 1: the cut gives $p(x, z_t=z) = \alpha_t(z) \cdot \beta_t(z)$

↳ Recall (p. 76): $p(x, z) = \text{before}_t \cdot \text{after}_t$ (factors up to flip t , factors past flip t); $z_{1:t-1} = (z_1, \dots, z_{t-1})$, $x_{1:t} = (x_1, \dots, x_t)$.

$$p(x, z_t=z) = \sum_{z_{1:t-1}} \sum_{z_{t+1:n}} p(x, z_{1:t-1}, z, z_{t+1:n})$$

(add over all the other coins)

$$= \sum_{z_{1:t-1}} \sum_{z_{t+1:n}} \text{before}_t(z_{1:t-1}, z) \cdot \text{after}_t(z, z_{t+1:n})$$

(one factor per arrow, split at flip t)

$$= \left(\sum_{z_{1:t-1}} \text{before}_t(z_{1:t-1}, z) \right) \cdot \left(\sum_{z_{t+1:n}} \text{after}_t(z, z_{t+1:n}) \right)$$

(before _{t} has no $z_{t+1:n}$, after _{t} no $z_{1:t-1}$:
distributive law)

$$= \left(\sum_{z_{1:t-1}} p(z_1) \cdot p(x_1 | z_1) \cdot \prod_{s=2}^t p(z_s | z_{s-1}) \cdot p(x_s | z_s) \right)$$

(write out before _{t} and after _{t} ;

$$\cdot \left(\sum_{z_{t+1:n}} \prod_{s=t+1}^n p(z_s | z_{s-1}) \cdot p(x_s | z_s) \right)$$

π, τ, E are these conditionals)

$$= \left(\sum_{z_{1:t-1}} p(x_{1:t}, z_{1:t-1}, z_t=z) \right) \cdot \left(\sum_{z_{t+1:n}} p(x_{t+1:n}, z_{t+1:n} | z_t=z) \right)$$

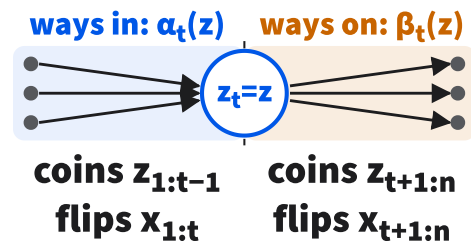
(left: chain rule. Right: chain rule + Markov:
given z_t , later coins and flips ignore $z_{1:t-1}, x_{1:t}$,
so before and after are independent given z_t)

$$= p(x_{1:t}, z_t=z) \cdot p(x_{t+1:n} | z_t=z) = \alpha_t(z) \cdot \beta_t(z)$$

(add out summed coins; definitions of α_t, β_t)

In words: a sequence through z at flip t = a way in • a way on; the ways on ignore how we arrived, so add each side, then multiply once.

Why it matters: adding over z gives $p(x) = \sum_z \alpha_t(z) \cdot \beta_t(z)$; dividing by $p(x)$ (Bayes' rule) gives $p(z_t=z | x)$.



Two short messages per coin replace the sum over all K^n sequences.

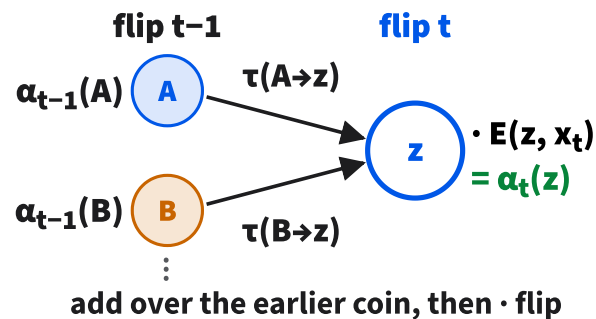
Proof, part 2: forward adds the ways in, then the flip

↳ Markov: coin t depends only on coin $t-1$, flip t only on coin t . Start: $\alpha_1(z) = p(x_1, z_1=z) = \pi(z) \cdot E(z, x_1)$.

$$\begin{aligned}
 \alpha_t(z) &= p(x_{1:t}, z_t=z) && \text{(definition)} \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z', z_t=z, x_t) && \text{(add over the coin before, } z_{t-1}=z') \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z') \cdot p(z_t=z \mid z_{t-1}=z', x_{1:t-1}) \cdot p(x_t \mid z_t=z, z_{t-1}=z', x_{1:t-1}) && \text{(chain rule)} \\
 &= \sum_{z'} p(x_{1:t-1}, z_{t-1}=z') \cdot p(z_t=z \mid z_{t-1}=z') \cdot p(x_t \mid z_t=z) && \text{(Markov: coin } t \text{ needs only coin } t-1; \text{ flip } t \text{ needs only coin } t) \\
 &= \sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \cdot E(z, x_t) && \text{(definitions of } \alpha_{t-1}, \tau, E) \\
 &= \left(\sum_{z'} \alpha_{t-1}(z') \cdot \tau(z' \rightarrow z) \right) \cdot E(z, x_t) && (E(z, x_t) \text{ does not depend on } z')
 \end{aligned}$$

In words: to stand on coin z at flip t , come from any coin z' , move, then show flip t .

Why it matters: each step reuses the K numbers α_{t-1} : K^2 products per flip, never the list of earlier sequences.



Forward: (old α $\cdot$ move, added over the old coin) $\cdot$ flip chance.

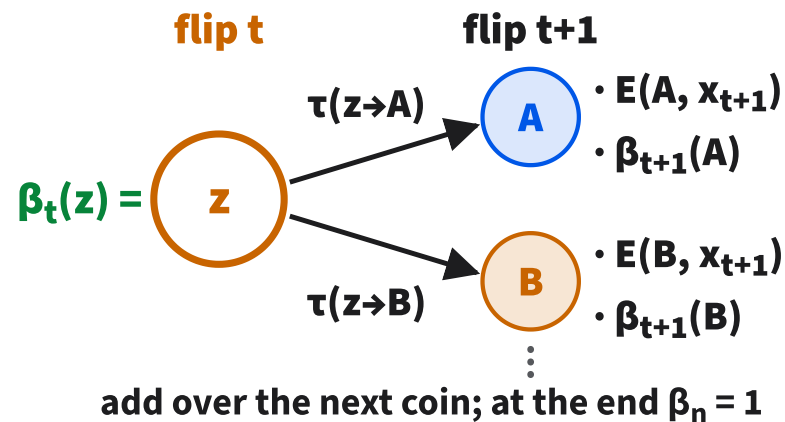
Proof, part 3: the backward message adds the ways on

↳ Markov: given the next coin z_{t+1} , the later flips forget z_t . Start: after flip n no flips remain, so $\beta_n(z) = 1$.

$$\begin{aligned}
 \beta_t(z) &= p(x_{t+1:n} \mid z_t = z) && \text{(definition)} \\
 &= \sum_{z'} p(z_{t+1} = z', x_{t+1}, x_{t+2:n} \mid z_t = z) && \text{(add over the next coin, } z_{t+1} = z') \\
 &= \sum_{z'} p(z_{t+1} = z' \mid z_t = z) \cdot p(x_{t+1} \mid z_{t+1} = z', z_t = z) \cdot p(x_{t+2:n} \mid z_{t+1} = z', x_{t+1}, z_t = z) && \text{(chain rule)} \\
 &= \sum_{z'} p(z_{t+1} = z' \mid z_t = z) \cdot p(x_{t+1} \mid z_{t+1} = z') \cdot p(x_{t+2:n} \mid z_{t+1} = z') && \text{(Markov: given } z_{t+1}, \text{ drop } z_t \text{ and } x_{t+1}) \\
 &= \sum_{z'} \tau(z \rightarrow z') \cdot E(z', x_{t+1}) \cdot \beta_{t+1}(z') && \text{(definitions of } \tau, E, \beta_{t+1})
 \end{aligned}$$

In words: from coin z at flip t , move to any coin z' , show flip $t+1$, then carry on from z' .

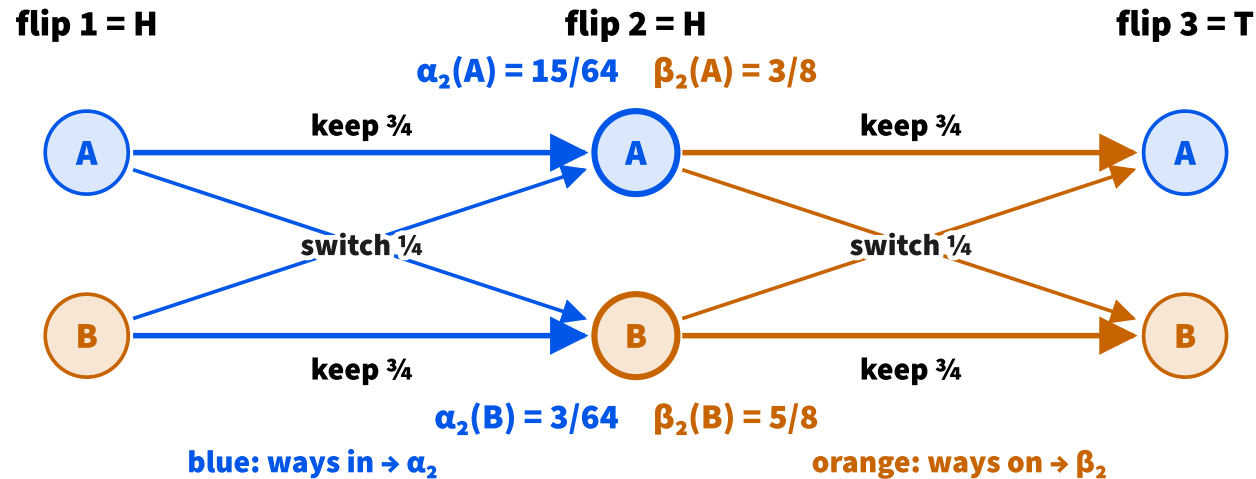
Why it matters: together with parts 1 and 2 this proves every line of the forward-backward theorem.



Backward: (move · next flip · next β), added over the next coin.

On H, H, T, forward · backward at flip 2 gives $p(\text{HHT}) = 15/128$

↳ Forward-backward theorem (p. 77) applied to the coins (first coin $\frac{1}{2}$; keep $\frac{3}{4}$, switch $\frac{1}{4}$; A shows H with $\frac{3}{4}$, B with $\frac{1}{4}$), cut at flip 2. The forward pass gives α_2 , the backward pass β_2 :



$$p(\text{HHT}) = \sum_z \alpha_2(z) \cdot \beta_2(z) = \underbrace{\frac{15}{64} \cdot \frac{3}{8}}_{\text{A at flip 2}} + \underbrace{\frac{3}{64} \cdot \frac{5}{8}}_{\text{B at flip 2}} = \frac{45}{512} + \frac{15}{512} = \frac{15}{128}$$

Share (Bayes' rule): coin A made flip 2 with chance $\frac{45}{45+15} = \frac{45}{60} = \frac{3}{4}$.

All steps: Appendix, Section 6 (p. 153).

Same as adding all $2^3 = 8$ sequences, but 2 numbers per coin per flip: Lecture 2's sum-product.

To learn the switching game, run the same EM as for two coins

↳ So far the rules were given. Now we see only the flips and must **learn** them: the heads chances θ_A, θ_B and the keep chance κ ; the start chance stays $\frac{1}{2}$.

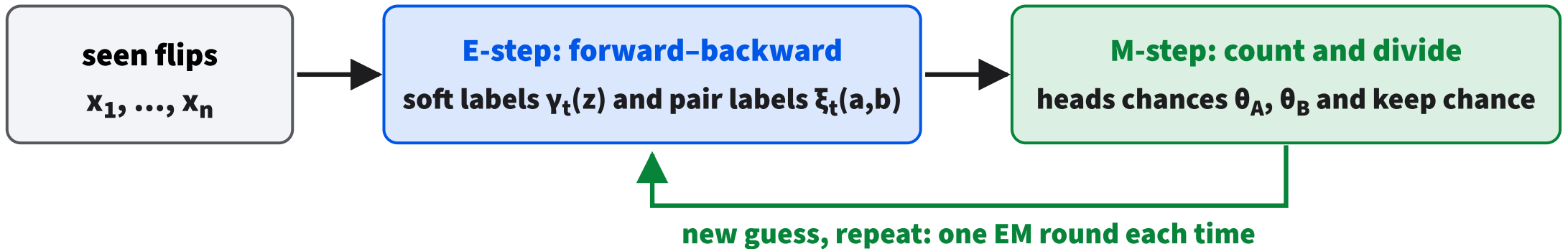
Why learn κ too? Nobody tells us how often the friend switches, and κ shapes the soft labels: with κ near 1 the coin rarely switches, so neighbouring flips share their coin; with $\kappa = \frac{1}{2}$ each flip is judged on its own. A wrong κ gives wrong soft labels, so EM must learn it along with θ_A, θ_B .

	two-coin game	switching game (hidden chain)
hidden	one coin per record	one coin per flip
E-step	$p(\text{coin} \mid \text{record})$ by Bayes (HH: 0.9) (p. 29)	$p(\text{coin at flip } t \mid \text{all flips})$ by forward-backward
extra soft label	none	$p(\text{coins at flips } t, t+1 \mid \text{all flips})$
M-step	heads to A ÷ flips to A	the same; keep chance = expected keeps ÷ moves

Same EM as for two coins, plus one new rule to learn: the keep chance κ .

Two soft labels: one flip's coin, and a neighbouring pair

Soft labels. $\gamma_t(z)$ = chance that coin z made flip t , given all the flips (one coin's soft label, like HH's 0.9 (p. 29)); $\xi_t(a, b)$ = chance that the coin is a at flip t and b at flip $t+1$, given all the flips (needed to count keeps: $\text{kept} = \xi_t(A, A) + \xi_t(B, B)$). On H, H, T: $\gamma_1(A) = 0.8$, $\gamma_2(A) = 0.75$, $\gamma_3(A) = 0.4$; kept between flips 1 and 2 with chance 0.8.



This EM for hidden chains is Baum–Welch (Baum et al., 1970), older than the name "EM".

Baum–Welch = forward–backward soft labels + counting

↳ Recall (p. 77): $\alpha_t(z)$ = chance of flips 1.. t and coin z at flip t ; $\beta_t(z)$ = chance of flips $t+1$.. n , given coin z at t ; $\tau(a \rightarrow b)$ = chance the coin moves from a to b ; $E(b, x)$ = chance coin b shows x ; θ_A = heads chance of coin A.

Theorem 6 (Baum–Welch; Baum, Petrie, Soules & Weiss, 1970). For a hidden Markov model with flips $x_1, \dots, x_n$, the rules to learn are the heads chances θ_A, θ_B and the **keep chance** κ (kappa): between two flips the friend keeps the same coin with chance κ and switches with chance $1 - \kappa$, i.e. $\tau(a \rightarrow a) = \kappa, \tau(a \rightarrow b) = 1 - \kappa$ ($a \neq b$); in our game $\kappa = \frac{3}{4}$ (keep $\frac{3}{4}$, switch $\frac{1}{4}$). One EM round is the E-step below and the M-step on the next slide, and it never lowers $p(x)$. Here the start chance stays $\frac{1}{2}$; with a separate chance per move, $\tau(a \rightarrow b) = \sum_{t < n} \xi_t(a, b) / \sum_{t < n} \gamma_t(a)$.

$$\gamma_t(z) = \frac{\alpha_t(z) \cdot \beta_t(z)}{p(x)}$$

$$\xi_t(a, b) = \frac{\alpha_t(a) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b)}{p(x)}$$

E-step, one coin: chance coin z made flip t , given all flips.

E-step, pairs: chance of coin a at t and b at $t+1$, given all flips.

E-step: forward–backward gives both soft labels, γ for one coin and ξ for a pair.

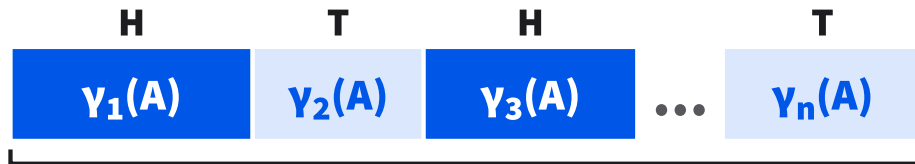
Baum–Welch, M-step: count with the soft labels

↳ Theorem 6 continued (p. 84): with the soft labels γ_t and ξ_t from the E-step, the M-step counts and divides.

$$\theta_A = \frac{\sum_{t=1}^n \gamma_t(A) \cdot [x_t=H]}{\sum_{t=1}^n \gamma_t(A)}, \quad \kappa = \frac{\sum_{t=1}^{n-1} (\xi_t(A, A) + \xi_t(B, B))}{n-1}$$

M-step: count and divide; $[x_t=H] = 1$ for heads, 0 for tails. κ : expected keeps $\div$ the $n-1$ moves ($\xi_t(A, A) + \xi_t(B, B)$ = chance the coin was kept between flips t and $t+1$).

Proof plan. The expected complete-data log-likelihood (coins and flips together) splits into $\gamma \cdot \ln(\text{heads chance})$ and $\xi \cdot \ln(\text{keep or switch chance})$ terms; counting maximizes each (Theorem 1) (p. 11); Theorem 4 (EM increases the likelihood) (p. 52): $p(x)$ never falls.



all blocks: $u + v = \sum_t \gamma_t(A)$, soft flips

solid: $u = \sum_t \gamma_t(A)[x_t=H]$, soft heads

$\theta_A = \text{solid length} \div \text{total length}$

	A at t+1	B at t+1	total
A at t	$\xi_t(A, A)$	$\xi_t(A, B)$	$\gamma_t(A)$
B at t	$\xi_t(B, A)$	$\xi_t(B, B)$	$\gamma_t(B)$
total	$\gamma_{t+1}(A)$	$\gamma_{t+1}(B)$	1

green = kept; the four cells add to 1

E-step: forward–backward soft labels. M-step: count with them (θ_B : the same with B).

Proof, E-step: cutting at a move gives the pair soft label

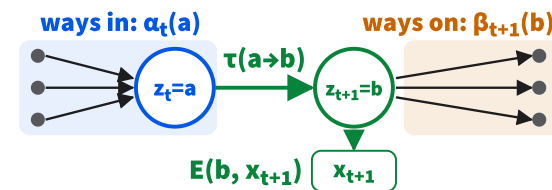
↳ **Why pairs:** keeping links two neighbouring coins, so we need each pair $(z_t, z_{t+1}) = (a, b)$; recall $p(x, z) = \text{one factor per arrow}$ (p. 76).

$$\begin{aligned}
 \xi_t(a, b) &= p(z_t=a, z_{t+1}=b \mid x) = \frac{p(x, z_t=a, z_{t+1}=b)}{p(x)} && \text{(definition; Bayes' rule)} \\
 &= \frac{1}{p(x)} \cdot \sum_{z_{1:t-1}} \sum_{z_{t+2:n}} \text{before}_t(z_{1:t-1}, a) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \text{after}_{t+1}(b, z_{t+2:n}) && \text{(add over other coins; split at the move)} \\
 &= \frac{1}{p(x)} \cdot \left(\sum_{z_{1:t-1}} \text{before}_t(z_{1:t-1}, a) \right) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \left(\sum_{z_{t+2:n}} \text{after}_{t+1}(b, z_{t+2:n}) \right) && \text{(distributive law; sums share no coins)} \\
 &= \frac{1}{p(x)} \cdot \underbrace{p(x_{1:t}, z_t=a)}_{\alpha_t(a)} \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \underbrace{p(x_{t+2:n} \mid z_{t+1}=b)}_{\beta_{t+1}(b)} && \text{(brackets are marginals: } \alpha_t(a), \beta_{t+1}(b))
 \end{aligned}$$

In words: a pair (a at flip t , b at flip $t+1$) is weighted by all the ways to arrive at a , times moving $a \rightarrow b$ and showing flip $t+1$, times all the ways to go on from b ; divide by $p(x)$ to get a chance.

Check: if we do not care which coin comes next (add over b), we get back the one-flip label $\gamma_t(a)$ (since $\sum_b \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b) = \beta_t(a)$).

Why it matters: forward-backward, run once, gives both labels γ and ξ .



Pair label: arrive at a , move to b and show the next flip, go on from b ; divide by $p(x)$.

Proof, M-step: start, move and flip terms separate

↳ EM's M-step maximizes the **expected complete-data log-likelihood** $Q(\theta)$; the posterior $q(z) = p_{\theta'}(z | x)$ from the current guess θ' is held fixed. $[\cdot] = 1$ if true, 0 if false.

$$\begin{aligned}
 Q(\theta) &= \sum_z p_{\theta'}(z | x) \cdot \ln p_{\theta}(x, z) = \sum_z q(z) \cdot \ln p_{\theta}(x, z) && \text{(the M-step objective)} \\
 &= \sum_z q(z) \cdot \left[\ln \pi(z_1) + \sum_{t=1}^{n-1} \ln \tau(z_t \rightarrow z_{t+1}) + \sum_{t=1}^n \ln E(z_t, x_t) \right] && \text{(log of a product = sum of logs)} \\
 &= \sum_z q(z) \cdot \ln \pi(z_1) + \sum_{t=1}^{n-1} \sum_z q(z) \cdot \ln \tau(z_t \rightarrow z_{t+1}) + \sum_{t=1}^n \sum_z q(z) \cdot \ln E(z_t, x_t) && \text{(swap the sums)} \\
 &= \sum_a \ln \pi(a) \cdot \sum_z q(z) \cdot [z_1=a] + \sum_{t=1}^{n-1} \sum_{a,b} \ln \tau(a \rightarrow b) \cdot \sum_z q(z) \cdot [z_t=a, z_{t+1}=b] && \text{(sort the sequences by the} \\
 &\quad + \sum_{t=1}^n \sum_c \ln E(c, x_t) \cdot \sum_z q(z) \cdot [z_t=c] && \text{coins each log factor uses)} \\
 &= \underbrace{\sum_a \gamma_1(a) \cdot \ln \pi(a)}_{\text{start}} + \underbrace{\sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a, b) \cdot \ln \tau(a \rightarrow b)}_{\text{moves}} + \underbrace{\sum_{t=1}^n \sum_z \gamma_t(z) \cdot \ln E(z, x_t)}_{\text{flips}} && \begin{aligned} &(\sum_z q(z) \cdot [z_t=c] \\ &= p_{\theta'}(z_t=c | x) = \gamma_t(c), \\ &\sum_z q(z) \cdot [z_t=a, z_{t+1}=b] = \xi_t(a, b)) \end{aligned}
 \end{aligned}$$

In words: each log factor touches one or two coins, so it needs only γ or ξ , never all K^n sequences.

Why it matters: the three terms share no parameter, so the M-step maximizes each on its own. **Start term:** π is fixed at $\frac{1}{2}$ here, so it needs no update.

Each log factor becomes one term, weighted by the soft label of the coins it touches.

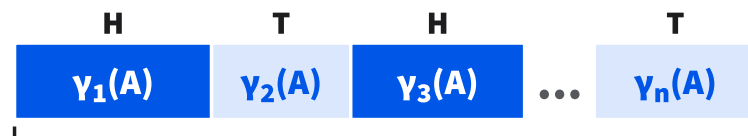
Proof, M-step: the flip terms give heads ÷ flips

↳ Recall Theorem 1 (counting) (p. 11): $u \cdot \ln \theta + v \cdot \ln(1 - \theta)$ is largest at $\theta = \frac{u}{u+v}$, heads ÷ flips. Coin A: $E(A, H) = \theta_A$, $E(A, T) = 1 - \theta_A$.

$$\begin{aligned}
 \text{flip terms of A} &= \sum_{t=1}^n \gamma_t(A) \cdot \ln E(A, x_t) && (\text{the } z=A \text{ flip terms; } z=B \text{ terms hold only } \theta_B) \\
 &= \underbrace{\left(\sum_t \gamma_t(A) \cdot [x_t=H] \right)}_{u : \text{ soft heads}} \cdot \ln \theta_A + \underbrace{\left(\sum_t \gamma_t(A) \cdot [x_t=T] \right)}_{v : \text{ soft tails}} \cdot \ln(1 - \theta_A) && (\text{H uses } \theta_A, \text{ T uses } 1 - \theta_A) \\
 \Rightarrow \theta_A &= \frac{u}{u+v} = \frac{\sum_t \gamma_t(A) \cdot [x_t=H]}{\sum_t \gamma_t(A) \cdot [x_t=H] + \sum_t \gamma_t(A) \cdot [x_t=T]} && (\text{Theorem 1}) \\
 &= \frac{\sum_{t=1}^n \gamma_t(A) \cdot [x_t=H]}{\sum_{t=1}^n \gamma_t(A)} && (\text{each flip is H or T})
 \end{aligned}$$

In words: credit each flip to coin A by its soft label, then divide heads by flips.

Why it matters: this is Theorem 6's θ_A (p. 85) (θ_B : the same with B).



all blocks: $u + v = \sum_t \gamma_t(A)$, soft flips

solid: $u = \sum_t \gamma_t(A)[x_t=H]$, soft heads

$\theta_A = \text{solid length} \div \text{total length}$

Soft counts in, Theorem 1 out: heads ÷ flips.

Proof, M-step: the move terms give keeps ÷ moves

↳ One keep chance κ for both coins: $\tau(a \rightarrow a) = \kappa, \tau(a \rightarrow b) = 1 - \kappa$ for $a \neq b$. Theorem 1 again (p. 11): $u \cdot \ln \kappa + v \cdot \ln(1 - \kappa)$ peaks at $\kappa = \frac{u}{u+v}$.

$$\begin{aligned}
 \text{move terms} &= \sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a, b) \cdot \ln \tau(a \rightarrow b) && \text{(the moves part)} \\
 &= \underbrace{\sum_{t < n} (\xi_t(A, A) + \xi_t(B, B))}_{u: \text{ expected keeps}} \cdot \ln \kappa + \underbrace{\sum_{t < n} (\xi_t(A, B) + \xi_t(B, A))}_{v: \text{ expected switches}} \cdot \ln(1 - \kappa) && \text{(kept uses } \kappa, \text{ switched } 1 - \kappa) \\
 \Rightarrow \kappa &= \frac{u}{u+v} = \frac{\sum_{t < n} (\xi_t(A, A) + \xi_t(B, B))}{\sum_{t < n} \sum_{a,b} \xi_t(a, b)} && \text{(Theorem 1)} \\
 &= \frac{\sum_{t=1}^{n-1} (\xi_t(A, A) + \xi_t(B, B))}{n-1} && \text{(each move's four } \xi \text{ add to 1)}
 \end{aligned}$$

In words: new keep chance = expected keeps ÷ number of moves.

If each coin had its own move chances, the same counting gives

$\tau(a \rightarrow b) = \sum_{t < n} \xi_t(a, b) / \sum_{t < n} \gamma_t(a)$: expected moves $a \rightarrow b$ ÷ expected visits to a .

Why it matters: this is Theorem 6's M-step for κ (p. 85); by Theorem 4 (p. 52), $p(x)$ never falls. ■

	A at t+1	B at t+1	total
A at t	$\xi_t(A, A)$	$\xi_t(A, B)$	$\gamma_t(A)$
B at t	$\xi_t(B, A)$	$\xi_t(B, B)$	$\gamma_t(B)$
total	$\gamma_{t+1}(A)$	$\gamma_{t+1}(B)$	1

green = kept; the four cells add to 1

Soft counts in, Theorem 1 out: keeps ÷ moves.

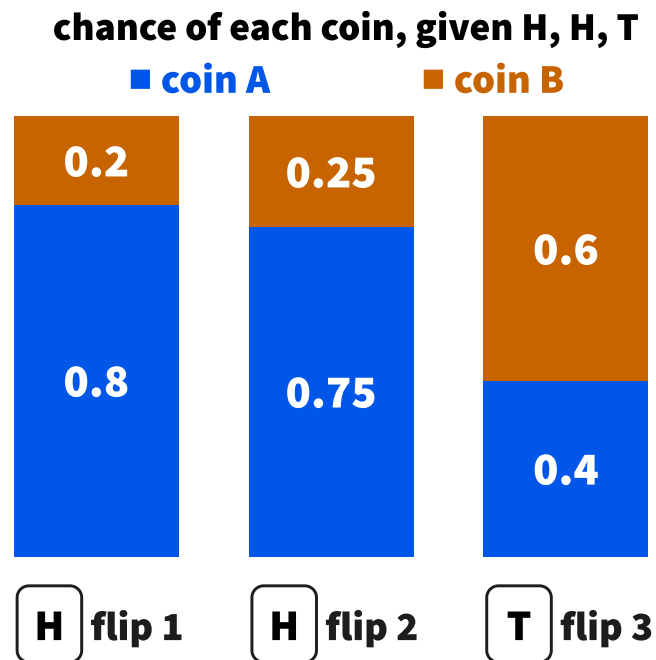
E-step: coin A's soft labels on H, H, T are 0.8, 0.75, 0.4

↳ Theorem 6's E-step (p. 84) applied (guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, keep $\frac{3}{4}$): $\gamma_t(A) = \alpha_t(A) \cdot \beta_t(A) / p(\text{HHT})$, the chance coin A made flip t , given all flips. Forward-backward gives the products; recall $p(\text{HHT}) = \frac{15}{128} = \frac{240}{2048}$ (p. 81).

flip	1 = H	2 = H	3 = T
$\alpha_t(A) \cdot \beta_t(A) \cdot 2048$	192	180	96
$\gamma_t(A) = \text{that} \div 240$	0.8	0.75	0.4
$\gamma_t(B) = 1 - \gamma_t(A)$	0.2	0.25	0.6

Flip 3 is a tail, so coin B (tails $\frac{3}{4}$) takes over there.

All steps: Appendix, Section 6 (p. 153).



Soft labels for coin A at flips 1, 2, 3: 0.8, 0.75, 0.4. The M-step counts with them.

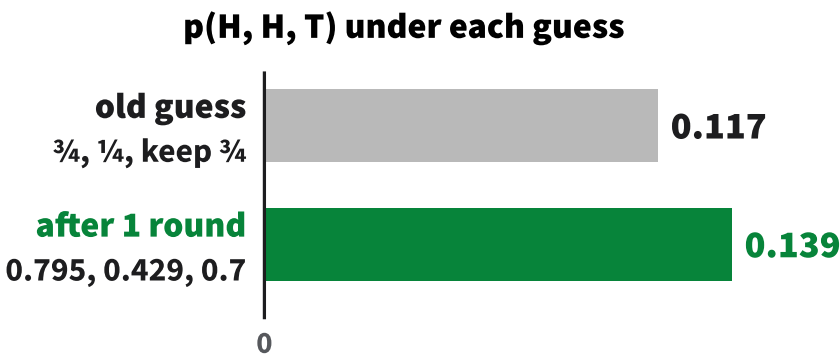
M-step: counting with soft labels raises $p(\text{HHT})$ to 0.139

↳ Theorem 6's M-step (p. 85) applied: recall coin A's soft labels 0.8, 0.75, 0.4 (p. 90) (B gets the rest), and the expected keeps 0.8 between flips 1 and 2 and 0.6 between flips 2 and 3 (from the pair labels ξ_t).

parameter	old	new = count ÷ total
θ_A	$\frac{3}{4}$	$\frac{0.8+0.75}{0.8+0.75+0.4} = \frac{1.55}{1.95} = 0.795$
θ_B	$\frac{1}{4}$	$\frac{0.2+0.25}{0.2+0.25+0.6} = \frac{0.45}{1.05} = 0.429$
keep κ	$\frac{3}{4}$	$\frac{0.8+0.6}{2} = 0.7$
$p(\text{HHT})$	$\frac{15}{128} = 0.117$	0.139 (forward pass)

θ : heads credited ÷ flips credited; κ : expected keeps ÷ 2 moves.

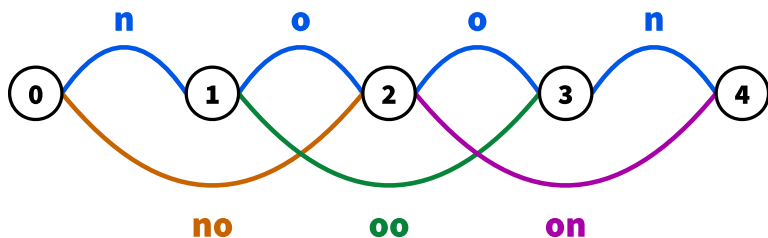
All steps: Appendix, Section 6 (p. 154).



One Baum–Welch round raises $p(\text{HHT})$, as Theorem 4 (EM increases the likelihood) promises.

Tokenizers learn where to cut words by EM over hidden cuts

every path from 0 to 4 is one cut of “noon”



node k = boundary after k letters

E-step: posterior weight of each cut



A language model reads text in pieces (**tokens**). No one labels where to cut "noon"; the **Unigram tokenizer** learns it by EM, starting with every piece equally likely.

COIN CHAIN

hidden coin sequence

flips H, H, T

heads chance

keep / switch chance

soft label γ_t

forward message α_t

TOKENIZER

hidden cut, e.g. no|on

letters n, o, o, n

piece probability

none: pieces drawn independently

posterior of a cut, given the letters

α_k = chance of all cuts of the first k letters

The likely cut no|on gets **0.61 of the weight**, so the M-step raises the pieces no and on and lowers oo. Speech recognition (CTC) runs the same sums over audio–letter alignments.

T. Kudo, “Subword regularization,” Association for Computational Linguistics (ACL) 2018; used in SentencePiece.

Worked example: Appendix, Section 6 (p. 155).

Cuts are hidden like coins; forward–backward adds over all cuts without listing them.

Next: when even the shortcut is too big



Forward-backward works because each coin touches only its neighbours. A hidden **chain of thought** (100 words a language model writes before its answer, each from a 50,000-word vocabulary) has $50,000^{100} \approx 10^{470}$ possible values, and each word depends on all earlier ones: **no small messages**.

How can we still get a soft label, such as HH's $p(A \mid \text{HH}) = 0.9$ (p. 29), without adding up every way?

guess 0.5

replay the game at random, count

skip the record

Answer. Replay and count: Monte Carlo, as in Lecture 4.

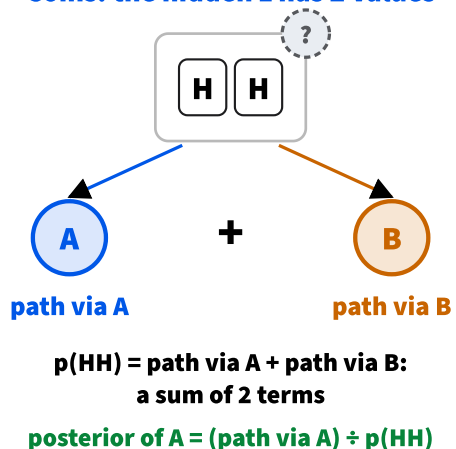
Replace the exact E-step by samples.

The E-step's sum can be impossible, yet running the model forward is cheap

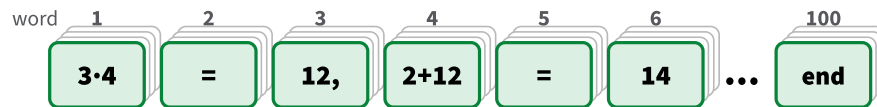
Goal: run EM when the hidden variable has far too many values to add up.

↳ Recall (p. 35): the E-step needs the **posterior** $p_{\theta}(z \mid x) = p_{\theta}(x, z) / \sum_{z'} p_{\theta}(x, z')$ of the hidden z given a record x such as HH, at the current guess θ . The sum has **one term per value** of z .

Coins: the hidden z has 2 values



Reasoning: the hidden z is a 100-word text



behind each word: a stack of 50,000 possible words

Sum over every z : $50,000 \cdot 50,000 \cdot \dots$ (100 times)
 $\approx 10^{470}$ terms, more than atoms in the visible universe

Run forward: pick one word at a time, left to right

100 easy steps; keep only the runs that reach the right answer, and the kept runs are posterior draws

All steps: Appendix, Section 7 (p. 156).

Adding up every z can be hopeless; running forward and keeping the matches is cheap. Monte Carlo EM uses such draws.

Monte Carlo EM: draw m hidden values per record, then fit as if they were told

↳ Recall (p. 35): EM (expectation–maximization) needs the posterior $p_{\theta_t}(z \mid x)$ of the hidden z at each guess θ_t . When its sum over z is too large, **draw** z instead.

Definition (Monte Carlo EM; G. C. G. Wei and M. A. Tanner, J. Amer. Statist. Assoc., 1990). Records $x^{(1)}, \dots, x^{(N)}$; start from θ_0 ; choose m , the number of draws per record. For $t = 0, 1, 2, \dots$:

E-step: for each record n , draw m hidden values, $z^{(n,j)}$ = the j -th draw for record n (for example by replaying the model and keeping the matching runs):

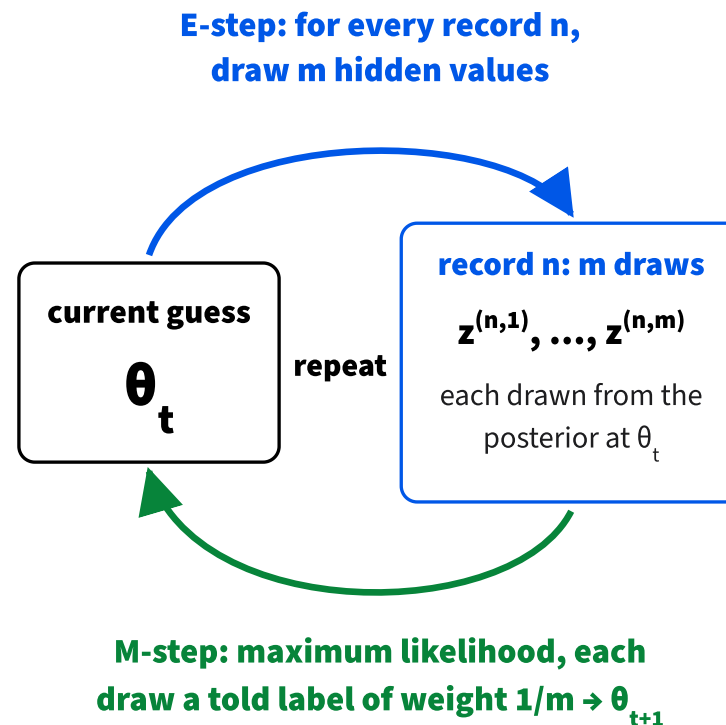
$$z^{(n,1)}, \dots, z^{(n,m)} \sim p_{\theta_t}(z \mid x^{(n)})$$

M-step: maximum likelihood with each draw as a told label of weight $\frac{1}{m}$:

$$\theta_{t+1} = \arg \max_{\theta} \sum_{n=1}^N \frac{1}{m} \cdot \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)})$$

Why it matters: only draws are needed, **never the sum** over all z .

Each draw counts $1/m$; with many draws, the fraction equal to z nears its posterior: exact EM.



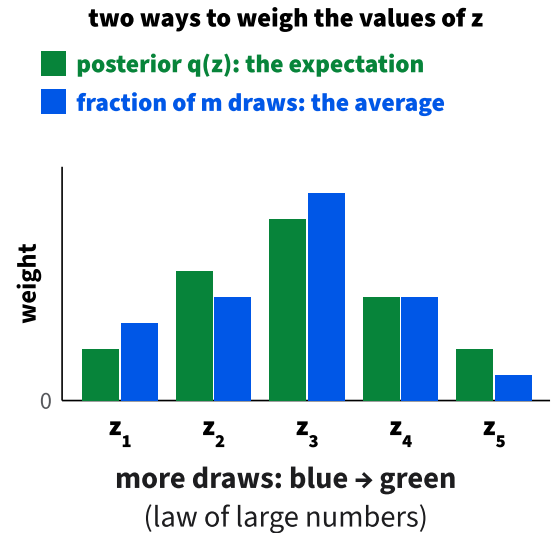
EM's M-step expectation becomes an average over m draws

↳ Recall Section 3 (p. 35) (EM = expectation–maximization): $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$ is record n 's posterior (its soft label) at the guess θ_t .

Monte Carlo rule (law of large numbers): for any function g , $\mathbb{E}_{z \sim q}[g(z)] \approx \frac{1}{m} \cdot \sum_{j=1}^m g(z^{(j)})$, $z^{(j)} \sim q$ independent.

$$\begin{aligned}
 & \sum_n \sum_z q_t^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z) && \text{EM's M-step objective} \\
 &= \sum_n \mathbb{E}_{z \sim q_t^{(n)}} [\ln p_{\theta}(x^{(n)}, z)] && \text{weights } q_t^{(n)}(z) \text{ sum to 1: an expectation} \\
 &\approx \sum_n \frac{1}{m} \cdot \sum_{j=1}^m \ln p_{\theta}(x^{(n)}, z^{(n,j)}) && \text{Monte Carlo rule, } g(z) = \ln p_{\theta}(x^{(n)}, z), z^{(n,j)} \sim q_t^{(n)} \\
 &= \sum_n \sum_z \hat{q}^{(n)}(z) \cdot \ln p_{\theta}(x^{(n)}, z) && \text{group equal draws}
 \end{aligned}$$

Line 3 is **Monte Carlo EM's M-step objective**; $\hat{q}^{(n)}(z) = \text{fraction of the } m \text{ draws equal to } z$.



In words: the same log-probability terms, with each posterior share $q_t^{(n)}(z)$ replaced by a fraction of draws $\hat{q}^{(n)}(z)$.

Why it matters: the M-step keeps EM's form, so whatever solved EM's M-step solves this one, with fractions in place of shares.

Same log-probability terms, new weights: each posterior share becomes a fraction of draws.

Draws act as told labels: Section 1's maximum likelihood

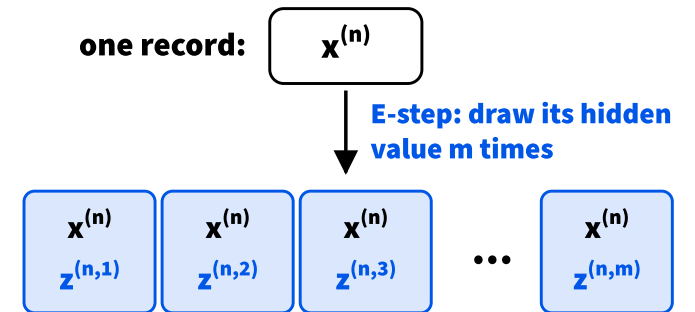
↳ Recall Section 1 (Theorem 1) (p. 11): when the friend tells every coin $z^{(n)}$, maximum likelihood maximizes $\sum_n \ln p_\theta(x^{(n)}, z^{(n)})$, and the answer is **count and divide**.

$$\begin{aligned}
 & \sum_{n=1}^N \frac{1}{m} \cdot \sum_{j=1}^m \ln p_\theta(x^{(n)}, z^{(n,j)}) && \text{Monte Carlo EM's M-step objective} \\
 &= \frac{1}{m} \cdot \sum_{\text{pairs } (n,j)} \ln p_\theta(x^{(n)}, z^{(n,j)}) && \text{one sum over the } N \cdot m \text{ pairs (record, draw)} \\
 &= \frac{1}{m} \cdot \ln \prod_{\text{pairs } (n,j)} p_\theta(x^{(n)}, z^{(n,j)}) && \text{a sum of logs is the log of a product} \\
 &= \frac{1}{m} \cdot \ln p_\theta(\text{pretend data}) && \text{the } N \cdot m \text{ pairs as independent told records}
 \end{aligned}$$

The factor $\frac{1}{m} > 0$ does not move the arg max: θ_{t+1} is **told-label maximum likelihood**.

In words: the drawn hidden values are used exactly as if a friend had told them.

Why it matters: no new optimizer. Any model we can fit with told labels, we can fit with Monte Carlo EM.



m told copies, each of weight 1/m

N records • m draws = Nm told pairs

M-step: fit as if the labels were told

(for the coins: count and divide)

Each draw is a told label, so the M-step is ordinary maximum likelihood on the pretend data.

Replay and keep: the kept runs are draws from the posterior

↳ Recall (p. 95): the E-step needs draws from the posterior $p_\theta(z \mid x) = p_\theta(x, z)/p_\theta(x)$, and $p_\theta(x) = \sum_{z'} p_\theta(x, z')$ is the sum that can be too large. Running the model **forward** needs no sum.

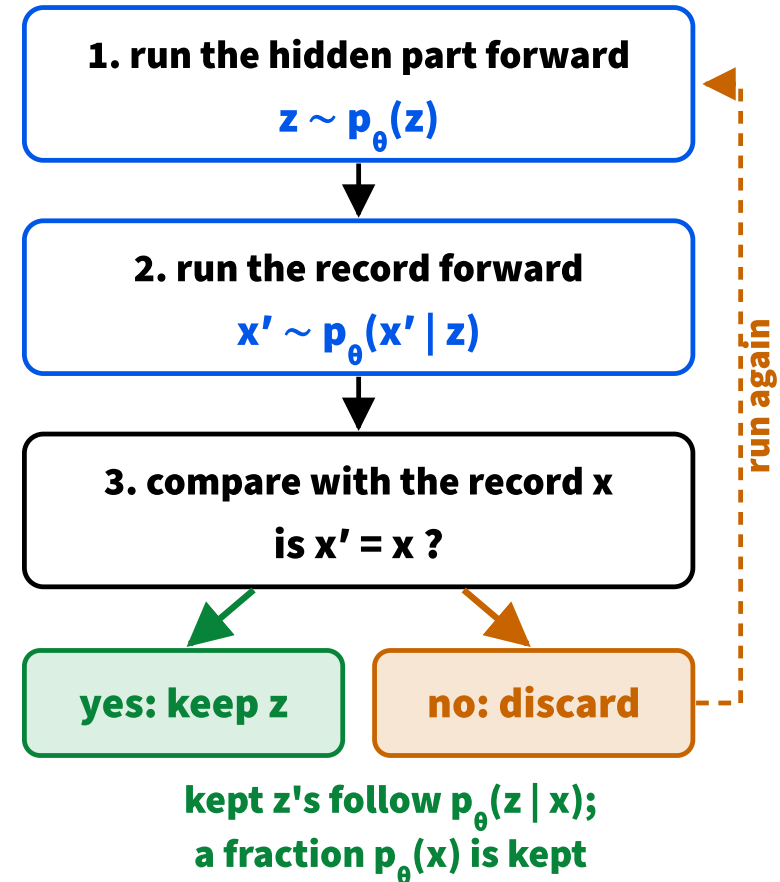
Proposition (replay and keep: rejection sampling). To draw the hidden z for a record x , repeat:

1. draw z from the model, $z \sim p_\theta(z)$;
2. generate a record from it, $x' \sim p_\theta(x' \mid z)$;
3. **keep** z only if $x' = x$.

The kept z 's are draws from the posterior $p_\theta(z \mid x)$.

Why: a value z is kept with chance $p_\theta(z) \cdot p_\theta(x \mid z)$ (be drawn, then reproduce x), which is proportional to its posterior.

Good: no sum over z is ever computed. **Cost:** only a fraction $p_\theta(x)$ of runs is kept, so **rare records waste most runs**.



Run forward, keep the matches: posterior draws, with no sum computed.

Proof: dividing by the keep rate gives the posterior

↳ **Proof of the proposition (replay and keep) (p. 98).** One run draws $z \sim p_\theta(z)$, then $x' \sim p_\theta(x' | z)$; it is **kept** when $x' = x$. For one run:

$$\begin{aligned}
 P(z \mid \text{kept}) &= \frac{P(z, \text{kept})}{P(\text{kept})} && \text{conditional probability} \\
 &= \frac{p_\theta(z) \cdot p_\theta(x \mid z)}{\sum_{z'} p_\theta(z') \cdot p_\theta(x \mid z')} && \text{draw } z; \text{ then } x' \text{ must equal } x; \text{ add over } z' \\
 &= \frac{p_\theta(x, z)}{\sum_{z'} p_\theta(x, z')} && \text{product rule} \\
 &= \frac{p_\theta(x, z)}{\underbrace{p_\theta(x)}} && \text{sum rule} \\
 &= \frac{p_\theta(x, z)}{P(\text{kept})} && \\
 &= p_\theta(z \mid x) && \text{definition} \blacksquare
 \end{aligned}$$

Runs are independent, so the kept values are too.

In words: a value z survives in proportion to how well it explains x , which is exactly the posterior.

Three textbook rules take "draw, then match" to the posterior, with $p_\theta(x)$ only in the proof.

The three rules used:

conditional probability

$$P(z \mid \text{kept}) = P(z, \text{kept}) / P(\text{kept})$$

product rule

$$p_\theta(z) \cdot p_\theta(x \mid z) = p_\theta(x, z)$$

sum rule

$$\sum_{z'} p_\theta(x, z') = p_\theta(x)$$

Theorem 7: the noise in a fraction of m draws shrinks like $1/\sqrt{m}$

↳ Monte Carlo EM replaces each posterior share by a **fraction of m draws**. How far from the share can that fraction be?

Theorem 7 (Monte Carlo error). Let q = the exact soft label (chance of coin A). Draw the coin m times, each time A with chance q , and let the **drawn share** = (number of A's) / m . Then

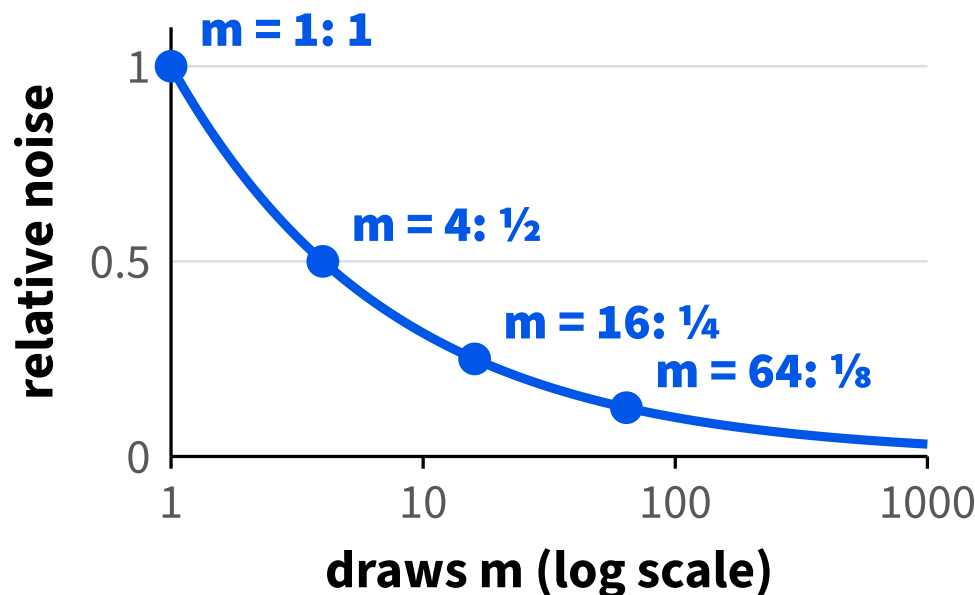
drawn share = q on average, typical error $\sqrt{q \cdot (1 - q) / m}$.

Four times the draws, half the error.

On HH ($q = 0.9$): 10 draws, e.g. A A A B A A A A A A, give share 0.9; repeating gives 0.8, 1.0, 0.9, ... around 0.9, typically off by $\sqrt{0.9 \cdot 0.1 / 10} \approx 0.095$. With 40 draws: ≈ 0.047 , half.

Why $\sqrt{m}$: averaging m independent draws divides the variance by m , so the typical error by $\sqrt{m}$. **Why it matters:** the M-step uses the drawn share, so more draws give a more precise EM step.

standard deviation $\div \sqrt{(q \cdot (1 - q))} = 1/\sqrt{m}$
any q : 4 times the draws, half the noise



Noise shrinks like $1/\sqrt{m}$: pick m for the precision you need; large m gives EM.

Proof of Theorem 7: indicators, then means and variances add

↳ **Setup.** Let $I_j = 1$ if draw j is that value, else 0; the fraction is $\frac{1}{m} \cdot \sum_j I_j$.

Mean:

$$\begin{aligned}\mathbb{E}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m} \cdot \sum_j \mathbb{E}[I_j] && \text{linearity of expectation} \\ &= \frac{1}{m} \cdot m \cdot (1 \cdot q + 0 \cdot (1 - q)) = q && \mathbb{E}[I_j] = 1 \cdot q + 0 \cdot (1 - q) = q\end{aligned}$$

Standard deviation:

$$\begin{aligned}\text{sd} &= \sqrt{\text{Var}\left[\frac{1}{m} \cdot \sum_j I_j\right]} && \text{definition} \\ &= \sqrt{\frac{1}{m^2} \cdot \sum_j \text{Var } I_j} && \text{independent: variances add; } \frac{1}{m} \text{ out squared} \\ &= \sqrt{\frac{1}{m} \cdot \text{Var } I_1} && \text{same law for every } j: m \text{ equal terms} \\ &= \sqrt{\frac{1}{m} \cdot q \cdot (1 - q)} && \text{Var } I_j = \mathbb{E}[I_j^2] - q^2 = q - q^2 = q \cdot (1 - q) \quad (I_j^2 = I_j) \\ &= \sqrt{q \cdot (1 - q)/m} && \blacksquare\end{aligned}$$

In words: the fraction is right on average, and its noise falls like $1/\sqrt{m}$.

variance; 0 or 1

$$\begin{aligned}\text{Var } X &= \mathbb{E}[X^2] - (\mathbb{E}X)^2; I_j^2 = I_j, \text{ so} \\ \mathbb{E}[I_j^2] &= q\end{aligned}$$

scaling

$$\text{Var}(c \cdot X) = c^2 \cdot \text{Var } X$$

independent draws

$$\text{Var } \sum_j I_j = \sum_j \text{Var } I_j$$

The two chains give

$$\begin{aligned}\mathbb{E}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m} \cdot m \cdot q = q \\ \text{Var}\left[\frac{1}{m} \cdot \sum_j I_j\right] &= \frac{1}{m^2} \cdot m \cdot q \cdot (1 - q) \\ &= \frac{q \cdot (1 - q)}{m}\end{aligned}$$

Mean q , standard deviation $\sqrt{q \cdot (1 - q)/m}$: right on average, noise $\propto 1/\sqrt{m}$.

Hard EM replaces each soft label by the single most likely hidden value

Definition (hard EM). At the guess θ_t , for each record n :

E-step: take the most likely hidden value (ties by a fixed rule)

$$\hat{z}^{(n)} = \arg \max_z p_{\theta_t}(z | x^{(n)}) = \arg \max_z p_{\theta_t}(x^{(n)}, z)$$

(same denominator $p_{\theta_t}(x^{(n)})$), and use the label 1 on $\hat{z}^{(n)}$, 0 elsewhere.

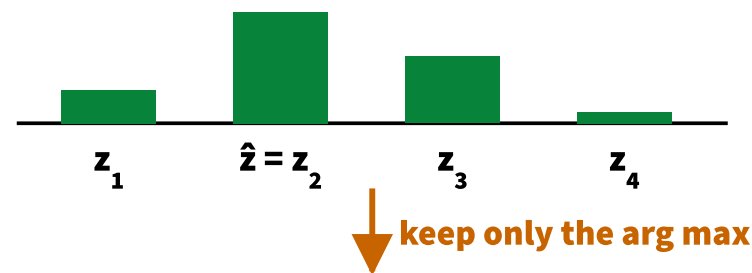
M-step: $\theta_{t+1} = \arg \max_{\theta} \sum_n \ln p_{\theta}(x^{(n)}, \hat{z}^{(n)})$, maximum likelihood with these told labels.

It is Monte Carlo EM with **one "draw" per record that is always the most likely value**, as in k-means (Section 5) (p. 62).

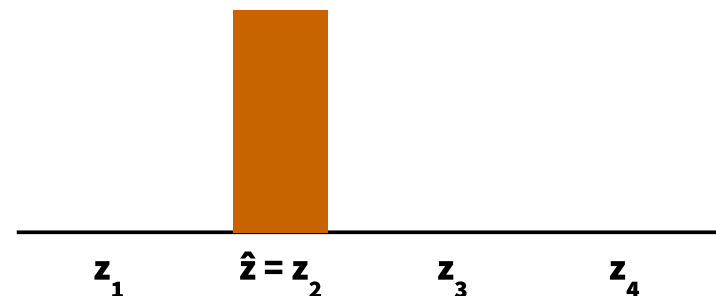
In words: each record goes entirely to its most likely hidden value.

Why it matters: no normalizing sum and no random draws are needed, but the label is **no longer the posterior**.

EM: soft label = the posterior of z



hard EM: weight 1 on $\hat{z}$, 0 elsewhere



Hard EM rounds each posterior to 0 or 1; it is cheap, but it throws the uncertainty away.

Theorem 4 does not cover hard EM: the log-likelihood can fall

↳ Recall Lemma 1 (p. 50): for any label q , $\ell(\theta) \geq \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$ per record, with equality only when q is the posterior. Here $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$.

$$\ell(\theta_{t+1}) \geq \sum_n \sum_z q^{(n)}(z) \cdot \ln \frac{p_{\theta_{t+1}}(x^{(n)}, z)}{q^{(n)}(z)}$$

Lemma 1, hard label $q^{(n)}$

$$= \sum_n \ln p_{\theta_{t+1}}(x^{(n)}, \hat{z}^{(n)})$$

$q^{(n)} = 1$ on $\hat{z}^{(n)}$; the 0 terms vanish

$$\geq \sum_n \ln p_{\theta_t}(x^{(n)}, \hat{z}^{(n)})$$

the M-step maximizes this sum

$$= \sum_n \left[\ln p_{\theta_t}(x^{(n)}) + \ln p_{\theta_t}(\hat{z}^{(n)} | x^{(n)}) \right]$$

product rule $p(x, z) = p(x) \cdot p(z | x)$

$$= \ell(\theta_t) - \sum_n (-\ln p_{\theta_t}(\hat{z}^{(n)} | x^{(n)}))$$

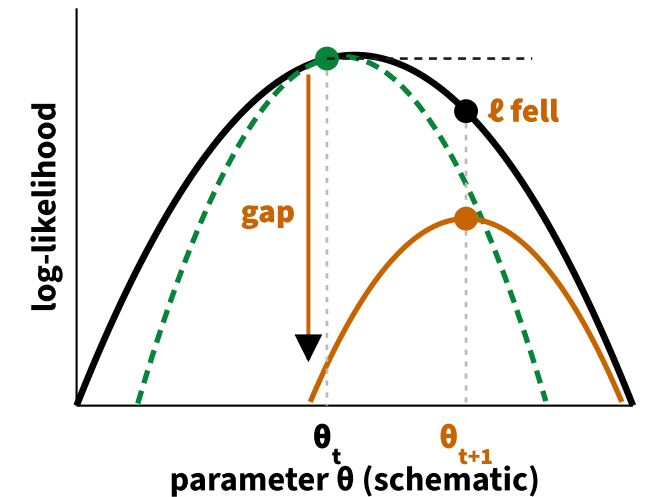
definition of ℓ ; orange = the gap

The **gap** is > 0 unless every posterior is certain. With EM's exact posterior it is 0, and the chain ends at $\ell(\theta_t)$: **Theorem 4 (p. 52)**.

In words: hard labels are not the posterior, so the bound no longer touches ℓ at the guess.

Why it matters: only $\ell(\theta_{t+1}) \geq \ell(\theta_t) - \text{gap}$ is guaranteed: hard EM can **end below its start**.

$\ell(\theta)$, the log-likelihood (black)
EM's bound (dashed): touches ℓ at θ_t
hard EM's bound: a gap at θ_t



Theorem 4's increase needs the exact posterior as the soft label; hard EM gives that up.

Replay the game and keep the runs that match the record: the kept runs follow the posterior

↳ **Proposition (replay and keep) (p. 98) applied to HH.** Replay the game at the guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ (fair toss for a coin, flip it twice) and **keep** a run only if it shows HH. Shown: 32 replays at their expected counts; a fair toss sends $32 \cdot \frac{1}{2} = 16$ runs to each coin.

picked A
16 runs



A shows HH with chance $\frac{3}{4} \cdot \frac{3}{4} = 9/16 \rightarrow 9$ of A's 16 runs are kept

picked B
16 runs



B shows HH with chance $\frac{1}{4} \cdot \frac{1}{4} = 1/16 \rightarrow 1$ of B's 16 runs is kept

kept: 9 + 1 = 10 runs; A made 9 of them $\rightarrow$ share $9/10 = 0.9$ = the posterior of A given HH

No sum was computed: we only ran the game forward and compared. This is **rejection sampling**. **Its cost:** $32 - 10 = 22$ of the 32 runs are thrown away. The keep rate $10/32 = 9/32 + 1/32$ is exactly $p(\text{HH})$, the sum of the two paths, which we never computed: the proposition's second claim. So **rare records waste most runs**.

Kept runs show A : B = 9 : 1, so each kept run's coin is a posterior draw: A with chance 0.9.

One Monte Carlo EM round with ten draws gives $\theta_A = 0.781$, close to EM's 0.767

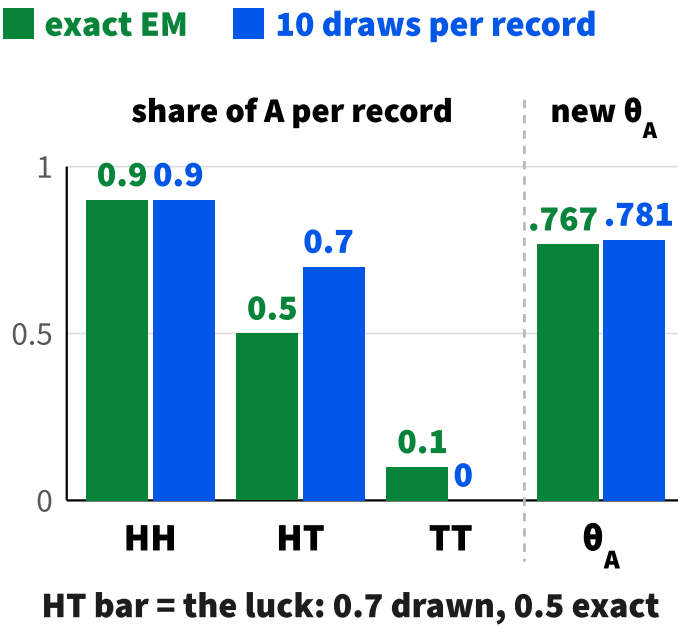
↳ **Definition (Monte Carlo EM) (p. 95)** applied to the coins, $m = 10$. At $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$, we drew each record's coin 10 times with a computer's random numbers (A with chance 0.9, 0.5, 0.1, the posteriors) (p. 43). In this one run A came up 9, 7, 0 times; another run gives other counts. M-step: count and divide (Theorem 1) (p. 11) with these fractions.

record	fraction of A's	heads to A	flips to A
HH	$9/10 = 0.9$	$0.9 \cdot 2 = 1.8$	$0.9 \cdot 2 = 1.8$
HT	$7/10 = 0.7$	$0.7 \cdot 1 = 0.7$	$0.7 \cdot 2 = 1.4$
TT	$0/10 = 0$	$0 \cdot 0 = 0$	$0 \cdot 2 = 0$
total		2.5	3.2

$$\theta_A = \frac{\text{heads to A}}{\text{flips to A}} = \frac{2.5}{3.2} = 0.781$$

EM (recall): $\frac{2.3}{3.0} = 0.767$

Coin B, from the remaining fractions 0.1, 0.3, 1: $\theta_B = 0.5/2.8 = 0.179$.



Then draw afresh at (0.781, 0.179) and repeat: EM with "share" replaced by "fraction of draws".

Theorem 7 on HT: ten draws give noise 0.158, and four times the draws halve it

↳ **Theorem 7 applied to HT.** At $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ the posterior of A given HT is $q = \frac{1}{2}$ (recall) (p. 43); we drew $m = 10$ times.

$$\begin{aligned} \text{standard deviation} &= \sqrt{q \cdot (1 - q) / m} && \text{Theorem 7} \\ &= \sqrt{\frac{1}{2} \cdot \frac{1}{2} / 10} = 0.158 && \frac{1}{4} / 10 = 0.025, \sqrt{0.025} = 0.158 \end{aligned}$$

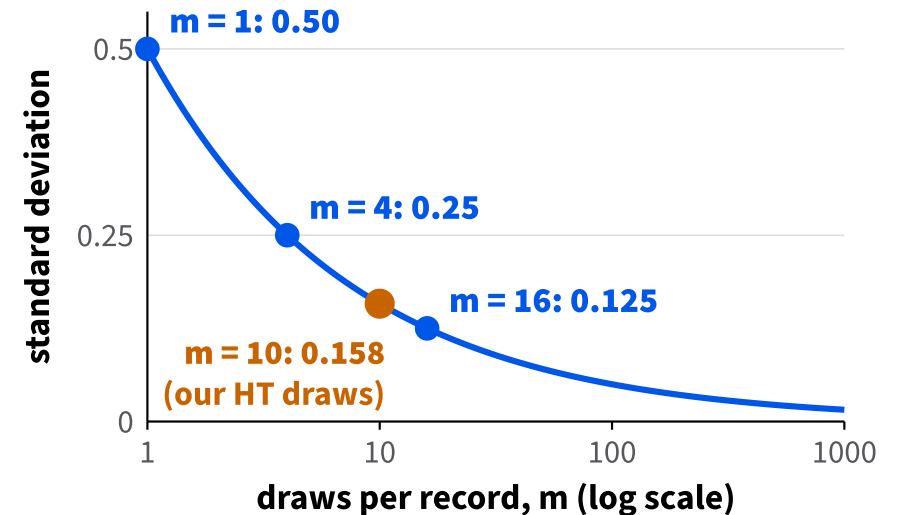
Ten draws typically miss by about 0.16, so HT's 0.7 (posterior 0.5) is **ordinary luck**, which the new guess inherits: $\theta_A = 0.781$, not EM's 0.767 (p. 44).

Hard EM instead: round, do not draw

Give each record wholly to its likelier coin: HH → A, HT → A (tie), TT → B. Counting then gives $\theta_A = \frac{3}{4}, \theta_B = 0$, and the labels stay A, A, B: hard EM is **stuck**. Its fit $\ell = -4.268$ is below the start's -4.000 (p. 29): rounding throws away information.

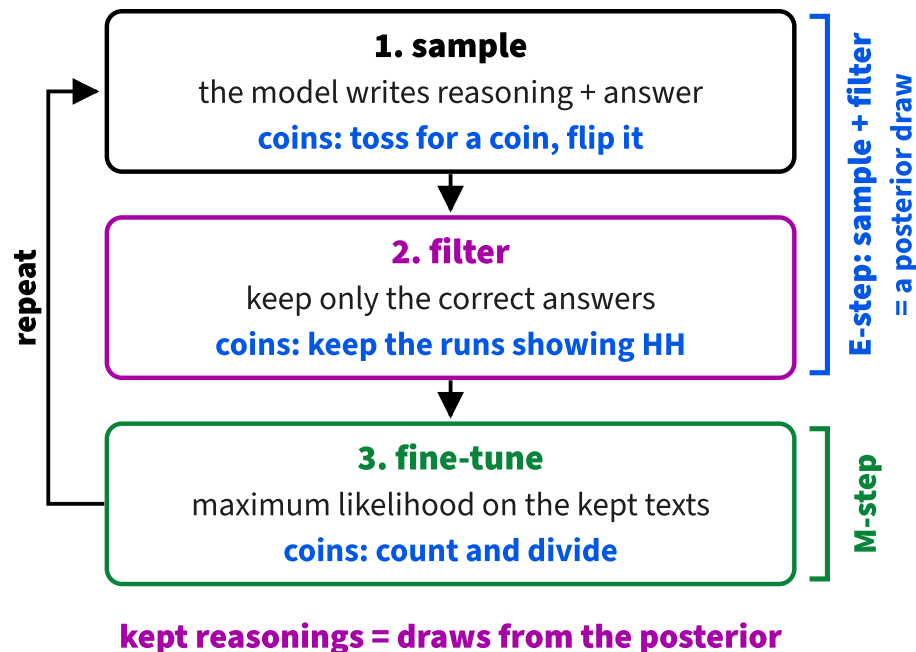
All steps: Appendix, Section 7 (p. 157).

HT, posterior $q = \frac{1}{2}$: standard deviation $0.5 / \sqrt{m}$
4 times the draws halves it: $0.50 \rightarrow 0.25 \rightarrow 0.125$



Draws add noise $\propto 1/\sqrt{m}$ that more draws remove; rounding to one coin can get stuck.

Teaching a model to reason from answers is replay-and-keep



The training data hold only questions and final answers; the **reasoning is hidden**, like the coin.

COIN GAME

hidden coin z

record HH

θ_A, θ_B (what EM learns)

toss chance π ($\frac{1}{2}$ in our game)

posterior of the coin given HH

replay, keep HH

count and divide

REASONING MODEL

$\leftrightarrow$ hidden reasoning trace z

$\leftrightarrow$ correct final answer

$\leftrightarrow$ the model's weights

$\leftrightarrow$ how often each kind of reasoning is written

$\leftrightarrow$ posterior of the reasoning, given the correct answer

$\leftrightarrow$ sample, keep correct

$\leftrightarrow$ fine-tune on kept texts

A tiny reasoner writes careful or hasty reasoning; one round raises the share of careful reasoning to $\frac{3}{4}$.

Self-Taught Reasoner, STaR (Zelikman et al., 2022) and **Reinforced Self-Training with EM, ReST-EM** (Singh et al., 2023) run this loop; ReST-EM names its two steps E and M.

Worked example: Appendix, Section 7 (p. 158).

Sample, keep the correct, fine-tune: Monte Carlo EM with a reasoning as the coin.

Next: the same records, three different answers. Which do we trust?

run on HH, HT, TT	start (θ_A, θ_B)	ends at	ℓ
EM (soft labels; recall Section 3) (p. 45)	$(\frac{3}{4}, \frac{1}{4})$	(0.789, 0.211)	-3.989, best
hard EM	$(\frac{3}{4}, \frac{1}{4})$	$(\frac{3}{4}, 0)$, stuck	-4.268
EM, identical start	$(\frac{1}{2}, \frac{1}{2})$	$(\frac{1}{2}, \frac{1}{2})$, never moves	$3 \ln \frac{1}{4} = -4.159$

Identical start: each record has chance $\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$ (any two flips at $\frac{1}{2}$), so $\ell = 3 \cdot \ln \frac{1}{4}$; equal coins split every record $\frac{1}{2} : \frac{1}{2}$, so the guess never moves.

Swap the names: $\theta_A = 0.211$, $\theta_B = 0.789$. Is the fit better, worse, or the same?

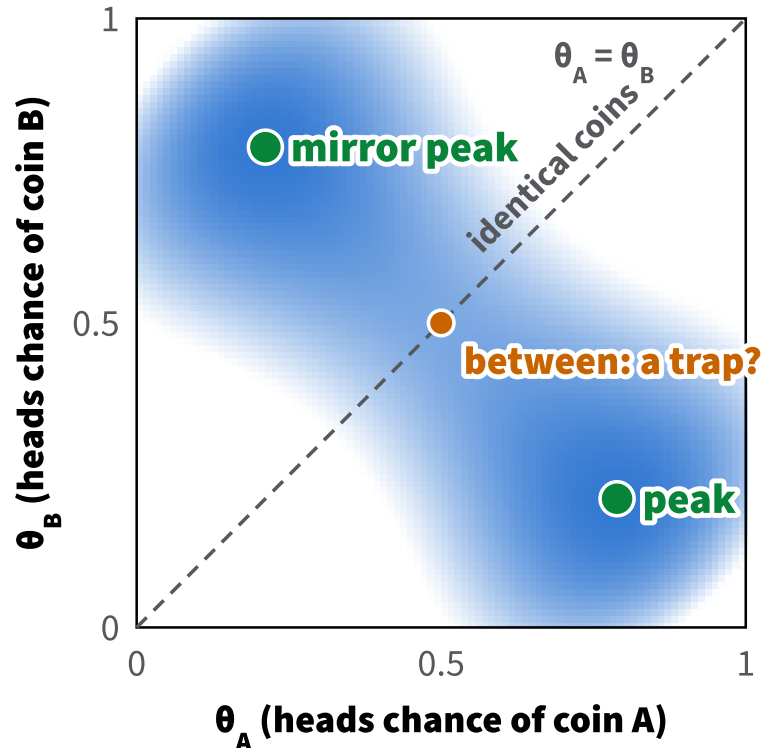
Answer. The same. $p(\text{HH}) = \frac{1}{2} \cdot 0.789^2 + \frac{1}{2} \cdot 0.211^2$ either way: the two paths just swap places, since each coin is picked with chance $\frac{1}{2}$. So every record's chance, and ℓ , is unchanged.

A fit can be stuck, or best yet not unique: which answer to trust is the next question.

The coin fit has two mirror-image peaks

ℓ : lower ℓ

higher ℓ



two equally high peaks, mirror images across the diagonal

Can a learned $\theta = (\theta_A, \theta_B)$ be trusted? Colour the **log-likelihood** ℓ of our records HH, HT, TT over every pair of heads chances.

Two equally high peaks appear, mirror images across the diagonal $\theta_A = \theta_B$ (identical coins), with an orange point between them.

Three questions for this section:

1. Why are the peaks mirror images?
2. Can EM get stuck between them?
3. Can the data always pick one answer?

First the general theory, then the key coin numbers.

Before trusting a fit, ask whether a different setting fits just as well.

Renaming the coins never changes the likelihood

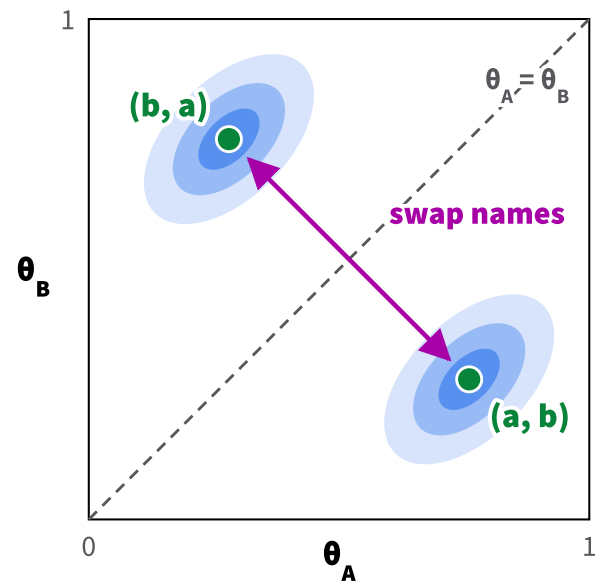
Label-switching lemma. If the friend picks each coin with chance $\frac{1}{2}$, then $\ell(\theta_A, \theta_B) = \ell(\theta_B, \theta_A)$ for every pair of heads chances.

Write $f(x | t)$ for the chance that one coin with heads chance t produces record x .

$$\begin{aligned}
 \ell(\theta_A, \theta_B) &= \sum_x \ln p_\theta(x) && \text{(independent records)} \\
 &= \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B) \right] && \text{(sum over hidden coin } z) \\
 &= \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_B) + \frac{1}{2} \cdot f(x | \theta_A) \right] && \text{(addition commutes)} \\
 &= \ell(\theta_B, \theta_A) && \text{(line 2, names swapped)}
 \end{aligned}$$

In words: calling coin A "B" and coin B "A" leaves every record's probability unchanged.

Why it matters: every peak (a, b) has an equally high **mirror peak** (b, a) . With an unknown pick chance π , the same steps give $\ell(\pi, \theta_A, \theta_B) = \ell(1 - \pi, \theta_B, \theta_A)$.



same ℓ at both points:
the hills are mirror images across the diagonal

Names are labels, not facts: the data see only the unordered pair of coins.

Identical coins stay identical under EM

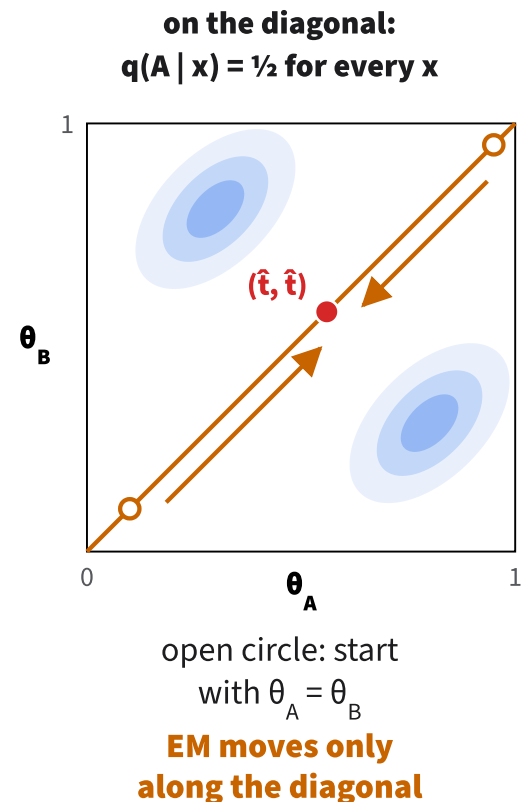
Setting. Pick chance $\frac{1}{2}$. Record x has h_x heads in n_x flips, so one coin with heads chance t gives it $f(x | t) = t^{h_x} \cdot (1 - t)^{n_x - h_x}$. Let $\hat{t} = \sum_x h_x / \sum_x n_x$: all heads over all flips.

Diagonal fact, part 1. From identical coins (t, t) , one EM step lands on $(\hat{t}, \hat{t})$; later steps stay there.

$$\begin{aligned}
 q(A | x) &= \frac{\frac{1}{2} \cdot f(x | t)}{\frac{1}{2} \cdot f(x | t) + \frac{1}{2} \cdot f(x | t)} = \frac{1}{2} && \text{(Bayes' rule, identical coins)} \\
 \theta_A^{\text{new}} &= \frac{\sum_x q(A | x) \cdot h_x}{\sum_x q(A | x) \cdot n_x} && \text{(M-step: credited heads } \div \text{ credited flips)} \\
 &= \frac{\frac{1}{2} \cdot \sum_x h_x}{\frac{1}{2} \cdot \sum_x n_x} = \hat{t} && (q(A | x) = \frac{1}{2} \text{ for every } x) \\
 &= \theta_B^{\text{new}} && (q(B | x) = \frac{1}{2} \text{ too: the same sums)} \\
 \Rightarrow (\hat{t}, \hat{t}) &\mapsto (\hat{t}, \hat{t}) && \text{(again identical coins: same steps, } t = \hat{t})
 \end{aligned}$$

In words: identical coins split every record half and half; both learn the pooled $\hat{t}$.

Why it matters: a start with $\theta_A = \theta_B$ never leaves the diagonal: EM cannot break the symmetry.



Identical starts stay identical: EM moves only along the diagonal.

Where EM stops on the diagonal, both slopes of ℓ are zero

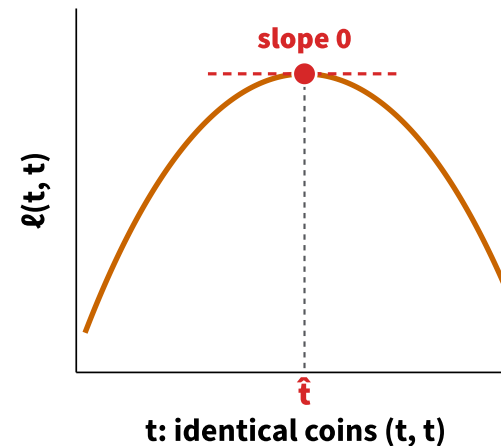
Recall (p. 110): $\ell = \sum_x \ln \left[\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B) \right]$ with $f(x | t) = t^{h_x} \cdot (1 - t)^{n_x - h_x}$, one coin's chance of record x ; $\hat{t} = \sum_x h_x / \sum_x n_x$.

Diagonal fact, part 2. $(\hat{t}, \hat{t})$ is a **stationary point** of ℓ : $\partial \ell / \partial \theta_A = \partial \ell / \partial \theta_B = 0$ there.

$$\begin{aligned}
 \left. \frac{\partial \ell}{\partial \theta_A} \right|_{(t,t)} &= \sum_x \frac{\frac{1}{2} \cdot f'(x | \theta_A)}{\frac{1}{2} \cdot f(x | \theta_A) + \frac{1}{2} \cdot f(x | \theta_B)} \Big|_{(t,t)} && \text{(chain rule for } \ln; \text{ only A's term has } \theta_A) \\
 &= \frac{1}{2} \cdot \sum_x \frac{f'(x | t)}{f(x | t)} && \text{(at } (t, t) \text{ the two halves add to } f) \\
 &= \frac{1}{2} \cdot \frac{d}{dt} \sum_x \ln f(x | t) && (f'/f = (\ln f)') \\
 &= 0 \text{ at } t = \hat{t} && \text{(Theorem 1: best fit = heads/flips)}
 \end{aligned}$$

In words: on the diagonal each slope is half the slope of one coin's log-likelihood, which is flat at $\hat{t}$. The same steps give $\partial \ell / \partial \theta_B = 0$.

Why it matters: stationary is not the same as best: whether $(\hat{t}, \hat{t})$ is a peak or a trap depends on how ℓ curves across the diagonal.



one coin's ℓ ,
top at $\hat{t}$ (Theorem 1)

both slopes vanish there

EM's resting point on the diagonal is flat in every direction, but not necessarily a peak.

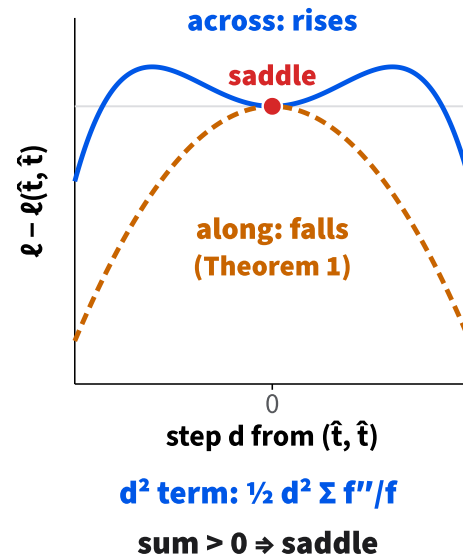
Across the diagonal, ℓ can curve upward: a saddle

Saddle test. Along the diagonal, $\ell(t, t) = \sum_x \ln f(x | t)$ is one coin's log-likelihood, largest at $t = \hat{t}$ (Theorem 1) (p. 11). Across it, ℓ rises if $\sum_x f''(x | \hat{t})/f(x | \hat{t}) > 0$; then $(\hat{t}, \hat{t})$ is a **saddle**: highest along, lowest across.

Taylor at $\hat{t}$: $f(x | \hat{t} \pm d) = f \pm f' \cdot d + \frac{1}{2} \cdot f'' \cdot d^2 \pm \frac{1}{6} \cdot f''' \cdot d^3 + O(d^4)$, with f, f', f'', f''' at $(x | \hat{t})$.

$$\begin{aligned} \ell(\hat{t} + d, \hat{t} - d) &= \sum_x \ln \left[\frac{1}{2} \cdot f(x | \hat{t} + d) + \frac{1}{2} \cdot f(x | \hat{t} - d) \right] && \text{(pick chance } \frac{1}{2}) \\ &= \sum_x \ln \left[f(x | \hat{t}) + \frac{1}{2} \cdot f''(x | \hat{t}) \cdot d^2 + O(d^4) \right] && \text{(average } \pm : \text{ odd powers cancel)} \\ &= \sum_x \left[\ln f(x | \hat{t}) + \frac{f''(x | \hat{t})}{2 \cdot f(x | \hat{t})} \cdot d^2 \right] + O(d^4) && (\ln(f + u) = \ln f + \frac{u}{f} + O(u^2)) \\ &= \ell(\hat{t}, \hat{t}) + \frac{d^2}{2} \cdot \sum_x \frac{f''(x | \hat{t})}{f(x | \hat{t})} + O(d^4) && \text{(collect the } d^2 \text{ terms)} \end{aligned}$$

In words: pulling the coins apart by $\pm d$ changes ℓ only at second order, upward when the curvature sum is positive.



Why it matters: a saddle is **not a peak**, yet by the diagonal fact (p. 111) EM started on the diagonal stops there.

A saddle is a worse fit, and a symmetric start makes EM stop exactly there.

Random restarts reach the top with high probability

Theorem 4 (p. 52) promises only that ℓ never goes down, $\ell(\theta^{(0)}) \leq \ell(\theta^{(1)}) \leq \dots$, not that EM reaches the top: it may stop at a lower peak or a saddle.

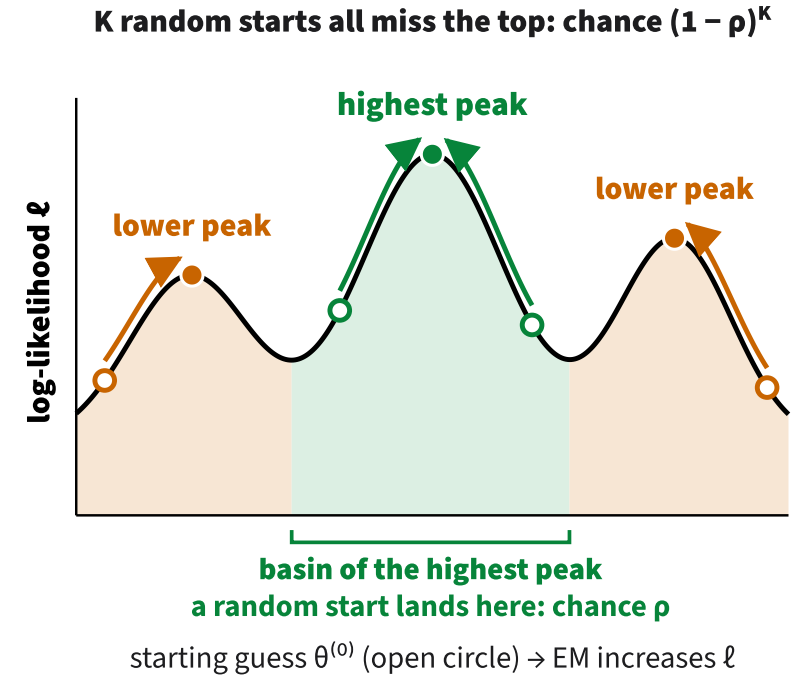
Definition. The **basin** of a peak: the starting guesses from which EM increases ℓ to that peak. ρ = the chance a random start lies in the highest peak's basin.

Restart principle. Run EM from K independent random starts; keep the run with the largest ℓ .

$$\begin{aligned} P(\text{all } K \text{ runs miss}) &= \prod_{k=1}^K P(\text{run } k \text{ misses}) && \text{(independent starts)} \\ &= (1 - \rho)^K && \text{(chance } 1 - \rho \text{ each)} \\ &\leq e^{-\rho \cdot K} && (1 - \rho \leq e^{-\rho}) \\ &\rightarrow 0 && (K \rightarrow \infty, \rho > 0) \end{aligned}$$

In words: each extra start multiplies the chance of failure by $1 - \rho$.

Why it matters: compare runs by ℓ , not names (by the lemma (p. 110) (a, b) and (b, a) are one answer); never start with $\theta_A = \theta_B$ (the diagonal fact) (p. 111).



Several random starts escape saddles and lower peaks; keep the largest ℓ .

More unknowns than free numbers means no unique fit

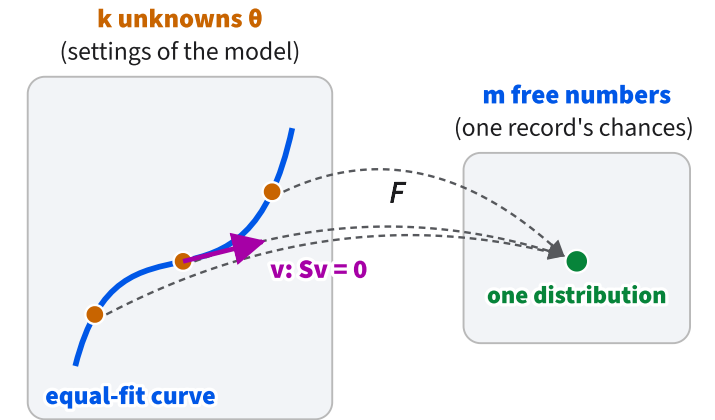
↳ Question 3: restarts find the highest ℓ , but can the data single out one setting at all, say with the pick chance π unknown too?

Definitions. A record with $m + 1$ outcomes has m **free numbers** (its chances sum to 1). Parameters are **identifiable** if different settings give different distributions of one record (up to renaming).

Theorem 8 (counting test). If a model has more unknowns than one record's distribution has free numbers, the parameters are **not identifiable**: near a typical setting, a whole curve of settings gives the same distribution.

Proof. F maps the k unknowns θ to the m free numbers; S , the slope table (the **Jacobian**), is its $m \times k$ matrix, of constant rank near a typical θ . A step v is invisible if $F(\theta + \varepsilon \cdot v) = F(\theta) + \varepsilon \cdot S \cdot v + O(\varepsilon^2)$ has $S \cdot v = 0$.

$$\begin{aligned} \dim\{\text{equal-fit settings near } \theta\} &= \dim\{v : S \cdot v = 0\} && \text{(constant-rank theorem)} \\ &= k - \text{rank } S && \text{(rank-nullity)} \\ &\geq k - m && (S \text{ has } m \text{ rows}) \\ &\geq 1 && (k > m) \quad \blacksquare \end{aligned}$$



$k > m$: a whole curve of settings
lands on the same distribution of one record

In words: more unknowns than free numbers: some change of the unknowns is invisible in the data. Passing (unknowns $\leq$ free numbers) does not prove uniqueness.

Count before you fit: no algorithm can choose between settings the data cannot tell apart.

With three flips per record, the two coins are pinned down

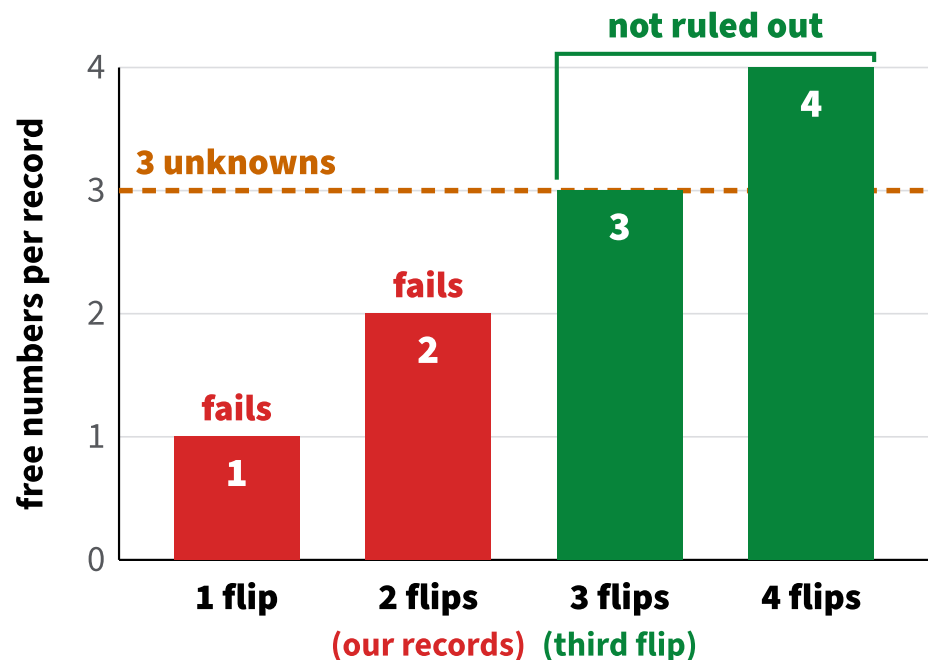
Now the friend flips the picked coin n times per record; the unknowns are π , θ_A , θ_B .

Counting: n flips give $n + 1$ head counts $0, \dots, n$ whose chances sum to 1, so n free numbers. With $n = 2$, Theorem 8 (p. 115) rules out a unique fit; with $n = 3$ it no longer does.

Theorem (Teicher). If $n \geq 3$, $\theta_A \neq \theta_B$ and $0 < \pi < 1$, the head-count distribution fixes $(\pi, \theta_A, \theta_B)$ up to swapping names (Teicher, 1963; Allman, Matias & Rhodes, 2009).

In words: two flips leave a curve of equal fits; three flips leave only a setting and its mirror.

Why it matters: passing the counting test is not a proof by itself (Theorem 8 says so); here a real theorem closes the gap.



n flips: $n + 1$ head counts, sum 1,
so n free numbers

Observe more per record, and the curve of equal fits shrinks back to two mirror peaks.

Proof, step 1: three flips reveal three averages

Proof of Teicher's theorem for $n = 3$. Write $m_j = \mathbb{E}[\theta^j] = \pi \cdot \theta_A^j + (1 - \pi) \cdot \theta_B^j$, the average of θ^j over the hidden coin ($m_0 = 1$), and $p(h)$ for the chance of h heads in 3 flips. For $n > 3$, look only at the first three flips of each record: their head-count chances are the same $p(h)$, so the same proof applies.

$$p(h) = \binom{3}{h} \cdot \mathbb{E}[\theta^h \cdot (1 - \theta)^{3-h}]$$

$$p(3) = \mathbb{E}[\theta^3] = m_3$$

$$p(2) = 3 \cdot \mathbb{E}[\theta^2 - \theta^3] = 3 \cdot (m_2 - m_3)$$

$$p(1) = 3 \cdot \mathbb{E}[\theta - 2 \cdot \theta^2 + \theta^3] = 3 \cdot (m_1 - 2 \cdot m_2 + m_3)$$

$\Rightarrow m_3, m_2, m_1$ are fixed by the head-count chances

(given the coin, the head count is binomial)

($h = 3$)

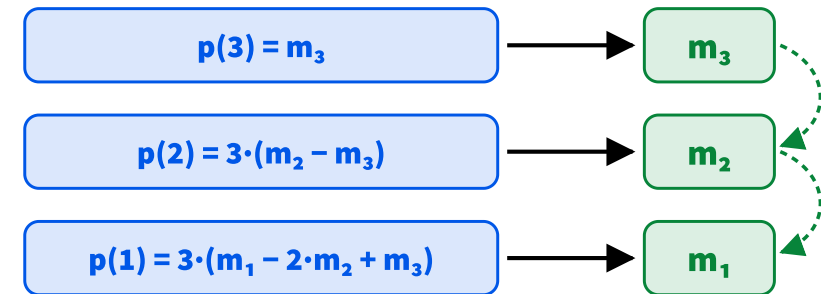
(expand $\theta^2 \cdot (1 - \theta)$, then average term by term)

(expand $\theta \cdot (1 - \theta)^2$, then average term by term)

(solve from the top down)

In words: each head-count chance is a fixed mix of the averages, and the mix is triangular: $p(3)$ gives m_3 ; then $p(2)$ gives m_2 ; then $p(1)$ gives m_1 .

What remains: show that m_1, m_2, m_3 fix θ_A, θ_B and π .



solve top-down: each line adds one new average

Three flips per record let the data reveal three averages m_1, m_2, m_3 .

Proof, step 2: three averages fix both coins

Recall (p. 117): the data fix m_1, m_2, m_3 , where $m_j = \mathbb{E}[\theta^j] = \pi \cdot \theta_A^j + (1 - \pi) \cdot \theta_B^j$ and $m_0 = 1$. Write $s = \theta_A + \theta_B$, $r = \theta_A \cdot \theta_B$.

$$\begin{aligned}
 \theta^2 &= s \cdot \theta - r \quad \text{for } \theta = \theta_A \text{ and } \theta = \theta_B && (\theta_A, \theta_B \text{ are the roots of } z^2 - s \cdot z + r) \\
 m_{j+2} &= s \cdot m_{j+1} - r \cdot m_j && (\text{times } \theta^j, \text{ average over the coin}) \\
 \begin{pmatrix} m_1 & -1 \\ m_2 & -m_1 \end{pmatrix} \begin{pmatrix} s \\ r \end{pmatrix} &= \begin{pmatrix} m_2 \\ m_3 \end{pmatrix} && (j = 0 \text{ and } j = 1) \\
 \det &= m_1 \cdot (-m_1) - (-1) \cdot m_2 = m_2 - m_1^2 && (2 \times 2 \text{ determinant}) \\
 &= \pi \cdot \theta_A^2 + (1 - \pi) \cdot \theta_B^2 - (\pi \cdot \theta_A + (1 - \pi) \cdot \theta_B)^2 && (\text{definition of } m_j) \\
 &= \pi \cdot (1 - \pi) \cdot \theta_A^2 + \pi \cdot (1 - \pi) \cdot \theta_B^2 - 2 \cdot \pi \cdot (1 - \pi) \cdot \theta_A \cdot \theta_B && (\text{expand; } \pi - \pi^2 = \pi \cdot (1 - \pi)) \\
 &= \pi \cdot (1 - \pi) \cdot (\theta_A - \theta_B)^2 > 0 && = (1 - \pi) - (1 - \pi)^2 \\
 & && (\text{the spread of } \theta; \theta_A \neq \theta_B, 0 < \pi < 1)
 \end{aligned}$$

$\det \neq 0$, so the 2×2 system has **exactly one solution** (s, r) ; $\{\theta_A, \theta_B\}$ are the two roots of $z^2 - s \cdot z + r$; and $m_1 = \pi \cdot \theta_A + (1 - \pi) \cdot \theta_B$ gives $\pi = (m_1 - \theta_B) / (\theta_A - \theta_B)$. ■

In words: three flips reveal three averages, which fix the sum and the product of the two heads chances.

Why it matters: two flips show only m_1, m_2 , so only the $j = 0$ equation: one equation for two unknowns s, r , a curve of equal fits.

The third flip supplies the missing equation.

Both mirror peaks have $\ell = -3.989$; the saddle only -4.159

↳ Label-switching lemma (p. 110) and saddle test (p. 113) applied to the coins. Recall EM's limit (p. 45) $\theta_A = \frac{3+\sqrt{3}}{6} = 0.789$, $\theta_B = 1 - \theta_A = 0.211$.

point (θ_A, θ_B)	$p(\text{HH})$	$p(\text{HT})$	$p(\text{TT})$	ℓ
peak (0.789, 0.211)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
mirror (0.211, 0.789)	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
saddle $(\frac{1}{2}, \frac{1}{2})$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	-4.159

$$p(\text{HT}) = \theta_A \cdot \theta_B = \frac{(3+\sqrt{3}) \cdot (3-\sqrt{3})}{36} = \frac{1}{6} \quad (1 - \theta_A = \theta_B)$$

$$p(\text{HH}) = p(\text{TT}) = \frac{1}{2} \cdot \left(1 - 2 \cdot \frac{1}{6}\right) = \frac{1}{3} \quad (\theta_A + \theta_B = 1)$$

$$\ell = 2 \cdot \ln \frac{1}{3} + \ln \frac{1}{6} = -3.989 \quad (\text{both peaks})$$

$$\hat{t} = \frac{3 \text{ heads}}{6 \text{ flips}} = \frac{1}{2} \quad (\text{diagonal fact})$$

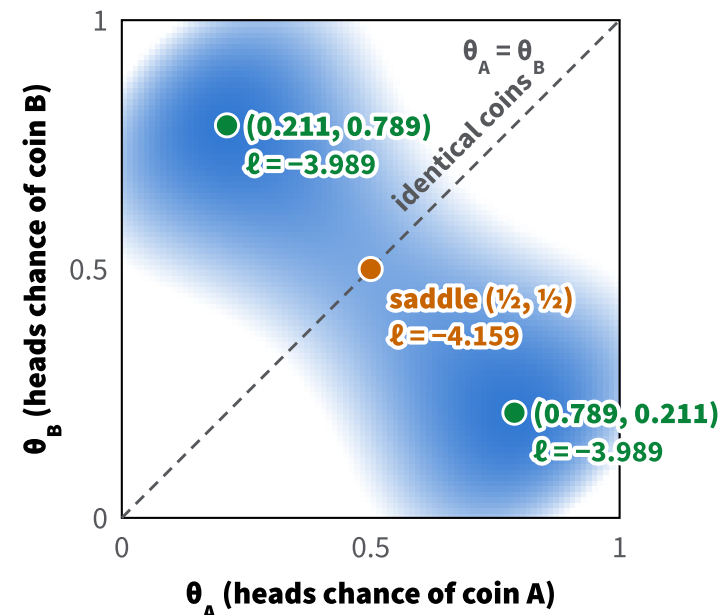
$$\ell\left(\frac{1}{2}, \frac{1}{2}\right) = 3 \cdot \ln \frac{1}{4} = -4.159 \quad (\text{each record } f = \frac{1}{4})$$

$$\sum_x f''/f = \frac{2-2+2}{1/4} = 8 > 0 \quad (f'' \text{ of } t^2, t - t^2, (1-t)^2)$$

$$\ell\left(\frac{1}{2} + d, \frac{1}{2} - d\right) \approx -4.159 + 4 \cdot d^2 \quad (\text{saddle test: } \frac{d^2}{2} \cdot 8)$$

ℓ : $-4.59 = \text{top} - 0.6$

$-3.99 = \text{top}$



equal peaks, mirror images; the saddle between is lower

All steps: Appendix, Section 8 (p. 159).

By the diagonal fact, a start with $\theta_A = \theta_B$ stays at the worse saddle; restart off the diagonal.

Two different coin settings give exactly the same data

↳ Theorem 8 (p. 115) applied to the coins: let the pick chance $\pi = p(A)$ be unknown too. Three unknowns π, θ_A, θ_B , but the chances $p(h)$ of $h = 0, 1, 2$ heads sum to 1: **2 free numbers** (recall $\theta_A + \theta_B = 1$, $\theta_A \cdot \theta_B = \frac{1}{6}$) (p. 46).

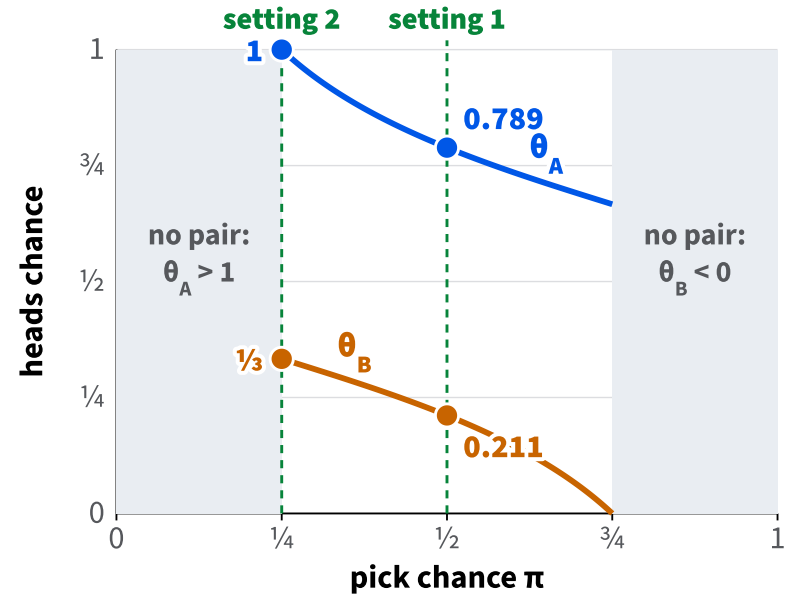
$$p(2) = \pi \cdot \theta_A^2 + (1 - \pi) \cdot \theta_B^2, \quad p(0) = \pi \cdot (1 - \theta_A)^2 + (1 - \pi) \cdot (1 - \theta_B)^2$$

π, θ_A, θ_B	setting 1 (EM's fit): $\frac{1}{2}, 0.789, 0.211$	setting 2: $\frac{1}{4}, 1, \frac{1}{3}$
$p(2)$	$\frac{1}{2} \cdot (\theta_A^2 + \theta_B^2) = \frac{1}{2} \cdot (1 - 2 \cdot \frac{1}{6}) = \frac{1}{3}$	$\frac{1}{4} \cdot 1 + \frac{3}{4} \cdot \frac{1}{9} = \frac{1}{3}$
$p(0)$	$= p(2) \quad (1 - \theta_A = \theta_B)$	$\frac{1}{4} \cdot 0 + \frac{3}{4} \cdot \frac{4}{9} = \frac{1}{3}$

Setting 2: try $\pi = \frac{1}{4}$ and a coin A that **always lands heads**; $p(0) = \frac{3}{4} \cdot (1 - \theta_B)^2 = \frac{1}{3}$ gives $\theta_B = \frac{1}{3}$. Both rows: $p(1) = 1 - \frac{1}{3} - \frac{1}{3} = \frac{1}{3}$.

Same readings, so the same $\ell = -3.989$. Every π from $\frac{1}{4}$ to $\frac{3}{4}$ has such a pair: a **ridge** of equal fits.

All steps: Appendix, Section 8 (p. 160).



every pair: $p(0 \text{ heads}) = p(2 \text{ heads}) = \frac{1}{3}$

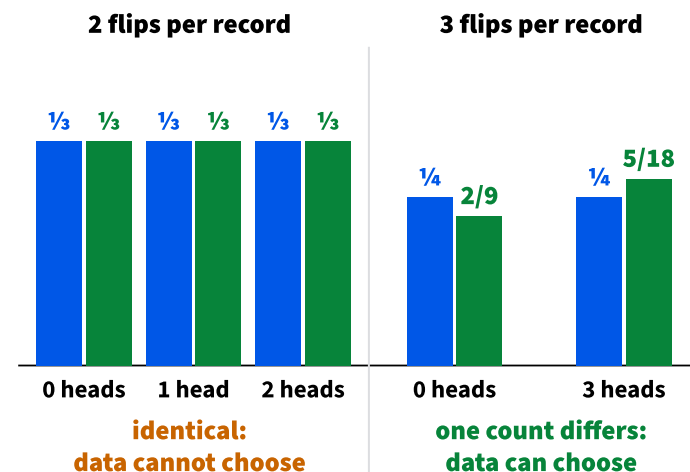
Different knobs, identical readings: the data cannot choose, so no algorithm can.

A third flip per record tells the two settings apart

↳ Teicher's theorem (p. 116) applied to the coins: 3 flips give 4 head counts summing to 1, so **3 free numbers = 3 unknowns**.

$$\begin{aligned}\theta_A^3 + \theta_B^3 &= (\theta_A + \theta_B)^3 - 3 \cdot \theta_A \cdot \theta_B \cdot (\theta_A + \theta_B) && \text{(cube of a sum)} \\ &= 1 - \frac{1}{2} = \frac{1}{2} && \text{(sum 1, product } \frac{1}{6}) \\ \text{setting 1: } p(3) &= \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} && (\pi = \frac{1}{2}) \\ \text{setting 1: } p(0) &= p(3) = \frac{1}{4} && (H \leftrightarrow T: 1 - \theta_A = \theta_B) \\ \text{setting 2: } p(3) &= \frac{1}{4} \cdot 1^3 + \frac{3}{4} \cdot \left(\frac{1}{3}\right)^3 = \frac{5}{18} \neq \frac{1}{4} && (\theta_A = 1, \theta_B = \frac{1}{3}) \\ \text{setting 2: } p(0) &= \frac{1}{4} \cdot 0^3 + \frac{3}{4} \cdot \left(\frac{2}{3}\right)^3 = \frac{2}{9} \neq \frac{1}{4} && (\text{tails: } 0, \frac{2}{3})\end{aligned}$$

Two flips gave both settings $\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$; with three flips the 3-heads and 0-heads chances already differ, as Teicher's theorem promises.

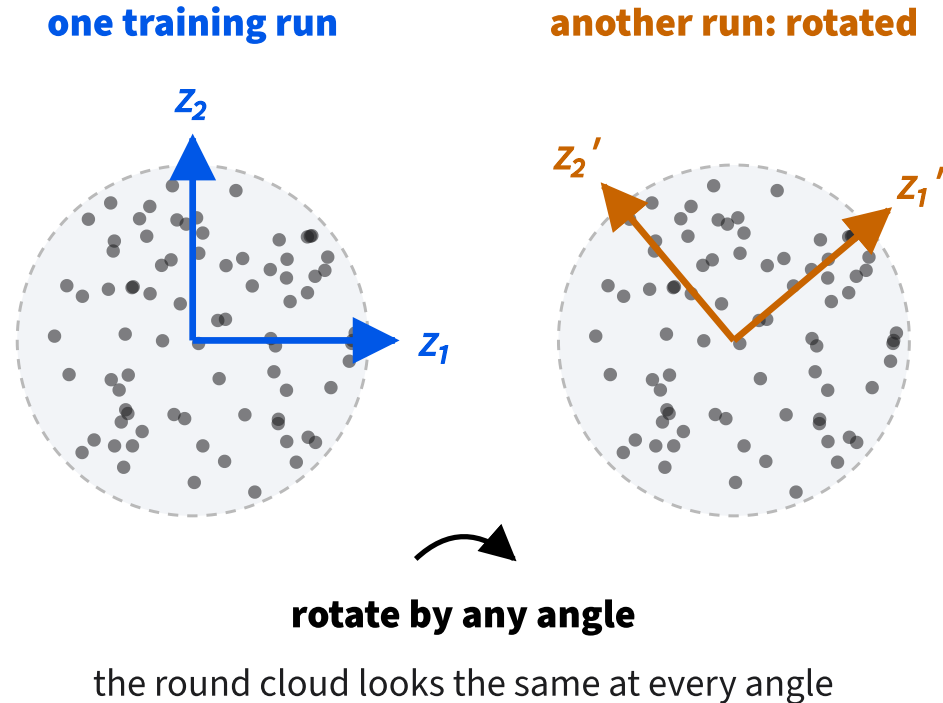


■ setting 1: $\pi = \frac{1}{2}, \theta_A = 0.789, \theta_B = 0.211$

■ setting 2: $\pi = \frac{1}{4}, \theta_A = 1, \theta_B = \frac{1}{3}$

One more flip per record turns the ridge back into two mirror peaks.

A learned code is a hidden coin, and its axes can be turned freely



A **variational autoencoder (VAE)** gives each image x a hidden code $z = (z_1, z_2)$; a decoder turns z into an image. We hope z_1 means pose and z_2 colour.

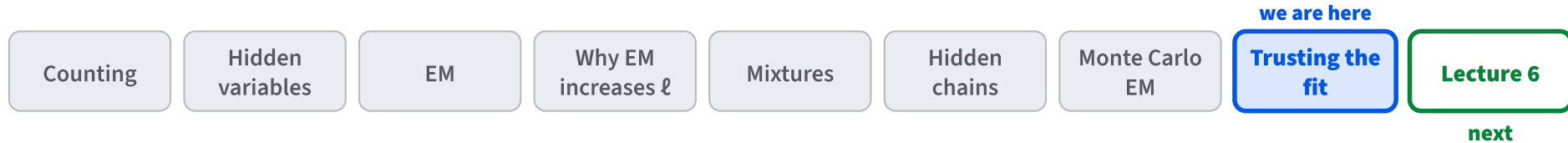
COIN GAME		VAE
coin A or B	$\Leftrightarrow$	code $z \in \mathbb{R}^2$
record HH	$\Leftrightarrow$	image x
θ_A, θ_B	$\Leftrightarrow$	decoder weights
swap A $\leftrightarrow$ B	$\Leftrightarrow$	flip or permute the code axes
ridge of equal fits	$\Leftrightarrow$	circle of equal fits, one per angle

With a round prior, a rotated code fits every image equally well, so training lands on **one angle by chance** (Locatello et al., 2019). A prior that changes with an observed label, such as day or night, **pins the axes** (Khemakhem et al., 2020).

Worked example: Appendix, Section 8 (p. 161).

Like the ridge, rotations give many equal codes; an extra observation pins one.

Next lecture: when the exact posterior costs too much



Exact E-step

The expectation step (E-step) for coins, mixtures, chains: **sum** over every hidden value.

Sampled E-step

Monte Carlo EM: **draw** hidden values instead of summing.

Lecture 6

Pick the guess q (a distribution over the hidden values) from a simple family and **maximize** Lemma 1's bound (p. 50) $B(q, \theta) \leq \ln p_\theta(x)$.

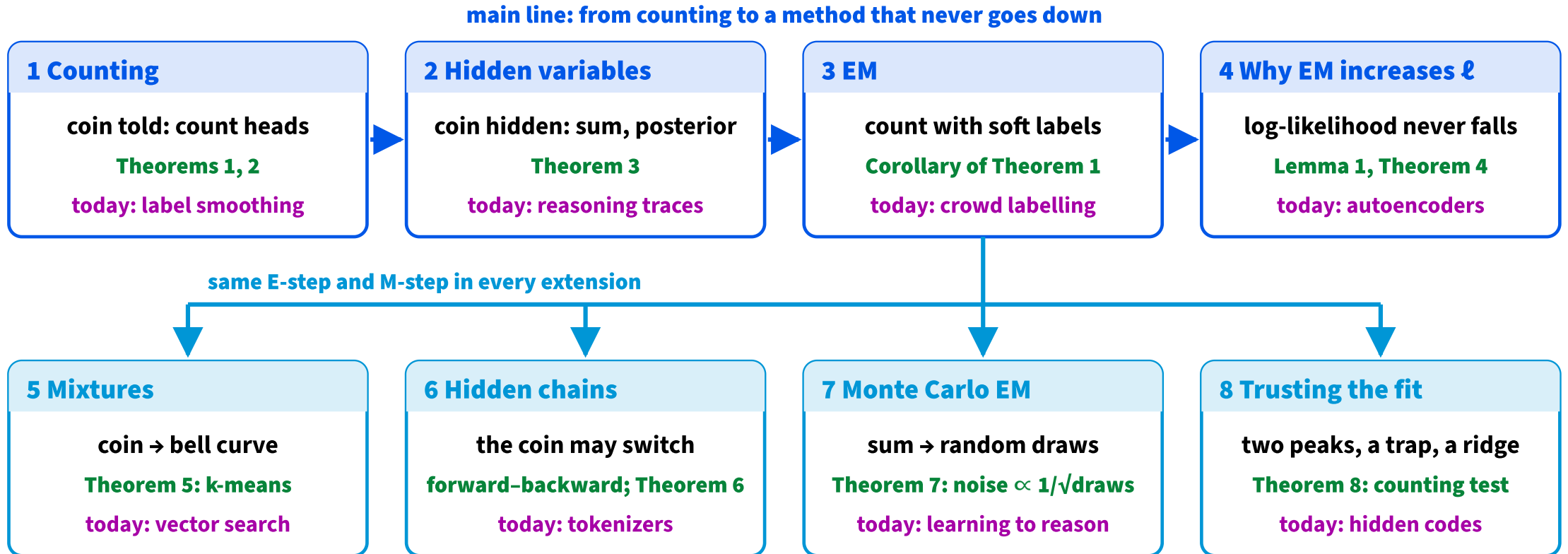
Coin game: the exact E-step sums 2 terms per record (coin A or B). With 16 hidden black-or-white pixels, how many?

32 65,536 16

Answer. $2^{16} = 65,536$: each pixel doubles the count. A 28×28 image has 784 pixels, so $2^{784} \approx 10^{236}$ terms, which **no computer can sum**. Lecture 6 approximates instead.

Today's bound $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$ **becomes Lecture 6's main tool.**

One coin game, one method, four extensions



black: what changes · green: the result (proved or computed) · magenta: where it runs today

Exact EM never lowers the log-likelihood ℓ (Theorem 4); Sections 5–8 keep its two steps.

Count when the coin is told; sum over it when it is hidden

Records HH, HT, TT; working guess $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$; each coin picked with chance $\frac{1}{2}$. ∂_{θ_A} = slope in θ_A .

Section	Result (the theory)	The result applied to the coins
1 Counting	Theorem 1 (maximum likelihood by counting) (p. 11): $\hat{\theta}$ = heads $\div$ flips. Theorem 2 (Laplace's rule) (p. 15): uniform prior $\Rightarrow$ posterior mean (heads + 1) $\div$ (flips + 2).	Coins told. A made HH, HT: 3 heads in 4 flips, $\frac{3}{4}$. B made TT: $\frac{0}{2} = 0$; Laplace $\frac{0+1}{2+2} = \frac{1}{4}$.
2 Hidden variables	Definition (p. 23): $p_{\theta}(x) = \sum_z p_{\theta}(x, z)$, a sum over the coin z. Posterior $p_{\theta}(z \mid x)$ = a path's share of the sum. Theorem 3 (Fisher's identity) (p. 27): slope of $\ln p_{\theta}(x)$ = posterior-weighted slopes of $\ln p_{\theta}(x, z)$, one per path.	$p(\text{HH}) = \underbrace{\frac{1}{2} \cdot \theta_A^2}_{\text{via A}} + \underbrace{\frac{1}{2} \cdot \theta_B^2}_{\text{via B}} = \frac{9}{32} + \frac{1}{32} = \frac{5}{16}.$ Posterior $p(A \mid \text{HH}) = \frac{9}{9+1} = 0.9$. $\partial_{\theta_A} \ln p(\text{HH}) = \frac{\partial_{\theta_A} (\frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot \theta_B^2)}{p(\text{HH})} = \frac{\theta_A}{5/16} = \frac{3/4}{5/16} = 2.4.$ Path slopes: $\partial_{\theta_A} \ln (\frac{1}{2} \cdot \theta_A^2) = \frac{2}{\theta_A} = \frac{8}{3}$ (A), 0 (B). Posterior-weighted: $0.9 \cdot \frac{8}{3} + 0.1 \cdot 0 = 2.4$, the same slope.

Told coin: count. Hidden coin: sum the paths, and the posterior splits each record between them.

EM counts with soft labels, and θ_A increases from 0.75 to 0.789

EM. E-step: soft labels $q_t = p_{\theta_t}(z \mid x)$, the posterior. **M-step:** count heads and flips with them, then divide (**Corollary** of Theorem 1) (p. 39).

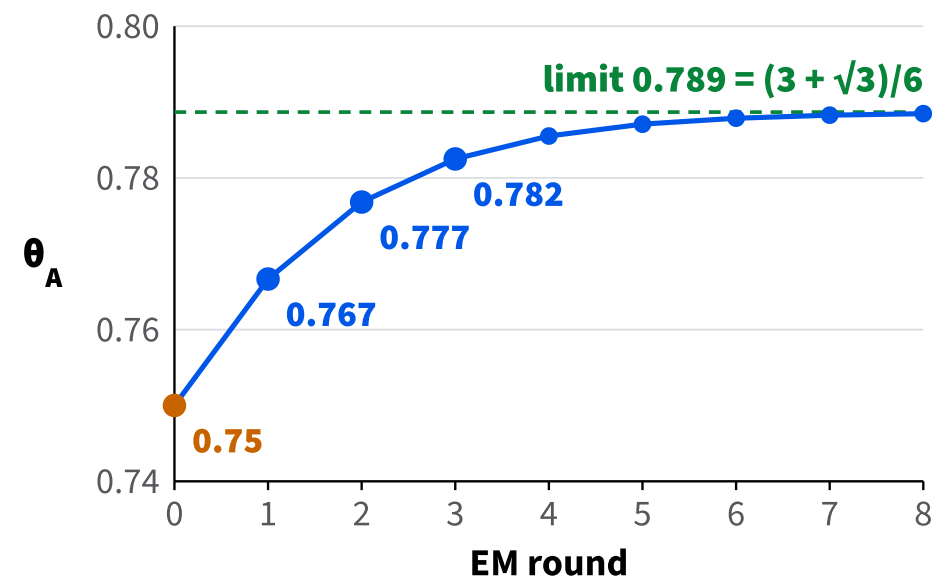
The definition (p. 35) applied to the coins, from $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$:

record	via A	via B	soft label for A
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{9}{9+1} = 0.9$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{32}$	$\frac{3}{3+3} = 0.5$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{1+9} = 0.1$

Heads to A: $0.9 \cdot 2 + 0.5 \cdot 1 + 0.1 \cdot 0 = 2.3$.

Flips to A: $2 \cdot (0.9 + 0.5 + 0.1) = 3$.

New $\theta_A = \frac{2.3}{3} = 0.767$; repeat the two steps. The rounds (figure) settle at the peak $(3 + \sqrt{3})/6 = 0.789$.



rounds 1, 2, 3: 0.767, 0.777, 0.782, ...
each step smaller than the last

E guesses the coin with the posterior, M counts with the guess; repeat until θ stops moving.

The bound touches $\ln p$ at the posterior, so EM never falls

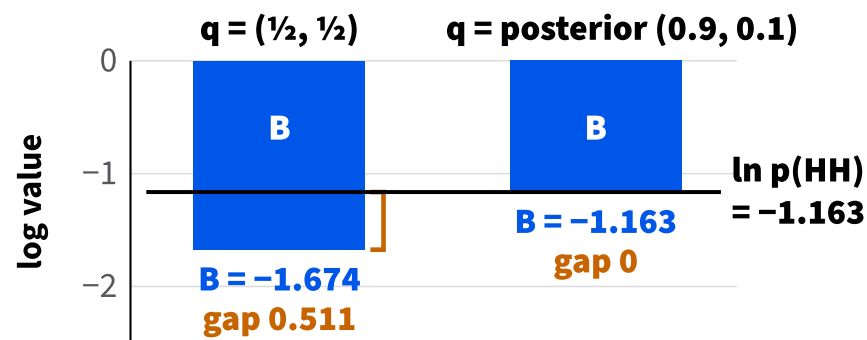
Lemma 1 (bound and gap) (p. 50). $\ln p_\theta(x) = B(q, \theta) + D_{\text{KL}}(q \parallel p_\theta(z \mid x)) \geq B$, with bound $B = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$.

Theorem 4 (EM increases the likelihood) (p. 52): $\ell(\theta_{t+1}) \geq \ell(\theta_t)$ in every round.

Lemma 1 applied to the coins, record HH, guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$. Recall $p(\text{HH}, A) = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}, p(\text{HH}, B) = \frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$; **log-likelihood** $\ell(\theta) = \ln p(\text{HH}) + \ln p(\text{HT}) + \ln p(\text{TT})$, with $p(\text{HH}) = p(\text{TT}) = \frac{5}{16}$ and $p(\text{HT}) = \frac{3}{16}$.

$$\begin{aligned}
 B(q = (\tfrac{1}{2}, \tfrac{1}{2})) &= \tfrac{1}{2} \cdot \ln \frac{9/32}{1/2} + \tfrac{1}{2} \cdot \ln \frac{1/32}{1/2} && \text{(definition of } B) \\
 &= \tfrac{1}{2} \cdot \ln \frac{9}{16} + \tfrac{1}{2} \cdot \ln \frac{1}{16} && \text{(divide)} \\
 &= \tfrac{1}{2} \cdot \ln \frac{9}{256} && (\ln a + \ln b = \ln ab) \\
 &= \ln \sqrt{9/256} = \ln \frac{3}{16} = -1.674 && (\tfrac{1}{2} \cdot \ln a = \ln \sqrt{a})
 \end{aligned}$$

$$\text{gap} = \ln p - B = \ln \frac{5}{16} - \ln \frac{3}{16} = \ln \frac{5}{3} = \mathbf{0.511} \quad (\text{Lemma 1})$$



bar: bound B · orange: gap up to $\ln p(\text{HH})$

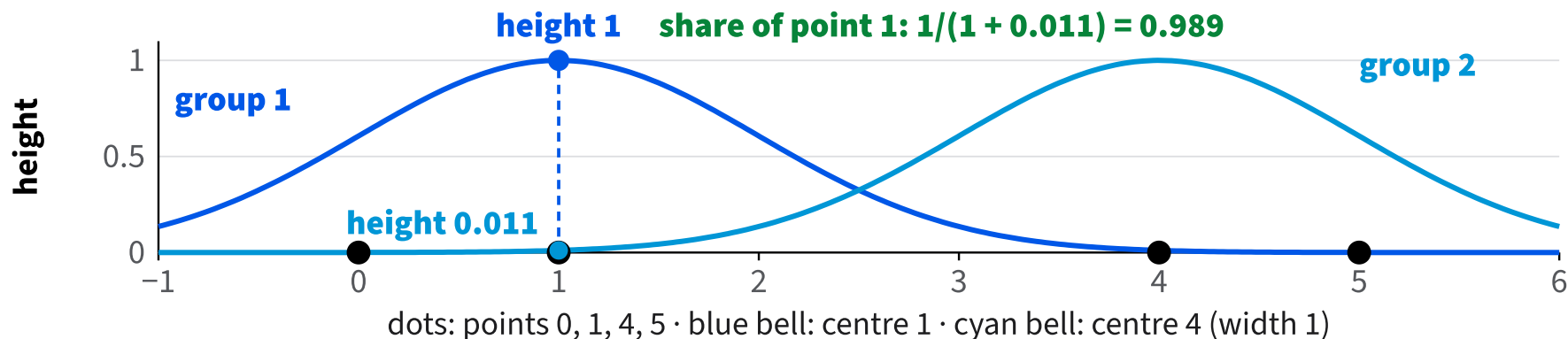
With $q = (0.9, 0.1)$, the posterior: **gap 0**.

Theorem 4 on the coins: $\ell(\frac{3}{4}, \frac{1}{4}) = 2 \cdot \ln \frac{5}{16} + \ln \frac{3}{16} = -4.000$; recall round 1 moves to $(0.767, 0.233)$ (p. 56) with $\ell = -3.993$.

The bound equals $\ln p$ at the posterior, so maximizing it can only raise ℓ .

Bells: E finds each group's share of a point, M averages with it

Section	Result (the theory)	The result applied to the example
5 Mixtures	E: a group's share of a point (its responsibility). M: each centre = share-weighted average. Theorem 5 (k-means as a limit) (p. 64): width $\sigma \rightarrow 0$ gives k-means.	Points 0, 1, 4, 5; centres 1 and 4; width $\sigma = 1$; equal weights. Point 1: distances $d = 0$ and 3. Heights $e^{-d^2/2}$: $e^0 = 1, e^{-9/2} = 0.011$. Group 1's share: $\frac{1}{1 + 0.011} = 0.989$.



A group's share is its height over the sum of heights; very thin bells give k-means.

Chains: forward-backward finds each share, M counts with it

Section	Result (the theory)	The result applied to the example
6 Hidden chains	Forward $\alpha_t(z)$ = chance of flips $1..t$ and coin z at flip t. Backward $\beta_t(z)$ = chance of flips $t+1..n$ given coin z at flip t. Forward-backward (any chain of n flips) (p. 77): $p(x) = \sum_z \alpha_t(z) \cdot \beta_t(z)$ at every t; posterior $\gamma_t(z) = \alpha_t(z) \cdot \beta_t(z) / p(x)$. Theorem 6 (Baum-Welch) (p. 84): E-step: soft labels γ_t and pair labels ξ_t (coins at flips $t, t+1$) from forward-backward; M-step: count with them.	Flips H, H, T; keep $\frac{3}{4}$, switch $\frac{1}{4}$. Recall at flip 2 (p. 81), for (A, B): $\alpha_2 = \left(\frac{15}{64}, \frac{3}{64}\right), \beta_2 = \left(\frac{3}{8}, \frac{5}{8}\right)$. Forward-backward at $t = 2$: $\alpha_2 \cdot \beta_2 = \frac{15}{64}$. $\frac{3}{8} = \frac{45}{512}$ (A), $\frac{3}{64} \cdot \frac{5}{8} = \frac{15}{512}$ (B). Sum: $p(\text{HHT}) = \frac{45 + 15}{512} = \frac{15}{128}$; A's share $\gamma_2(A) = \frac{45}{60} = \frac{3}{4}$.

Bell or chain, the E-step is a share of a sum; the M-step counts with the shares.

Draws shrink noise like $1/\sqrt{m}$; three flips pin the coins down

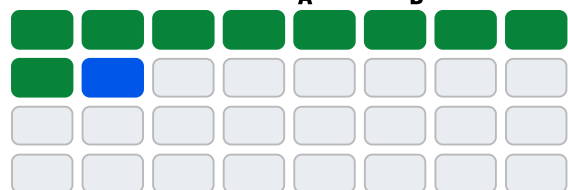
Section	Result (the theory)	The result applied to the coins
7 Monte Carlo EM	E: draw the coin m times from the posterior. Proposition (replay and keep) (p. 98): kept runs are posterior draws, kept with chance $p_\theta(x)$. Theorem 7 (Monte Carlo error) (p. 100): the noise (standard deviation) of the drawn share is $\sqrt{q \cdot (1 - q)/m}$, q the exact soft label.	HH: soft label $q = 0.9$, $m = 100$ draws. $\sqrt{0.9 \cdot 0.1/100} = 0.03$: drawn share 0.90 ± 0.03. Four times the draws, $m = 400$: $\sqrt{0.09/400} = 0.015$, half the noise.
8 Trusting the fit	Restart principle (p. 114): run from several guesses; keep the best log-likelihood ℓ. Label-switching lemma (p. 110): swapping the names keeps ℓ. Theorem 8 (counting test) (p. 115): more unknowns than free numbers $\Rightarrow$ no unique fit (a ridge). Theorem (Teicher) (p. 116): with $n \geq 3$ flips, $\theta_A \neq \theta_B$ and $0 < \pi < 1$, the fit is unique up to a swap.	Unknowns $\pi = p(A)$, θ_A, θ_B: 3. Two flips: $p(h)$ = chance of h heads. $p(0) + p(1) + p(2) = 1$: 2 free numbers < 3, a ridge. Three flips: $p(0) + \dots + p(3) = 1$: 3 free = 3: the test passes, and Teicher's theorem gives pinned (up to swapping names).

Draws buy precision slowly; a third flip pins the coins down (Teicher's theorem).

Open questions: today's coin game, far larger

Wasted draws

32 reruns at $\theta_A = 3/4, \theta_B = 1/4$



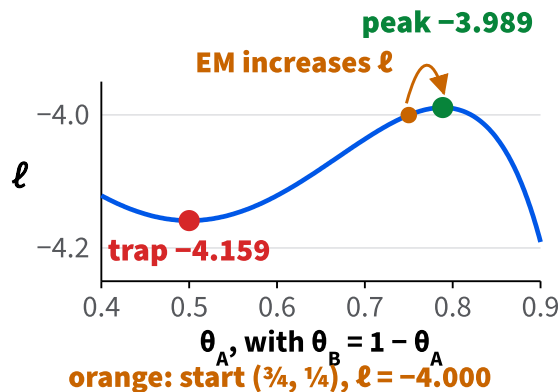
9 HH via A 1 via B 22 wasted

1 kept (widened) 999 wasted

same idea: 1 correct trace in 1,000

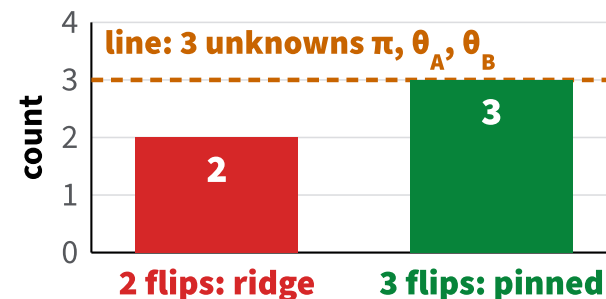
To draw the coin given HH, keep only reruns showing HH: chance $1/2 \cdot 3/4 \cdot 3/4 + 1/2 \cdot 1/4 \cdot 1/4 = 10/32$. Rerun $\leftrightarrow$ reasoning trace; HH $\leftrightarrow$ correct answer. **Can we sample correct traces without the waste?** (Phan et al., 2023)

Trap or peak



At $(1/2, 1/2)$ each record has chance $1/4$, so $\ell = 3 \cdot \ln(1/4) = -4.159$; every soft label is $1/2$, so EM never moves. **When does a spectral estimate (one pass over data averages) start EM near the peak?** (Zhang et al., 2016)

Meaningful codes



bars: free numbers in the data

Two flips: 3 head counts whose chances sum to 1, so 2 free numbers. Three flips: 4 counts, 3 free. Extra flip $\leftrightarrow$ extra observed label (e.g. time). **Which extra observations pin down a network's hidden code?** (Hyvärinen & Morioka, 2016)

Today's coin game, scaled up: wasted draws, good starts, pinning down the hidden code.

Four questions: counting, posterior, soft counts, the increase

1. A coin shows 3 heads, 0 tails. Laplace's rule estimates its heads chance as...

1 4/5 3/4

Answer. 4/5: $\frac{h+1}{h+t+2} = \frac{3+1}{3+0+2}$. Plain counting says $3/3 = 1$: certainty from 3 flips.

2. $\theta_A = \frac{3}{4}$, $\theta_B = \frac{1}{4}$, fair pick. The record HHH came from coin A with chance...

9/10 27/28 3/4

Answer. 27/28: paths $\frac{1}{2} \cdot \left(\frac{3}{4}\right)^3 = \frac{27}{128}$ and $\frac{1}{2} \cdot \left(\frac{1}{4}\right)^3 = \frac{1}{128}$; A's share $27/(27+1)$.

3. Soft labels for coin A: HH 0.9, HT 0.5, TT 0.1. What θ_B does the M-step give?

0.7/3 2.3/3 0.1

Answer. 0.233. B's shares = 1-A's: 0.1, 0.5, 0.9.
Heads: $0.1 \cdot 2 + 0.5 \cdot 1 + 0.9 \cdot 0 = 0.7$.
Flips (2 per record): $0.1 \cdot 2 + 0.5 \cdot 2 + 0.9 \cdot 2 = 3$.
 $\theta_B = 0.7/3$ (heads $\div$ flips, Corollary of Theorem 1) (p. 39).

4. One exact EM round takes ℓ from -5.20 to -5.31 . What happened?

a bug normal converged

Answer. A bug: Theorem 4 (p. 52) says $\ell(\theta_{t+1}) \geq \ell(\theta_t)$ in every exact round; ℓ fell by 0.11.

Bayes for the label, counting for the coin, and ℓ must never fall.

One question per section: bells, chains, draws, trust

5. Centres 1 and 4, width 1, equal weights; height $e^{-d^2/2}$ at distance d . Group 1's share of the point 3?

0.18 0.82 0.50

Answer. 0.18: distances 2 and 1, heights $e^{-4/2} = 0.135$ and $e^{-1/2} = 0.607$. Share = $\text{height}_1 \div (\text{height}_1 + \text{height}_2) = \frac{0.135}{0.135 + 0.607} = \frac{0.135}{0.742} = 0.18$.

7. HT's soft label is $q = 0.5$. With $m = 100$ posterior draws, the noise (standard deviation) of the drawn share is...

0.05 0.03 0.0025

Answer. 0.05: Theorem 7 (p. 100), $\sqrt{q \cdot (1 - q)/m} = \sqrt{0.5 \cdot 0.5/100} = \sqrt{0.0025}$. The noise is largest at $q = \frac{1}{2}$; HH's $q = 0.9$ gives only 0.03.

6. Chain: keep $\frac{3}{4}$, switch $\frac{1}{4}$. $\alpha_1(A) = \frac{3}{8}$, $\alpha_1(B) = \frac{1}{8}$ (chance of flip 1 and the coin). Flip 2 is T; A shows T with $\frac{1}{4}$. $\alpha_2(A)$?

5/64 15/64 10/32

Answer. 5/64: $\left(\underbrace{\frac{3}{8} \cdot \frac{3}{4}}_{\text{keep A}} + \underbrace{\frac{1}{8} \cdot \frac{1}{4}}_{\text{switch}} \right) \cdot \underbrace{\frac{1}{4}}_{\text{T by A}} = \frac{10}{32} \cdot \frac{1}{4} = \frac{5}{64}$.

8. Two EM runs end at the same ℓ but different θ . Most likely?

a bug swap or ridge not converged

Answer. Swapped names or a ridge. Swap: (0.789, 0.211) and (0.211, 0.789) both give $\ell = -3.989$. Ridge (p. 120): with 2 flips, 3 unknowns > 2 free numbers, so a whole curve of $(\pi, \theta_A, \theta_B)$ fits equally.

Every answer is a rule from today applied to one new number.

References (1 of 3): counting, hidden variables, EM, the bound

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[Encoder gap] C. Cremer, X. Li, D. Duvenaud. Inference suboptimality in variational autoencoders. *International Conference on Machine Learning (ICML)* 2018.

Start here: Dempster, Laird and Rubin (1977) named EM and proved it increases the likelihood.

References (2 of 3): mixtures, k-means, hidden chains

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[Theorem 5] B. Kulis, M. I. Jordan. Revisiting k-means: new algorithms via Bayesian nonparametrics. *International Conference on Machine Learning (ICML)* 2012.

[Spread-out starts] D. Arthur, S. Vassilvitskii. k-means++: the advantages of careful seeding. *Symposium on Discrete Algorithms (SODA)*, 1027–1035, 2007.

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[Theorem 6] L. E. Baum, T. Petrie, G. Soules, N. Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *Annals of Mathematical Statistics* 41(1), 164–171, 1970.

[Tokenizer] T. Kudo. Subword regularization: improving neural network translation models with multiple subword candidates. *Association for Computational Linguistics (ACL)* 2018.

[SentencePiece] T. Kudo, J. Richardson. SentencePiece: a simple and language independent subword tokenizer and detokenizer for neural text processing. *Empirical Methods in Natural Language Processing (EMNLP)*, demonstrations, 2018.

For one book on today's lecture, read Bishop, Chapter 9: mixtures, k-means and EM.

References (3 of 3): draws, reasoning, identifiability

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Every frontier paper here is expectation-maximization at scale: infer, then count.

The appendix follows: the few calculations the lecture skipped

The lecture showed one key calculation per idea. The appendix keeps **only the steps behind results the slides state without their arithmetic**, three slides at most per section, in lecture order.

1 • Counting

coin A's slope; Laplace by Beta integrals; label-smoothing loss

2 • Hidden variables

Theorem 3's 2.4 on HH; a log of a sum; the trace's 0.81

3 • EM

count and divide; why the limit is 0.789; crowd labelling

4 • Why EM increases the likelihood

the floor on HH; round 1 in one table; the encoder's gap

5 • Mixtures

narrowing bells; vector-search cells; top-2 routing

6 • Hidden chains

the messages on H, H, T; keep chance 0.7; the cut no|on

7 • Monte Carlo EM

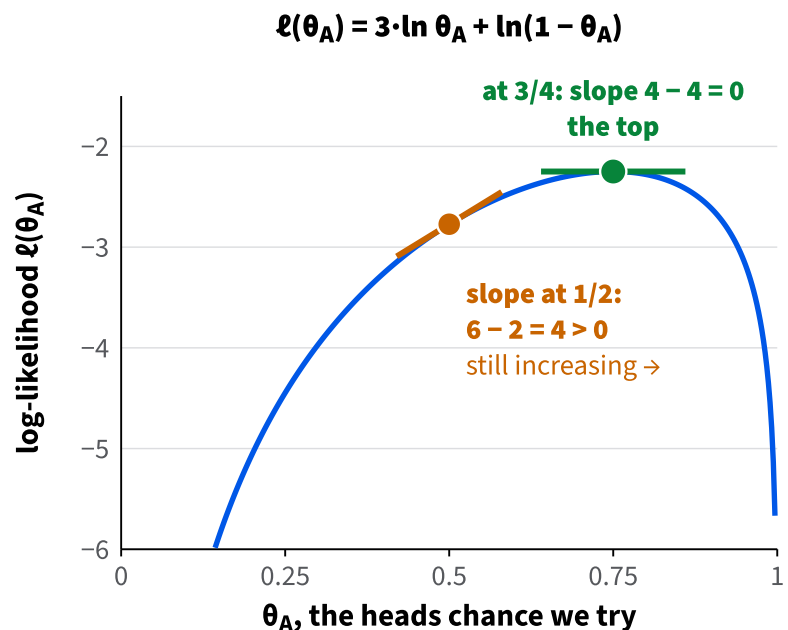
10^{470} texts; hard EM's fall; tiny reasoner

8 • Trusting the fit

the saddle in full; the ridge; pinning the code axes

Every number in the lecture is derived on its own slide or in the appendix.

Taking logs and setting the slope to zero gives $\theta_A = 3/4$



↳ Details the main result $\theta_A = \frac{3}{4}$ (p. 17): Theorem 1's proof (p. 11) on coin A (3 heads, 1 tail). **Maximum-likelihood estimate**: the θ_A with the largest likelihood.

$$\ell(\theta_A) = \ln L = 3 \cdot \ln \theta_A + \ln(1 - \theta_A)$$

$$\ell'(\theta_A) = \frac{3}{\theta_A} - \frac{1}{1 - \theta_A}$$

$$\ell' = 0 \Rightarrow 3 \cdot (1 - \theta_A) = \theta_A$$

$$\Rightarrow 3 - 3 \cdot \theta_A = \theta_A$$

$$\Rightarrow 3 = 4 \cdot \theta_A \Rightarrow \theta_A = \frac{3}{4}$$

$$\ell'\left(\frac{1}{2}\right) = \frac{3}{1/2} - \frac{1}{1/2} = 6 - 2 = 4 > 0$$

$$\ell'\left(\frac{3}{4}\right) = \frac{3}{3/4} - \frac{1}{1/4} = 4 - 4 = 0$$

Log: product $\rightarrow$ sum.

Slope: $(\ln u)' = u'/u$,
 $u = 1 - \theta_A$, $u' = -1$.

Zero at the top; times
 $\theta_A \cdot (1 - \theta_A) > 0$.

Expand.

Add $3 \cdot \theta_A$; divide by 4.

Check: > 0 , increasing.

Check at $\frac{3}{4}$: the top.

Why logs? A sum is easier to differentiate than a product, and $\ln$ only rises, so ℓ and L peak at the same θ_A .

Maximum likelihood gives $\theta_A = \frac{3}{4}$, the counted fraction.

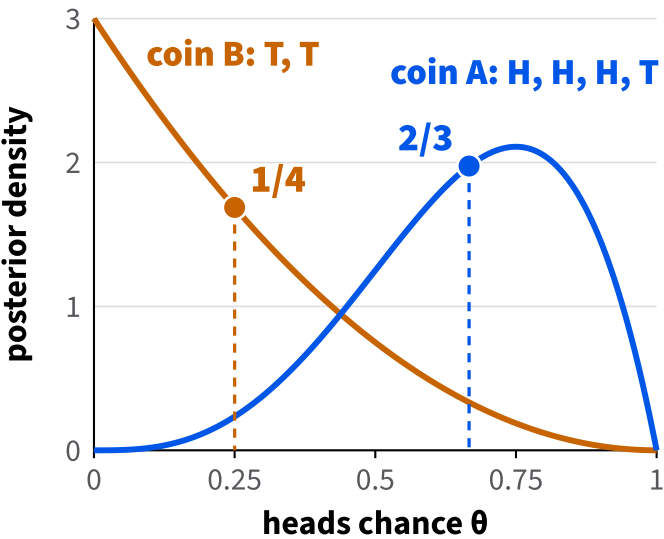
Laplace's rule on the coins: B rises to 1/4, A drops to 2/3

↳ Details $\frac{1}{4}$ and $\frac{2}{3}$ (p. 19): Theorem 2's proof (p. 15) per coin, $\mathbb{E}[\theta] = \frac{\int_0^1 \theta \cdot \theta^h \cdot (1 - \theta)^t d\theta}{\int_0^1 \theta^h \cdot (1 - \theta)^t d\theta}$,
each integral by the Beta integral $I(a, b) = a! b! / (a + b + 1)!$.

step	coin B: $h = 0, t = 2$	coin A: $h = 3, t = 1$
denominator $I(h, t)$	$\frac{0! 2!}{3!} = \frac{1}{3}$	$\frac{3! 1!}{5!} = \frac{1}{20}$
posterior = likelihood ÷ it	$3 \cdot (1 - \theta)^2$	$20 \cdot \theta^3 \cdot (1 - \theta)$
numerator $I(h + 1, t)$	$\frac{1! 2!}{4!} = \frac{2}{24} = \frac{1}{12}$	$\frac{4! 1!}{6!} = \frac{24}{720} = \frac{1}{30}$
mean = numerator ÷ denominator	$\frac{1/12}{1/3} = \frac{3}{12} = \frac{1}{4}$	$\frac{1/30}{1/20} = \frac{20}{30} = \frac{2}{3}$
Laplace $\frac{h + 1}{h + t + 2}$	$\frac{0 + 1}{2 + 2} = \frac{1}{4}$	$\frac{3 + 1}{4 + 2} = \frac{2}{3}$

Check (Beta integral, $a = 0, b = 2$): $\frac{0! 2!}{3!} = \frac{1}{3}$ matches the direct $\int_0^1 (1 - \theta)^2 d\theta = \frac{1}{3}$; coin A's $\frac{3! 1!}{5!} = \frac{1}{20}$ gives the density $20 \cdot \theta^3 \cdot (1 - \theta)$.

posterior densities, uniform prior:
B: $3 \cdot (1 - \theta)^2$, A: $20 \cdot \theta^3 \cdot (1 - \theta)$



Coin B: $\frac{0+1}{2+2} = \frac{1}{4}$. **Coin A:** $\frac{3+1}{4+2} = \frac{2}{3}$, pulled from $\frac{3}{4}$ toward $\frac{1}{2}$.

Under the smoothed target, the loss is lowest at $p_{\text{cat}} = 0.9$, not 1

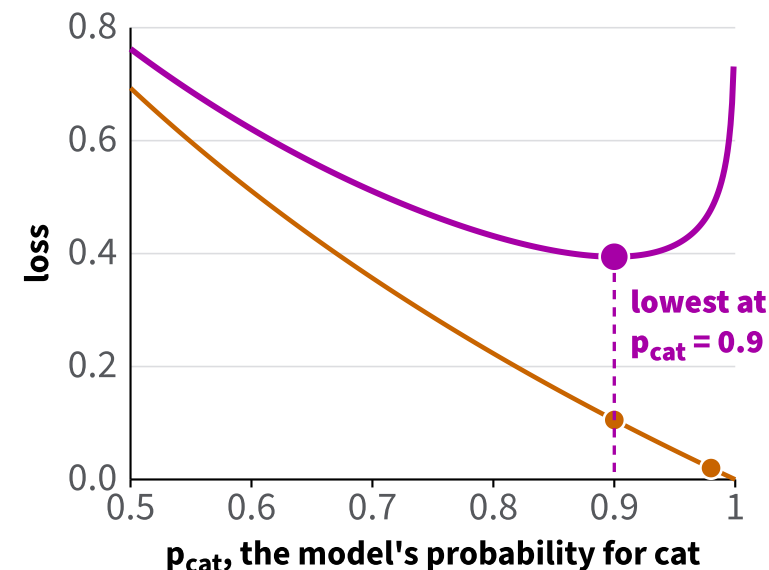
↳ Details the main-flow example label smoothing (p. 20): the cat photo's target, and why the smoothed loss is lowest at $p_{\text{cat}} = 0.9$ (Theorem 1) (p. 11).

Smoothed target $\text{target}_k = (1 - \varepsilon) \cdot [k = \text{cat}] + \varepsilon/K$, with $K = 3$ classes (cat, dog, fox) and $\varepsilon = 0.15$: cat $0.85 \cdot 1 + 0.05 = 0.9$, dog and fox $0.85 \cdot 0 + 0.05 = 0.05$.

Cross-entropy loss $-\sum_k \text{target}_k \cdot \ln p_k$ (lower is better); p_k is the model's probability for class k . Dog and fox merge into weight $0.05 + 0.05 = 0.1$. With the **hard target** (1, 0, 0) the loss $-\ln p_{\text{cat}}$ **keeps falling** toward 1 (orange).

smoothed target (0.9, 0.05, 0.05)

hard target (1, 0, 0)



$$f(p) = -0.9 \cdot \ln p - 0.1 \cdot \ln \frac{1-p}{2}$$

smoothed loss; $p = p_{\text{cat}}$,
dog, fox: $\frac{1-p}{2}$ each

$$-f(p) = 0.9 \cdot \ln p + 0.1 \cdot \ln(1-p) - 0.1 \cdot \ln 2$$

log-likelihood with
counts $h = 0.9$, $t = 0.1$

$$\Rightarrow p = \frac{h}{h+t} = \frac{0.9}{0.9+0.1} = 0.9$$

Theorem 1

Smoothed loss is lowest at $p_{\text{cat}} = 0.9$, not 1: imagined counts stop the push to certainty.

Theorem 3 on HH: both sides give 2.4

↳ Details the Theorem 3 (p. 27) line (slope 2.4) of the main-flow key calculation (p. 29): $x = \text{HH}$, $z \in \{A, B\}$, $\theta_j = \theta_A$, at $\theta = (\frac{3}{4}, \frac{1}{4})$. Recall the paths $\frac{9}{32}, \frac{1}{32}$ and $p(\text{HH}) = \frac{9}{32} + \frac{1}{32} = \frac{10}{32}$.

$$\underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH})}_{\text{left side} = 2.4 \text{ (directly)}} = \underbrace{p(A \mid \text{HH})}_{\text{weight } 9/10} \cdot \underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH}, A)}_{\text{path A slope} = 2/\theta_A = 8/3} + \underbrace{p(B \mid \text{HH})}_{\text{weight } 1/10} \cdot \underbrace{\frac{\partial}{\partial \theta_A} \ln p(\text{HH}, B)}_{\text{path B slope} = 0 \text{ (no } \theta_A \text{)}}$$

Left side, directly (chain rule, no posterior):

$$\begin{aligned} \ln p(\text{HH}) &= \ln \left(\frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot \theta_B^2 \right) \\ \frac{\partial}{\partial \theta_A} \ln p(\text{HH}) &= \frac{\frac{1}{2} \cdot 2 \cdot \theta_A}{p(\text{HH})} = \frac{\theta_A}{p(\text{HH})} = \frac{3/4}{10/32} \\ &= \frac{3}{4} \cdot \frac{32}{10} = \frac{12}{5} = 2.4 \end{aligned}$$

Right side, path by path:

$$\begin{aligned} \ln p(\text{HH}, A) &= \ln \left(\frac{1}{2} \cdot \theta_A \cdot \theta_A \right) = \ln \frac{1}{2} + \ln \theta_A + \ln \theta_A = \ln \frac{1}{2} + 2 \cdot \ln \theta_A \\ \ln p(\text{HH}, B) &= \ln \left(\frac{1}{2} \cdot \theta_B \cdot \theta_B \right) = \ln \frac{1}{2} + \ln \theta_B + \ln \theta_B = \ln \frac{1}{2} + 2 \cdot \ln \theta_B \\ \text{slopes in } \theta_A: &\frac{2}{\theta_A} = \frac{8}{3} \text{ at } \theta_A = \frac{3}{4} \text{ (path A), } 0 \text{ (path B: no } \theta_A \text{)} \\ \text{weights } p(A \mid \text{HH}) &= \frac{9/32}{10/32} = \frac{9}{10}, \quad p(B \mid \text{HH}) = \frac{1/32}{10/32} = \frac{1}{10} \\ \text{sum: } \frac{9}{10} \cdot \frac{8}{3} &= \frac{72}{30}, \text{ so } 2.4 = \frac{9}{10} \cdot \frac{8}{3} + \frac{1}{10} \cdot 0 = \frac{24}{10} \end{aligned}$$

left 2.4 = right 2.4 ✓ both sides agree.

Why check? The right side used only one-path logs, which split like counting; **no log of a sum** was differentiated. EM builds on this.

2.4 > 0 both ways: raising θ_A makes HH more likely, and posterior-weighted path slopes find it.

On the coins: hiding the coin ties θ_A to θ_B

↳ Details the main-flow "coin hidden" result (p. 31): the log of a sum on HH, in numbers.

The theory's chain for z hidden (p. 25) (a log of a sum does not split) applied to the coins. Recall: with the coin told, counting gave $\theta_A = \frac{3}{4}$ whatever θ_B is.

Coin hidden: one log, two coins

$$\ln p(\text{HH}) = \ln \underbrace{\left(\frac{1}{2} \cdot \theta_A^2\right)}_{\text{path A}} + \underbrace{\left(\frac{1}{2} \cdot \theta_B^2\right)}_{\text{path B}}$$

Both θ_A and θ_B sit in **one** log; a log of a sum is not a sum of logs:

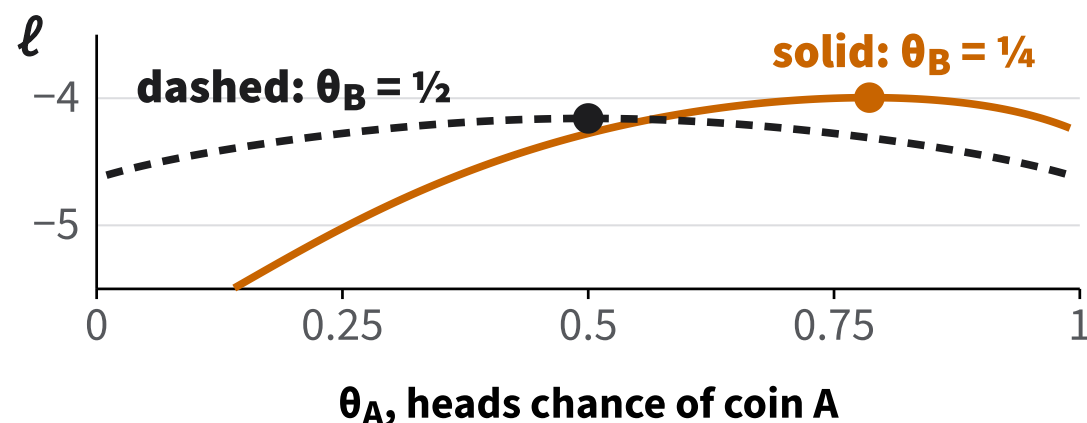
$$\ln \left(\frac{9}{32} + \frac{1}{32} \right) = \ln \frac{10}{32} = -1.163$$

$$\ln \frac{9}{32} + \ln \frac{1}{32} = \ln \left(\frac{9}{32} \cdot \frac{1}{32} \right) = \ln \frac{9}{1024} = -4.734$$

$$\text{so } -1.163 \neq -4.734.$$

Why it matters: in the figure, which plots the full ℓ (all three records), changing θ_B from $\frac{1}{4}$ to $\frac{1}{2}$ moves the best θ_A (dot), so θ_A cannot be fitted alone.

Coin hidden: the best θ_A (dot) moves with θ_B



Hidden coin $\Rightarrow$ a log of a sum $\Rightarrow$ the best θ_A depends on θ_B : no counting formula.

Training on the answer raises the trace that explains it

↳ Details the main-flow reasoning trace (p. 32): how the posterior 0.81 weights the training slope (Theorem 3) (p. 27).

Toy model: one parameter s , the chance of writing z_1 ; now $s = 0.3$.

Training increases $\ln p(14)$, with

$$p(14) = s \cdot \underbrace{1}_{\substack{\text{says } 14 \\ \text{after } z_1}} + (1 - s) \cdot \underbrace{0.1}_{\substack{\text{says } 14 \text{ after} \\ z_2 \text{ (a slip)}}} = 0.30 + 0.07 = 0.37.$$

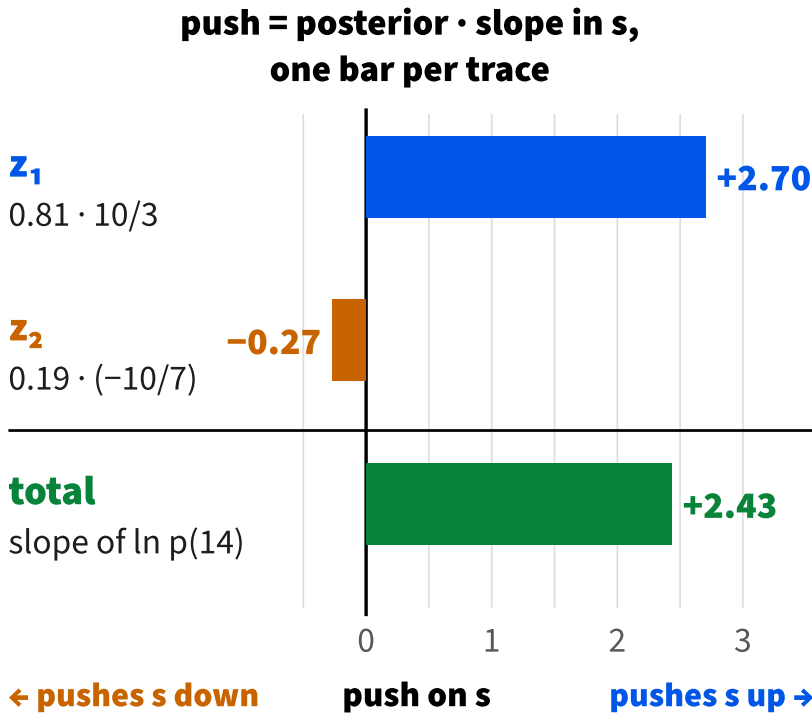
Theorem 3 with $\theta_j = s$:

trace	$\ln p(z, 14)$	slope in s	posterior $p(z \mid 14)$	posterior · slope
z_1	$\ln s + \ln 1$	$\frac{1}{s} = \frac{10}{3}$	$\frac{0.30}{0.37} = 0.81$	+2.70
z_2	$\ln(1 - s) + \ln 0.1$	$\frac{-1}{1-s} = -\frac{10}{7}$	$\frac{0.07}{0.37} = 0.19$	-0.27
total				+2.43

Check: $\frac{d}{ds} \ln p(14) = \frac{1}{0.37} \cdot \frac{d}{ds} [s + 0.1 \cdot (1 - s)] = \frac{1-0.1}{0.37} = 2.43 \checkmark$

Then raise s , recompute $p(z \mid 14)$, and repeat: $p(14)$ increases toward 1.

Slope +2.43 > 0: training on answers alone raises the chance of the trace that explains them.



Phan et al., Training Chain-of-Thought via Latent-Variable Inference, Neural Information Processing Systems (NeurIPS) 2023.

Slope zero gives count-and-divide: $\theta_A = 2.3 / (2.3 + 0.7)$

↳ Details the main-flow M-step $\theta_A = 2.3/3 = 0.767$ (p. 44): the slope-zero proof with $H_A = 2.3, T_A = 0.7$.

$$f = 2.3 \cdot \ln \theta_A + 0.7 \cdot \ln(1 - \theta_A)$$

$$f' = \frac{2.3}{\theta_A} - \frac{0.7}{1 - \theta_A}$$

$$= 0 \Rightarrow 2.3 \cdot (1 - \theta_A) = 0.7 \cdot \theta_A$$

$$\theta_A = \frac{2.3}{2.3 + 0.7} = 0.767$$

$$\theta_B = \frac{0.7}{0.7 + 2.3} = 0.233$$

Coin A's part of the M-step objective.

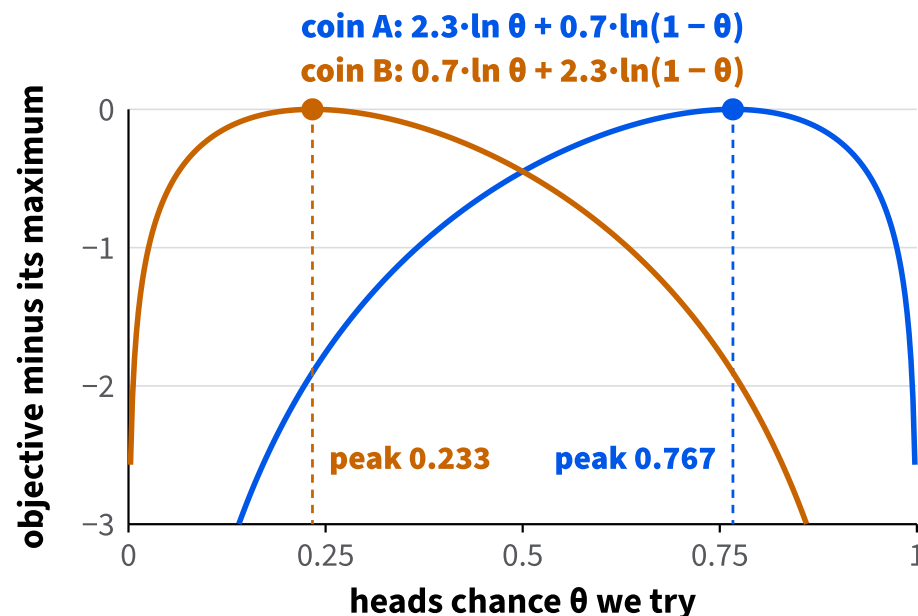
Slope of each log term: $\frac{d}{d\theta} \ln \theta = \frac{1}{\theta}$,
 $\frac{d}{d\theta} \ln(1 - \theta) = -\frac{1}{1 - \theta}$.

Slope zero; multiply by $\theta_A \cdot (1 - \theta_A)$. f is concave (each $\ln$ curves down), so this is the peak.

Expected heads ÷ expected flips.

Same steps for coin B's part.

Why count-and-divide: Theorem 1's peak (p. 11), heads ÷ (heads + tails), holds for any counts, whole or fractional.



The arg max needs no search: weighted counts, then divide.

Why 0.789: there the model gives the data's own shares

↳ Details the main-flow limit 0.789 (p. 45): solve for the θ where the model matches the data's shares (2, 1, 0 heads once each: shares $1/3, 1/3, 1/3$), and its ℓ .

1 head has chance $2 \cdot p(\text{HT})$ (HT or TH, equally likely). Matching: $p(\text{HH}) = p(\text{TT}) = 1/3, p(\text{HT}) = 1/2 \cdot 1/3 = 1/6$.

$$\frac{1}{2} \cdot (\theta_A^2 + (1 - \theta_A)^2) = \frac{1}{3}$$

$$1 - 2 \cdot \theta_A \cdot (1 - \theta_A) = \frac{2}{3} \Rightarrow \theta_A \cdot (1 - \theta_A) = \frac{1}{6}$$

$$\theta_A^2 - \theta_A + \frac{1}{6} = 0 \Rightarrow \theta_A = \frac{1 \pm \sqrt{1 - 4 \cdot \frac{1}{6}}}{2} = \frac{3 \pm \sqrt{3}}{6}$$

$$\theta_A = \frac{3 + \sqrt{3}}{6} = 0.789$$

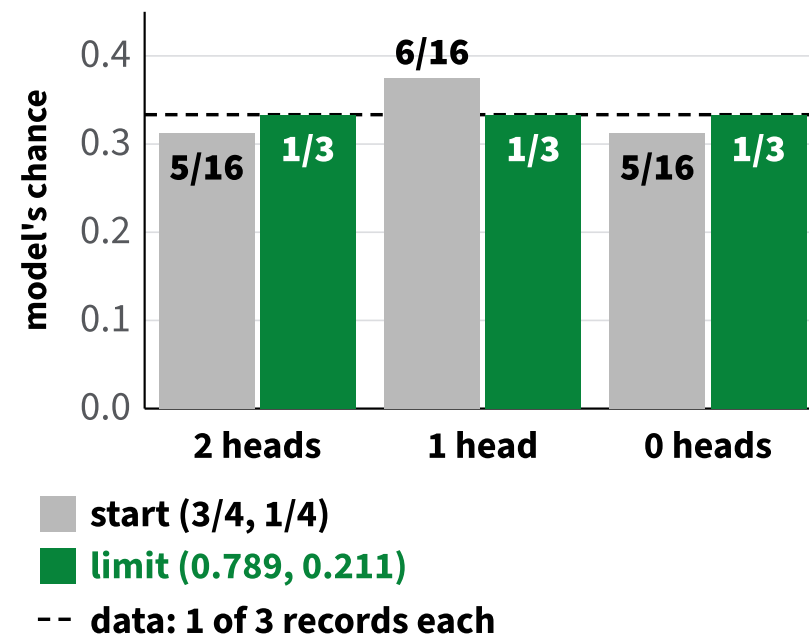
Model $p(\text{HH}) = \frac{1}{3}, \theta_B = 1 - \theta_A$.

Expand, multiply by 2.

Move to one side; quadratic formula;
 $\sqrt{1/3} = \frac{\sqrt{3}}{3}$.

Larger root of
 $\theta^2 - \theta + \frac{1}{6} = 0$;
the smaller root, 0.211, is θ_B .

each head count: model vs data



At $\theta_A = 0.789$ the model gives 2, 1, 0 heads the data's shares $\frac{1}{3}$ each: $\ell = 2 \cdot \ln \frac{1}{3} + \ln \frac{1}{6} = -3.989$.

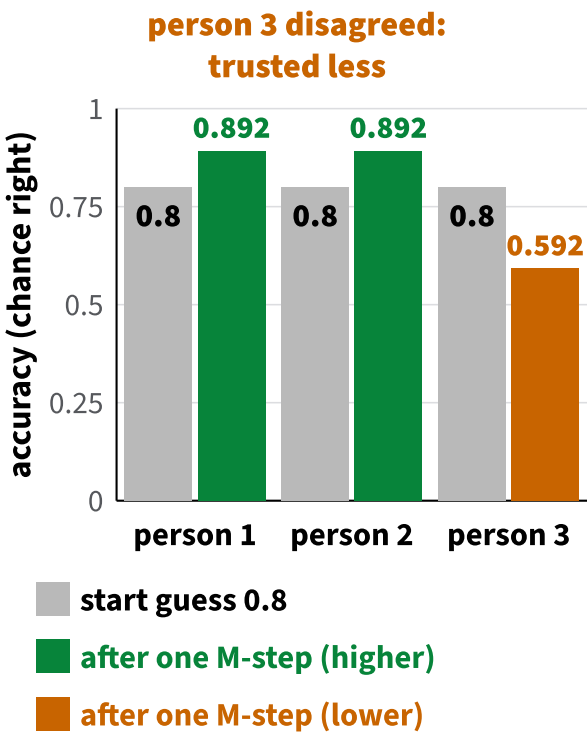
Crowd labelling: photo 1 is cat with 0.8, then each accuracy is counted like θ_A

↳ Details the main-flow crowd labelling (p. 47). Start accuracies $a_k = 0.8$, truth cat or dog with $\frac{1}{2}$ each. **Photo 1**, labels cat, cat, dog: cat path $\frac{1}{2} \cdot 0.8^2 \cdot 0.2 = 0.064$, dog path $\frac{1}{2} \cdot 0.2^2 \cdot 0.8 = 0.016$; posterior of cat = $\frac{0.064}{0.064 + 0.016} = 0.8$. **Photo 2**, labels dog, dog, dog: dog path $\frac{1}{2} \cdot 0.8^3 = 0.256$ (all right), cat path $\frac{1}{2} \cdot 0.2^3 = 0.004$ (all wrong); posterior of dog = $\frac{0.256}{0.256 + 0.004} = 0.9846$.

M-step. Each cell: the label given, and the posterior chance it is right.

person	photo 1	photo 2	new accuracy a_k
1	cat: 0.8	dog: 0.9846	$(0.8 + 0.9846)/2 = 0.892$
2	cat: 0.8	dog: 0.9846	$(0.8 + 0.9846)/2 = 0.892$
3	dog: $1 - 0.8 = 0.2$	dog: 0.9846	$(0.2 + 0.9846)/2 = 0.592$

Like $\theta_A = \text{heads} \div \text{flips}$: **expected right answers $\div$ photos labelled**. Person 3, outvoted on photo 1, is trusted less.



With no true labels, EM learns whom to trust: person 3 drops to 0.592.

On HH, the guess $(\frac{1}{2}, \frac{1}{2})$ gives the floor -1.674 , with gap 0.511

↳ Details the main result (p. 55): Lemma 1 (p. 50), guess $\frac{1}{2}$: bound -1.674 , gap 0.511 .

HH at $\theta = (\theta_A, \theta_B) = (\frac{3}{4}, \frac{1}{4})$	pick · H · H	path
via coin A	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4}$	$\frac{9}{32}$
via coin B	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4}$	$\frac{1}{32}$
$p(\text{HH})$	sum	$\frac{10}{32} = \frac{5}{16}$

A **path**: the coin is picked, then shows H, H. A **guess** $q = (q(A), q(B))$ splits HH between the coins; its **bound** averages $\ln(\text{path} \div \text{guess})$:

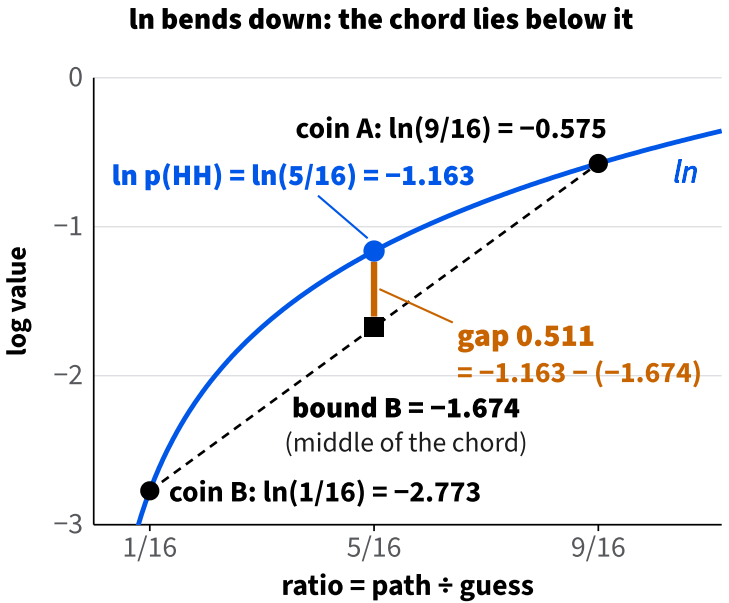
$$B(q) = q(A) \cdot \ln \frac{9/32}{q(A)} + q(B) \cdot \ln \frac{1/32}{q(B)}$$

Guess $(\frac{1}{2}, \frac{1}{2})$: ratios $\frac{9}{32} \div \frac{1}{2} = \frac{9}{16}$ and $\frac{1}{32} \div \frac{1}{2} = \frac{1}{16}$, so
 $B = \frac{1}{2} \cdot \ln \frac{9}{16} + \frac{1}{2} \cdot \ln \frac{1}{16} = \frac{1}{2} \cdot (-0.575) + \frac{1}{2} \cdot (-2.773) = -1.674$.

Jensen's inequality on HH: the q 's cancel for **any** guess.

$$B \leq \ln \left(q(A) \cdot \frac{9/32}{q(A)} + q(B) \cdot \frac{1/32}{q(B)} \right) = \ln \frac{10}{32} = \ln \frac{5}{16} = -1.163$$

gap $= \ln \frac{5}{16} - B = -1.163 - (-1.674) = 0.511$.



Because ln bends down, every guess gives a floor below $\ln p(\text{HH})$.

Round 1 in one table: $-4.00028 \leq -3.99577 \leq -3.99293$

↳ Details the main result (p. 56): round 1 (Theorem 4 (p. 52), $t = 0$) increases ℓ . Recall round 1 from $\theta_0 = (\frac{3}{4}, \frac{1}{4})$ (p. 44): E-step labels q_0 (chance of A) 0.9, 0.5, 0.1 for HH, HT, TT; M-step $\theta_1 = (\frac{2.3}{3}, \frac{0.7}{3}) = (\frac{23}{30}, \frac{7}{30})$; B keeps the old guess q_0 .

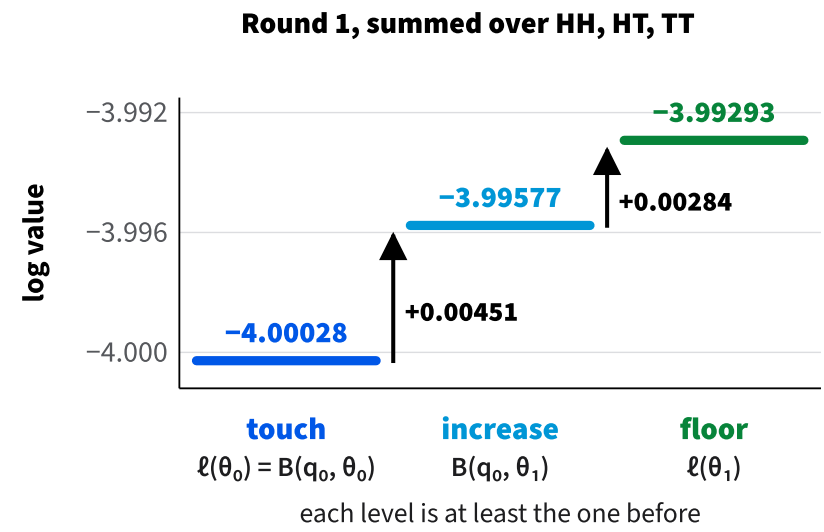
$$p_{\theta}(\text{HH}) = \frac{1}{2} \cdot \theta_A^2 + \frac{1}{2} \cdot \theta_B^2, \text{ at } \theta_1: \frac{1}{2} \cdot \frac{529+49}{900} = \frac{289}{900};$$

$$p_{\theta}(\text{HT}) = \frac{1}{2} \cdot \theta_A \cdot (1 - \theta_A) + \frac{1}{2} \cdot \theta_B \cdot (1 - \theta_B) = 2 \cdot \frac{1}{2} \cdot \frac{23}{30} \cdot \frac{7}{30} = \frac{161}{900}.$$

record	$\ln p_{\theta_0}(x)$	B at θ_1	$\ln p_{\theta_1}(x)$
HH	$\ln \frac{5}{16} = -1.16315$	-1.13739^*	$\ln \frac{289}{900} = -1.13597$
HT	$\ln \frac{3}{16} = -1.67398$	gap 0 [†] , so -1.72099	$\ln \frac{161}{900} = -1.72099$
TT	mirrors HH (swap H↔T, A↔B), so sum = 2·HH + HT		
sum	-4.00028	-3.99577	-3.99293

* B for HH, old labels 0.9, 0.1: $0.9 \cdot \ln \frac{529}{1620} + 0.1 \cdot \ln \frac{49}{180} = -1.13739$,
 with path A ÷ label = $\frac{1}{2} \cdot (\frac{23}{30})^2 \div 0.9 = \frac{529}{1620}$, path B ÷ label = $\frac{1}{2} \cdot (\frac{7}{30})^2 \div 0.1 = \frac{49}{180}$.
[†] HT: paths $\frac{1}{2} \cdot \frac{23}{30} \cdot \frac{7}{30} = \frac{1}{2} \cdot \frac{7}{30} \cdot \frac{23}{30}$ at θ_1 , so its posterior stays 0.5 = old label: gap 0.

$$\begin{array}{ccccccc} \ell(\theta_0) & = & B(q_0, \theta_0) & \leq & B(q_0, \theta_1) & \leq & \ell(\theta_1) \\ \text{touch} & & & \text{increase} & & \text{floor} & \\ -4.000 & = & -4.000 & \leq & -3.996 & \leq & -3.993 \\ \text{rises: } & -3.99577 - (-4.00028) = 0.00451, & & & -3.99293 - (-3.99577) = 0.00284 & & \end{array}$$



Summed over the records: $-4.000 \leq -3.996 \leq -3.993$, so ℓ increased.

Coin by coin, the guess 0.8 gives the bound -1.2076

↳ Details the main result (p. 57): the encoder guess 0.8 has bound -1.2076 and gap 0.0444. Lemma 1's bound (p. 50) for the guess $q = (0.8, 0.2)$ on HH at $\theta = (\frac{3}{4}, \frac{1}{4})$. Recall the paths $\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$ and $\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$; exact posterior $\frac{9}{32} \div \frac{10}{32} = 0.9$.

Its bound, coin by coin:

coin	guess	path	ratio = path ÷ guess	ln ratio	guess · ln
A	0.8	9/32	0.3516	-1.0454	-0.8363
B	0.2	1/32	0.1563	-1.8563	-0.3713
bound (ELBO) = sum					-1.2076

Check: $-1.2076 + 0.0444 = -1.1632 = \ln \frac{5}{16} \checkmark$.

The gap, directly as a Kullback–Leibler divergence:

$$\begin{aligned}
 \text{Gap} &= D_{\text{KL}}((0.8, 0.2) \parallel (0.9, 0.1)) = 0.8 \cdot \ln \frac{0.8}{0.9} + 0.2 \cdot \ln \frac{0.2}{0.1} \\
 &= 0.8 \cdot (-0.1178) + 0.2 \cdot (0.6931) \\
 &= -0.0942 + 0.1386 = 0.0444.
 \end{aligned}$$

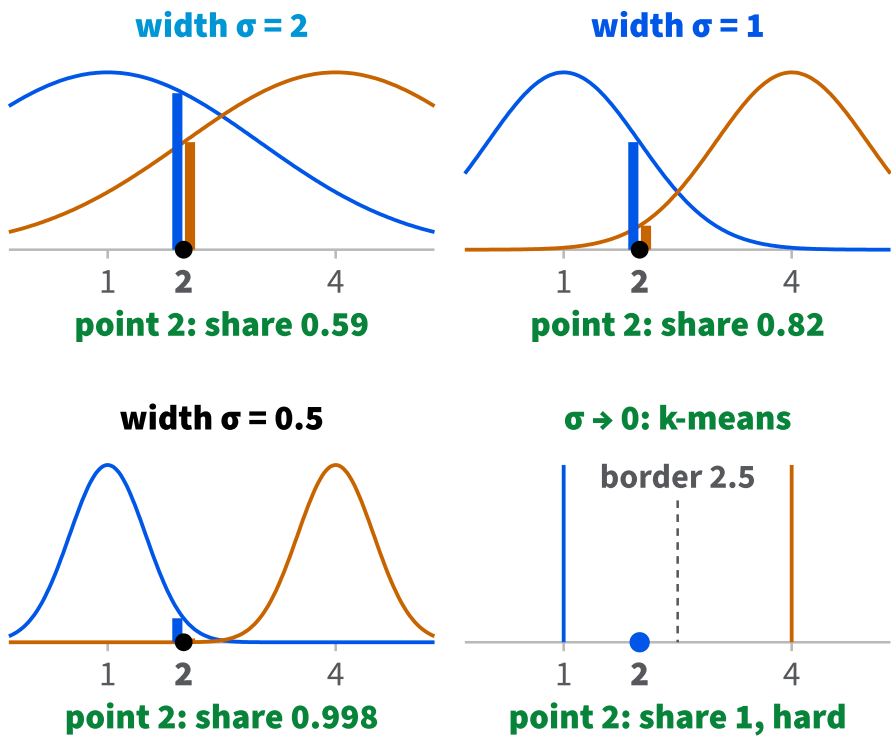
And the bound again, from Lemma 1 as $\ln p(\text{HH})$ minus the gap:

$$\begin{aligned}
 \text{Evidence lower bound} &= \ln p(\text{HH}) - \text{gap} = \ln \frac{5}{16} - 0.0444 \\
 &= -1.1632 - 0.0444 = -1.2076.
 \end{aligned}$$

Coin by coin, the bound is -1.2076 ; adding the gap 0.0444 gives back $\ln p(\text{HH})$.

Narrow the bells, and EM's soft share hardens into a hard choice

↳ Details the main-flow narrowing of the bells (p. 71): point 2's share 0.59, 0.82, 0.998 at $\sigma = 2, 1, 0.5$. Theorem 5 (p. 64) applied to one new point: watch its E-step share as the width σ shrinks.



A new point at **2**: distance $d = 1$ to centre 1, $d = 2$ to centre 4. With width σ , a bell's height is $e^{-d^2/2\sigma^2}$ (the shared factor $\frac{1}{\sqrt{2\pi}\cdot\sigma}$ cancels, so heights drop it).

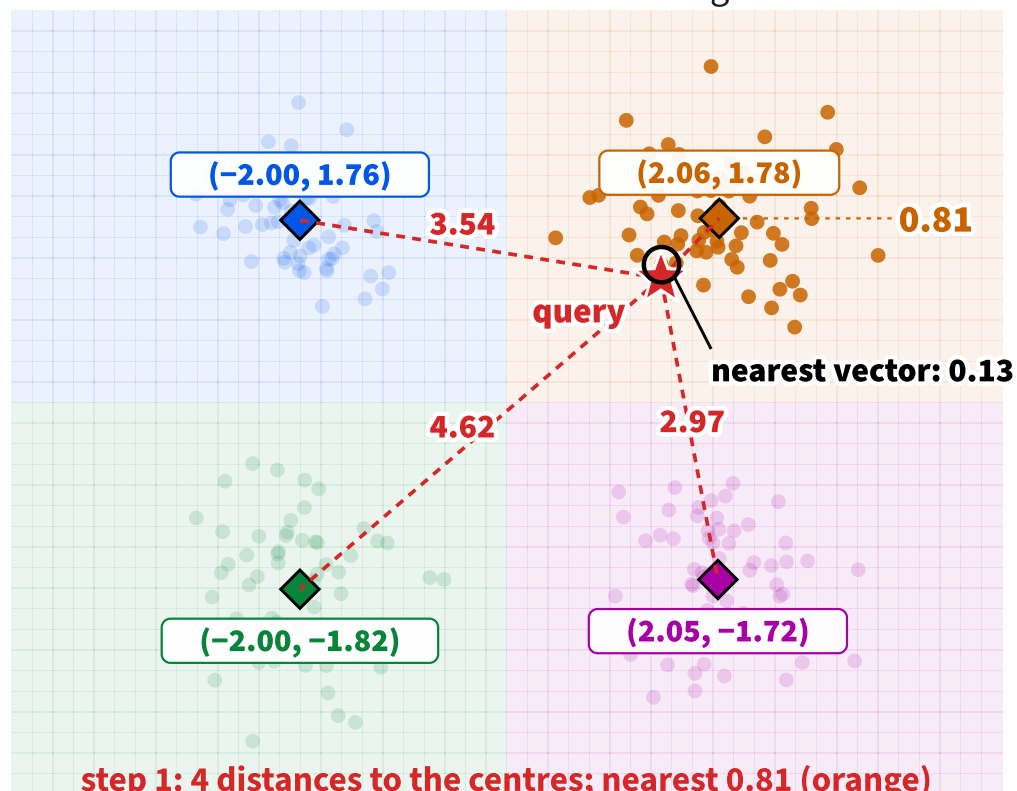
σ	near bell ($d = 1$) $e^{-1^2/2\sigma^2}$	far bell ($d = 2$) $e^{-2^2/2\sigma^2}$	share
2	0.882	0.607	$\frac{0.882}{0.882 + 0.607} = 0.59$
1	0.607	0.135	$\frac{0.607}{0.607 + 0.135} = 0.82$
0.5	0.135	0.000335	$\frac{0.135}{0.135 + 0.000335} = 0.998$

Why: the far bell shrinks faster than the near one, so **the share $\rightarrow 1$** as $\sigma \rightarrow 0$: k-means' hard label.

Narrower bells $\Rightarrow$ the nearer centre wins more surely $\Rightarrow$ in the limit, k-means.

k-means cells: 64 distance computations instead of 240

diamonds: 4 k-means centres · shading: their cells



step 2: only the 60 vectors of that cell: $4 + 60 = 64$ of 240

↳ Details the main-flow example vector search (p. 72): one search on the 240 points, and the "500 times fewer" at scale.

Built once: the 240 points of the four-group animation (p. 70) are the stored vectors; k-means gives 4 centres, and each vector is filed in its nearest centre's cell, 60 per cell.

Step 1: distances from the query (1.5, 1.2) to the 4 centres (blue 3.54, **orange 0.81**, green 4.62, purple 2.97). Orange centre (2.06, 1.78): $\sqrt{0.56^2 + 0.58^2} = \sqrt{0.65} = 0.81$, the nearest.

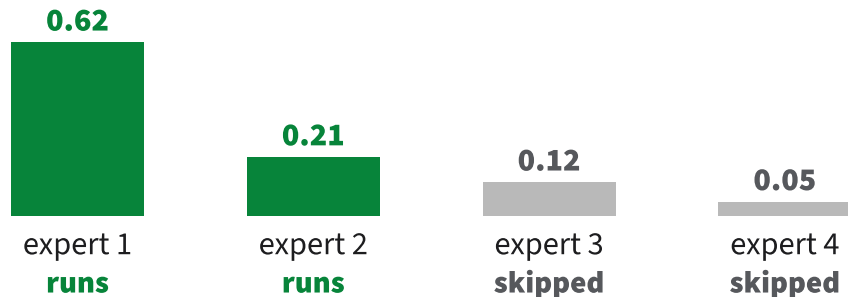
Step 2: compare with that cell's 60 vectors only; the nearest (ringed) lies at distance 0.13. Total $4 + 60 = 64$, not 240.

At scale: 1,000,000 vectors in 1,000 cells of about 1,000: $1,000 + 1,000 = 2,000$ distances; $1,000,000 \div 2,000 = 500$ times fewer (Jégou et al., 2011). **Price:** a neighbour just across a cell border can be missed.

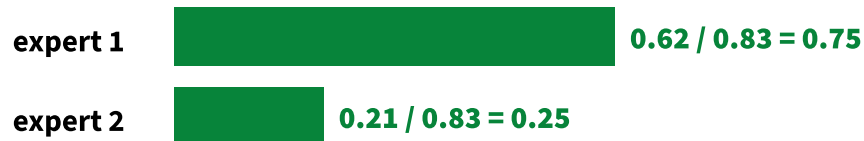
k-means' nearest-centre step, reused as an index: search a cell, not the library.

Top-2 routing on one token: run two experts, skip two

1 · router weights (sum 1)



2 · keep the top 2, divide each by 0.83



3 · mix the two experts' predictions

p(next token = "Paris"): expert 1: 0.9, expert 2: 0.5

$$0.75 \cdot 0.9 + 0.25 \cdot 0.5 = 0.80$$

experts 3 and 4 are never computed

↳ Details the main-flow example mixture of experts (p. 73): top-2 routing on one token, and Mixtral's 2 of 8.

Step 1: the text is "The capital of France is". For the token "is", the router gives the 4 experts weights 0.62, 0.21, 0.12, 0.05 (illustrative; they sum to 1).

Step 2: keep the two largest, $0.62 + 0.21 = 0.83$, and rescale to sum to 1: $\frac{0.62}{0.83} = 0.75$, $\frac{0.21}{0.83} = 0.25$.

Step 3: run only experts 1 and 2. Their chances that the next token is "Paris" (illustrative) are 0.9 and 0.5, so the layer outputs $0.75 \cdot 0.9 + 0.25 \cdot 0.5 = 0.80$.

At scale: Mixtral (Jiang et al., 2024) has 8 experts per layer and runs 2 per token: $\frac{2}{8}$, a quarter of the expert computation. There each expert outputs a vector, and the layer mixes those vectors with the same kind of weights (here 0.75, 0.25).

Top-2 hardens the router's weights, as $\sigma \rightarrow 0$ hardened EM's shares into k-means.

Forward and backward messages on H, H, T give $p(\text{HHT}) = 15/128$

↳ Details the main results (p. 81) $p(\text{HHT}) = \frac{15}{128}$, A at flip 2 with chance $\frac{3}{4}$, and the products behind the soft labels 0.8, 0.75, 0.4 (p. 90). Rules: first coin $\frac{1}{2}$; keep $\frac{3}{4}$, switch $\frac{1}{4}$; A shows H with $\frac{3}{4}$, B with $\frac{1}{4}$.

↳ **Forward** $\alpha_t(z)$ = chance of flips 1 to t **and** coin z at flip t , left to right. **Backward** $\beta_t(z)$ = chance of the flips after t , given coin z at t , right to left from $\beta_3 = 1$.

$$\alpha_1(A) = \frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}, \quad \alpha_1(B) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$$

$$\alpha_2(A) = \left(\frac{3}{8} \cdot \frac{3}{4} + \frac{1}{8} \cdot \frac{1}{4}\right) \cdot \frac{3}{4} = \frac{15}{64}, \quad \alpha_2(B) = \left(\frac{3}{8} \cdot \frac{1}{4} + \frac{1}{8} \cdot \frac{3}{4}\right) \cdot \frac{1}{4} = \frac{3}{64}$$

$$\alpha_3(A) = \left(\frac{15}{64} \cdot \frac{3}{4} + \frac{3}{64} \cdot \frac{1}{4}\right) \cdot \frac{1}{4} = \frac{48}{256} \cdot \frac{1}{4} = \frac{3}{64}$$

$$\beta_2(A) = \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{8}, \quad \beta_2(B) = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{3}{4} = \frac{5}{8}$$

$$\beta_1(A) = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{8} + \frac{1}{4} \cdot \frac{1}{4} \cdot \frac{5}{8} = \frac{27+5}{128} = \frac{1}{4}$$

$$p(\text{HHT}) = \frac{15}{64} \cdot \frac{3}{8} + \frac{3}{64} \cdot \frac{5}{8} = \frac{45+15}{512} = \frac{15}{128}, \quad p(A \text{ at flip 2} \mid \text{HHT}) = \frac{45}{60} = \frac{3}{4}$$

$$\alpha_t(A) \cdot \beta_t(A) \cdot 2048: \frac{3}{8} \cdot \frac{1}{4} \rightarrow 192, \quad \frac{15}{64} \cdot \frac{3}{8} \rightarrow 180, \quad \frac{3}{64} \cdot 1 \rightarrow 96$$

Flip 1 = H: start · flip.

Into each coin: keep or switch, then H.

Into A, then T.

Out of each coin: move, then T.

Out of A: move, H, then β_2 .

Cut at flip 2: add; share (Bayes).

Flips 1, 2, 3; $\div 240$ gives 0.8, 0.75, 0.4.

Two numbers per coin per flip replace the list of all $2^3 = 8$ sequences: Lecture 2's sum-product.

Kept pairs give the keep chance 0.7; the forward pass then gives $p(\text{HHT}) = 0.139$

↳ Details the main results (p. 91): keep chance 0.7 and the rise $0.117 \rightarrow 0.139$. **Theorem 6's pair label (p. 84):** $\xi_t(a, b) = \alpha_t(a) \cdot \tau(a \rightarrow b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b) \div p(\text{HHT})$; the keep chance needs only the kept pairs ($a = b$, keep $\frac{3}{4}$). Messages from the previous slide; $p(\text{HHT}) = \frac{240}{2048}$.

kept pair	A → A: $\alpha_t(A) \cdot \text{keep} \cdot E(A, x_{t+1}) \cdot \beta_{t+1}(A)$	B → B: the same with B	kept: $\xi_t(A, A) + \xi_t(B, B)$
flips 1 → 2 (H)	$\frac{3}{8} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{8} = \frac{162}{2048}$	$\frac{1}{8} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot \frac{5}{8} = \frac{30}{2048}$	$\frac{162+30}{240} = 0.8$
flips 2 → 3 (T)	$\frac{15}{64} \cdot \frac{3}{4} \cdot \frac{1}{4} \cdot 1 = \frac{90}{2048}$	$\frac{3}{64} \cdot \frac{3}{4} \cdot \frac{3}{4} \cdot 1 = \frac{54}{2048}$	$\frac{90+54}{240} = 0.6$

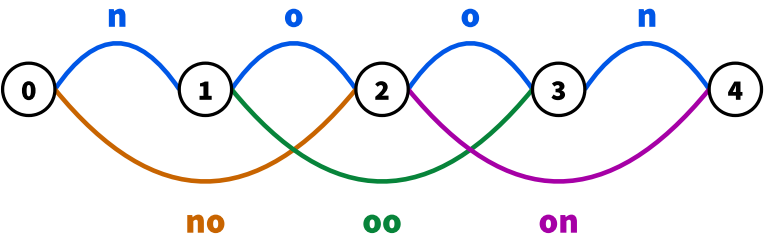
New keep chance (expected keeps ÷ moves): $\kappa = \frac{0.8+0.6}{2} = 0.7$. **New guess** $\theta_A = 0.795$, $\theta_B = 0.429$, keep 0.7, switch 0.3; tails A 0.205, B 0.571.

flip	$\alpha_t(A)$: into A, then the flip	$\alpha_t(B)$: into B, then the flip
1 = H	$\frac{1}{2} \cdot \frac{1.55}{1.95} = 0.397$	$\frac{1}{2} \cdot \frac{0.45}{1.05} = 0.214$
2 = H	$(0.397 \cdot 0.7 + 0.214 \cdot 0.3) \cdot 0.795 = 0.272$	$(0.397 \cdot 0.3 + 0.214 \cdot 0.7) \cdot 0.429 = 0.115$
3 = T	$(0.272 \cdot 0.7 + 0.115 \cdot 0.3) \cdot 0.205 = 0.046$	$(0.272 \cdot 0.3 + 0.115 \cdot 0.7) \cdot 0.571 = 0.093$

Add at flip 3: $p(\text{HHT}) = 0.046 + 0.093 = 0.139 > \frac{15}{128} = 0.117$, as **Theorem 4 (p. 52) (EM increases the likelihood) promises.**

With every piece at 1/5, the cut no|on has posterior 0.610

every path from 0 to 4 is one cut of “noon”



node k = boundary after k letters

posterior of each cut = chance · 625 ÷ 41

n o o n	1/41
n o on	5/41
n oo n	5/41
no o n	5/41
no on	25/41

↳ Details the main-flow example tokenizer (p. 92): where the weight 0.61 of the cut no|on comes from.

Pieces n, o, no, on, oo start at 1/5 each.

A cut's chance = product of its pieces: 4 pieces 1/625, 3 pieces 5/625, 2 pieces 25/625; all 5 cuts: (1+5+5+5+25)/625 = 41/625.

E-step: a cut's posterior = its chance ÷ total, e.g. no|on: $\frac{25}{41} = 0.610$. A piece's expected uses add its uses over the cuts, weighted by the posteriors.

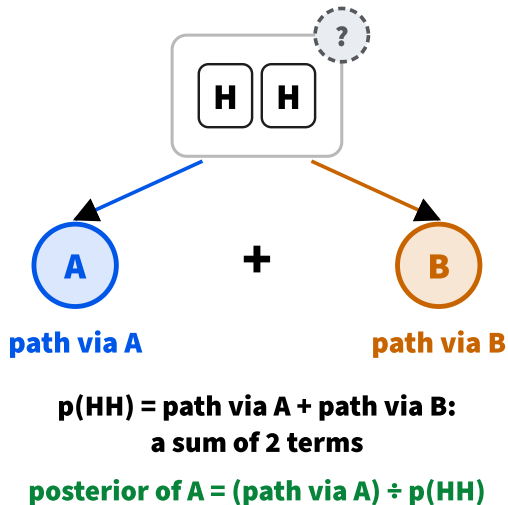
M-step: piece probability = its uses ÷ all uses, like heads ÷ flips; the 41 cancels.

Cuts are hidden like coins; forward-backward adds over all cuts without listing them.

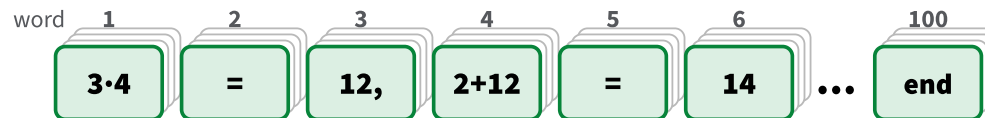
A 100-word reasoning has about 10^{470} possible texts, so its E-step sum cannot be done

↳ Details the main-flow count (p. 93) " $\approx 10^{470}$ terms". The hidden z is a 100-word text, and each word is one of 50,000; the posterior's sum has one term per text.

Coins: the hidden z has 2 values



Reasoning: the hidden z is a 100-word text



behind each word: a stack of 50,000 possible words

Sum over every z : $50,000 \cdot 50,000 \cdot \dots$ (100 times) $\approx 10^{470}$ terms, more terms than atoms in the visible universe (about 10^{80})

$50,000 \approx 10^{4.7}$, so 100 words give $(10^{4.7})^{100} = 10^{470}$

Run forward: pick one word at a time, left to right

100 easy steps; keep only the runs that reach the right answer, and the kept runs are posterior draws

The count: $50,000^{100}$ texts. Since $\log_{10} 50,000 = 4.699 \approx 4.7$, $50,000 \approx 10^{4.7}$, so $50,000^{100} \approx (10^{4.7})^{100} = 10^{4.7 \cdot 100} = 10^{470}$, far above the about 10^{80} atoms in the visible universe.

One term per text: 10^{470} terms, while running forward needs only 100 word choices.

Hard EM then stops at once, at a fit worse than where it started

↳ Details the main-flow result (p. 106) "stuck at $(\frac{3}{4}, 0)$, $\ell = -4.268$ ". Round 1 from $(\frac{3}{4}, \frac{1}{4})$: hard labels A, A (tie), B, so A gets 3 heads in 4 flips, $\theta_A = \frac{3}{4}$, and B 0 heads in 2 flips, $\theta_B = 0$. Round 2, at $(\frac{3}{4}, 0)$: the two paths pick the labels, and their sum $p(x)$ scores the fit.

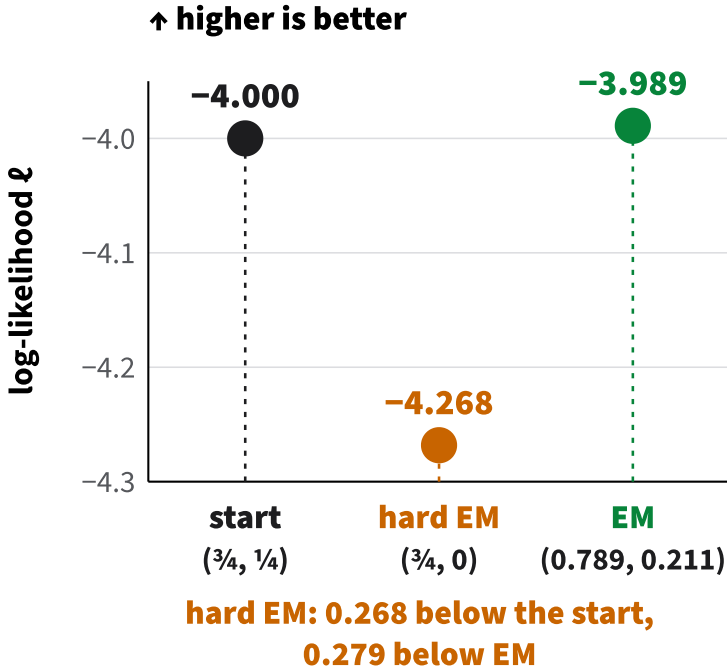
record	path via A	path via B	label	$p(x)$
HH	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{32}$	$\frac{1}{2} \cdot 0 \cdot 0 = 0$	A	$\frac{9}{32}$
HT	$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{32}$	$\frac{1}{2} \cdot 0 \cdot 1 = 0$	A	$\frac{3}{32}$
TT	$\frac{1}{2} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{32}$	$\frac{1}{2} \cdot 1 \cdot 1 = \frac{16}{32}$	B	$\frac{17}{32}$

Labels A, A, B again $\Rightarrow$ same counts $\Rightarrow$ **stuck** at $(\frac{3}{4}, 0)$.

Score: $\ell = \ln \frac{9}{32} + \ln \frac{3}{32} + \ln \frac{17}{32} = \ln \frac{9 \cdot 3 \cdot 17}{32^3} = \ln \frac{459}{32768} = -4.268$.

Start: $\ell(\frac{3}{4}, \frac{1}{4}) = \ln \frac{10}{32} + \ln \frac{6}{32} + \ln \frac{10}{32} = \ln \frac{600}{32768}$, so $\ell = -4.000$ (recall (p. 29) the start's path sums $\frac{9}{32} + \frac{1}{32}, \frac{3}{32} + \frac{3}{32}, \frac{1}{32} + \frac{9}{32}$); EM's limit -3.989 (recall) (p. 46).

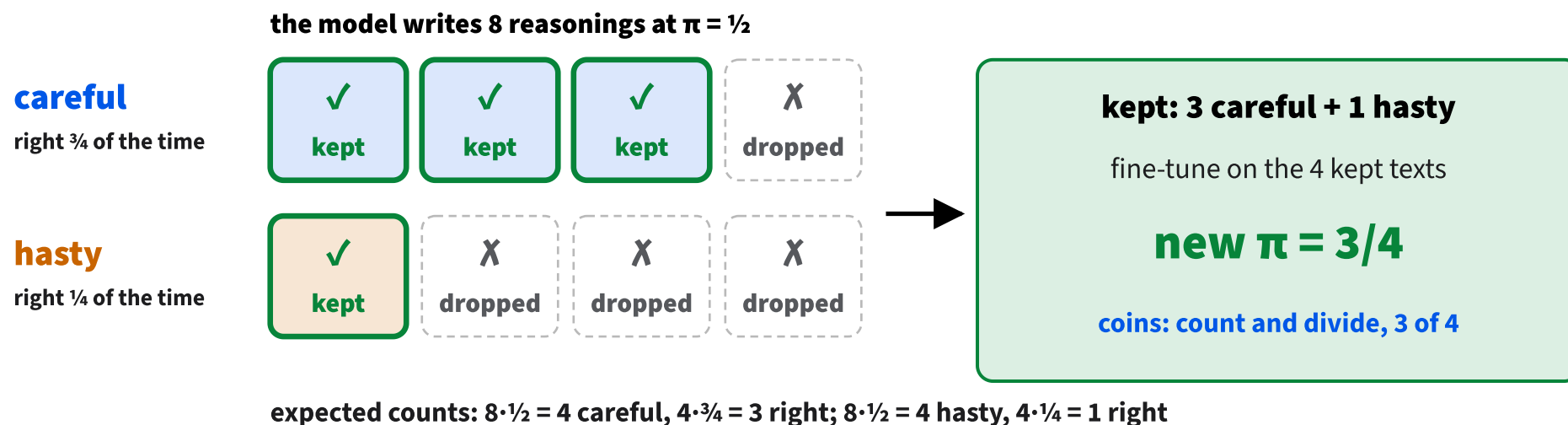
So hard EM ends **0.268 below its start**.



Hard labels are not the posterior, so Theorem 4 cannot protect ℓ : hard EM ends below its start.

A tiny reasoner: keeping the correct answers raises careful reasoning from $\frac{1}{2}$ to $\frac{3}{4}$

↳ Details the main-flow example tiny reasoner (p. 107) (careful reasoning $\frac{1}{2} \rightarrow \frac{3}{4}$): **round 1**. Toy model: it writes a **careful** reasoning with chance π (right with chance $\frac{3}{4}$) or a **hasty** one (right with chance $\frac{1}{4}$). These are coins A and B with one flip; "correct" plays heads. Start $\pi = \frac{1}{2}$. Here EM learns the toss chance π ; the heads chances $\frac{3}{4}, \frac{1}{4}$ stay fixed.



Why $\frac{3}{4}$: new π = share of careful among kept = $\frac{\frac{1}{2} \cdot \frac{3}{4}}{\frac{1}{2} \cdot \frac{3}{4} + \frac{1}{2} \cdot \frac{1}{4}} = \frac{3}{4}$, the **posterior** of "careful" given "correct".

One round of STaR on the toy model = one Monte Carlo EM step for π (shown at expected counts).

Between the peaks, $(\frac{1}{2}, \frac{1}{2})$ is a saddle that can trap EM

↳ Details the main result (p. 119): the saddle, $\ell = -4.159$. Saddle test (p. 113) in full.

Between the peaks lies $(\frac{1}{2}, \frac{1}{2})$, identical coins: each record has chance $\frac{1}{4}$, so $\ell = 3 \cdot \ln \frac{1}{4} = -4.159$.

step d from $(\frac{1}{2}, \frac{1}{2})$	$p(\text{HH})$	$p(\text{HT})$	$p(\text{TT})$
along: $(\frac{1}{2}+d, \frac{1}{2}+d)$	$(\frac{1}{2}+d)^2$	$\frac{1}{4} - d^2$	$(\frac{1}{2}-d)^2$
across: $(\frac{1}{2}+d, \frac{1}{2}-d)$	$\frac{1}{4} + d^2$	$\frac{1}{4} - d^2$	$\frac{1}{4} + d^2$

Along, both coins have $\theta = \frac{1}{2} + d$: $p(\text{HH}) = \frac{1}{2} \cdot \theta^2 + \frac{1}{2} \cdot \theta^2 = \theta^2$,
 $p(\text{HT}) = \theta \cdot (1 - \theta) = \frac{1}{4} - d^2$.

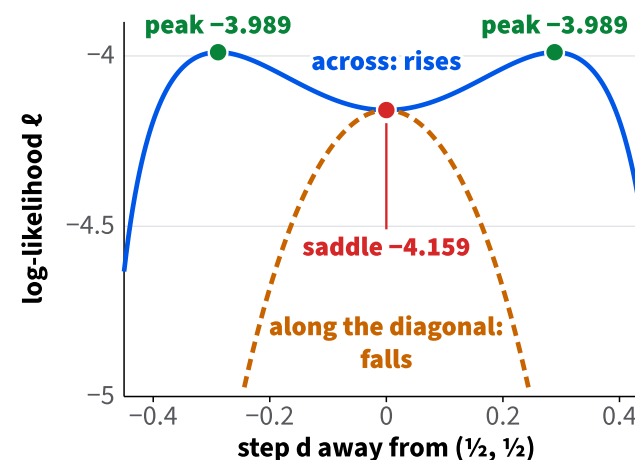
Across, the $\pm d$ terms cancel: $\frac{1}{2} \cdot (\frac{1}{2} + d)^2 + \frac{1}{2} \cdot (\frac{1}{2} - d)^2 = \frac{1}{4} + d^2$.

along: $\ell = 2 \cdot \ln(\frac{1}{2} + d) + 2 \cdot \ln(\frac{1}{2} - d) + \ln(\frac{1}{4} - d^2)$

since $(\frac{1}{2} + d) \cdot (\frac{1}{2} - d) = \frac{1}{4} - d^2$: $\ell = 3 \cdot \ln(\frac{1}{4} - d^2)$, largest at $d = 0$

across: $\ell = 2 \cdot \ln(\frac{1}{4} + d^2) + \ln(\frac{1}{4} - d^2)$, smallest at $d = 0$:

$$\begin{aligned}
 &= 3 \cdot \ln \frac{1}{4} + 2 \cdot \ln(1 + 4 \cdot d^2) + \ln(1 - 4 \cdot d^2) \quad (\text{factor out } \frac{1}{4}) \\
 &= 3 \cdot \ln \frac{1}{4} + 8 \cdot d^2 - 4 \cdot d^2 + O(d^4) \quad (\ln(1 + u) = u + O(u^2)) \\
 &= 3 \cdot \ln \frac{1}{4} + 4 \cdot d^2 + O(d^4) \quad (\text{the test's } 4 \cdot d^2)
 \end{aligned}$$



the same point is lowest across and highest along

Highest along, lowest across: a saddle, a worse fit than the mirror peaks.

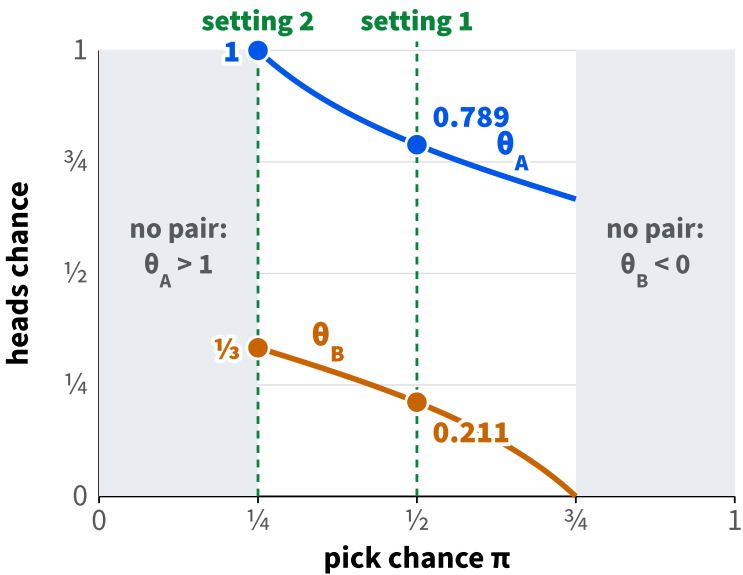
Every pick chance from $\frac{1}{4}$ to $\frac{3}{4}$ has its own equal-fit pair: a ridge

↳ Details the main result (p. 120): the ridge of equal fits. Fix the pick chance π and match $p(2) = p(0) = \frac{1}{3}$. Subtracting the two equations forces the average heads chance $\pi \cdot \theta_A + (1 - \pi) \cdot \theta_B = \frac{1}{2}$, so $\theta_A = \frac{1}{2} + (1 - \pi) \cdot d$, $\theta_B = \frac{1}{2} - \pi \cdot d$. Then $p(2) = \frac{1}{4} + \pi \cdot (1 - \pi) \cdot d^2 = \frac{1}{3}$ gives $d = 1/\sqrt{12 \cdot \pi \cdot (1 - \pi)}$.

π	d	θ_A	θ_B
$\frac{1}{4}$	$1/\sqrt{9/4} = \frac{2}{3}$	$\frac{1}{2} + \frac{3}{4} \cdot \frac{2}{3} = 1$	$\frac{1}{2} - \frac{1}{4} \cdot \frac{2}{3} = \frac{1}{3}$
$\frac{1}{2}$	$1/\sqrt{3} = 0.577$	$\frac{1}{2} + \frac{1}{2 \cdot \sqrt{3}} = 0.789$	$\frac{1}{2} - \frac{1}{2 \cdot \sqrt{3}} = 0.211$
$\frac{3}{4}$	$1/\sqrt{9/4} = \frac{2}{3}$	$\frac{1}{2} + \frac{1}{4} \cdot \frac{2}{3} = \frac{2}{3}$	$\frac{1}{2} - \frac{3}{4} \cdot \frac{2}{3} = 0$

$\pi = \frac{1}{2}$: $d = 0.577$, $\theta_A = 0.789$ is setting 1; $\pi = \frac{1}{4}$: $d = \frac{2}{3}$, $\theta_A = 1$, $\theta_B = \frac{1}{3}$ is setting 2.

Each π gets its own pair (plus its mirror, with $\theta_A < \theta_B$): a **ridge** of equal fits, all with $\ell = -3.989$. Outside $\frac{1}{4}$ to $\frac{3}{4}$, $\theta_A > 1$ or $\theta_B < 0$ (shaded).



every pair: $p(0 \text{ heads}) = p(2 \text{ heads}) = \frac{1}{3}$

The data cannot choose between settings on the ridge, so no algorithm can.

A prior that changes with an observed label pins the code axes

↳ Details the main-flow example learned codes (p. 122): the day/night prior on one code and its 90° turn.

Tiny instance. Rotate every code by 90°, $(2, 0) \rightarrow (0, 2)$ (the decoder undoes it). Images keep their likelihood; only the prior can notice.

Round prior $\frac{1}{2\pi} \cdot e^{-(z_1^2+z_2^2)/2}$: both codes get $\frac{e^{-2}}{2\pi} = 0.0215$. Images alone cannot fix the axes (Locatello et al., 2019).

Prior set by a label u , seen with each image: here, photo taken by day or by night.

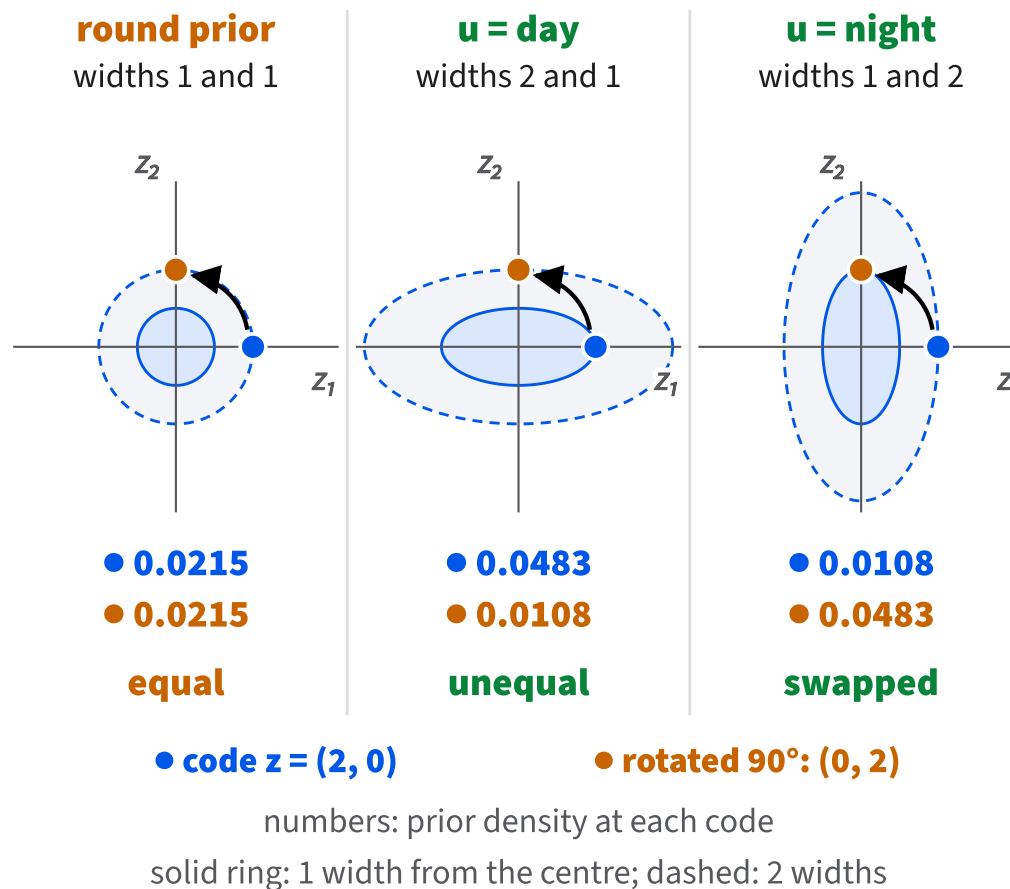
day, standard deviations (widths) 2 and 1:

$$p(z) = \frac{1}{2\pi \cdot 2 \cdot 1} \cdot e^{-(z_1^2/2^2 + z_2^2/1^2)/2}$$

$$p(2, 0) = \frac{e^{-1/2}}{4\pi} = 0.0483, p(0, 2) = \frac{e^{-2}}{4\pi} = 0.0108$$

night (widths 1 and 2) swaps them

Only **sign flips** of the axes keep both priors, so the axes are pinned (Khemakhem et al., 2020).



An extra observation, like the third flip, makes the hidden code identifiable.